

COSE471: Data Science

Classification (2)

Department of Computer Science and Engineering

Korea University

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May 1, 2026

Course Schedule

Week	Day	Content	HW/Project	Note
1	Mar 06	Introduction		
2	Mar 13	Data		
3	Mar 20	Pattern mining		
4	Mar 27	Tutorial 1		<i>Latex, Preprocessing/Pattern mining</i>
5	Apr 03	Clustering		
6	Apr 10	Outlier Detection & Invited Talk		<i>An invited talk from Meta</i>
7	Apr 17	Tutorial 2	● HW (1&2 merged)	<i>Clustering, Outlier detection</i>
8	Apr 24	Classification		
9	May 01	Classification	●	
10	May 08	Tutorial 3		<i>Classification, Feature engineering</i>
11	May 15	Mid-term exam		<i>Offline, Closed-book, No electronic devices</i>
12	May 22	Review & Project Notice	● Project	
13	May 29	Project discussion		
14	Jun 05	Project discussion		
15	Jun 12	Final presentation		
16	Jun 19	Wrap-up and Final report	●	

Announcement

❑ Homework is released!

- You can submit your responses through Google Form
 - Part 1: Data and Pattern Mining
 - Part 2: Clustering
 - Part 3: Outlier Detection and Classification
- Due on May 8 11:59PM
 - Scores and answers will be out on May 12

❑ Mid-term Exam

- May 15 9:00AM
- **Woojung Hall B101 and 205 (two classrooms)**
 - **Not the regular classroom!**

Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - Overview
 - GINI index → CART
 - **Entropy** → ID3
 - Gain ratio → C4.5
- ❑ Classification evaluation
- ❑ Ensemble classifiers

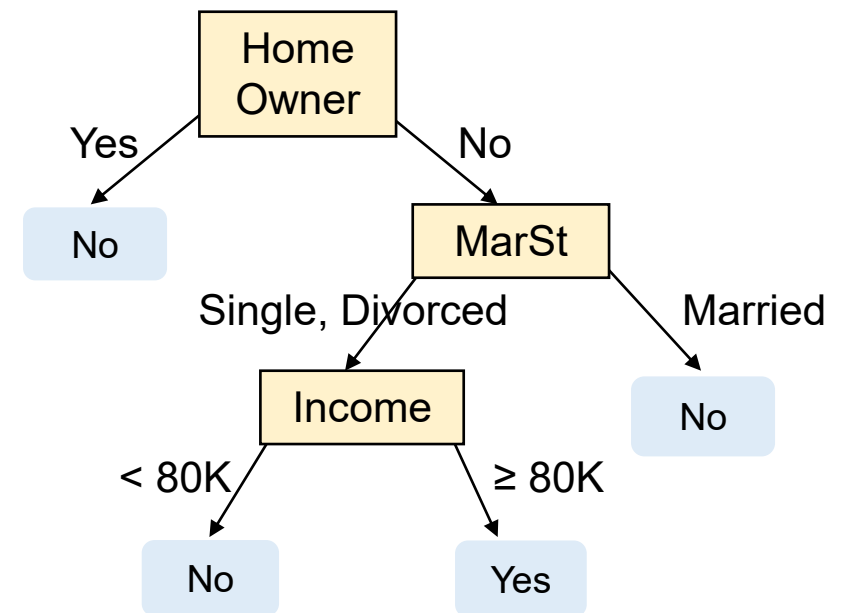
Classification > Methodologies > Decision Tree

- A top-down, recursive, divide-and-conquer process to classify data into labels

	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

Tree
induction
➔
Learning
a model

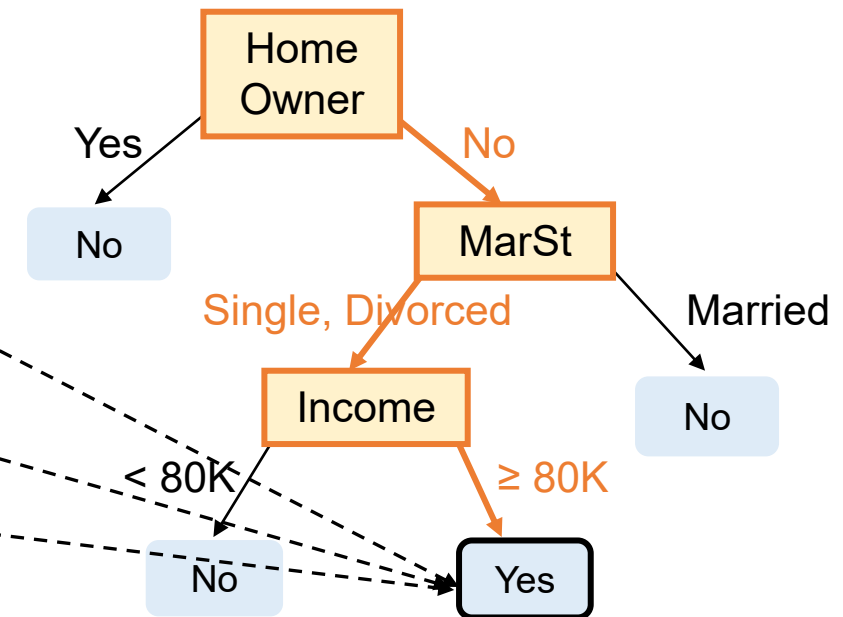
Example decision tree



Classification > Methodologies > Decision Tree

- A top-down, recursive, divide-and-conquer process to classify data into labels

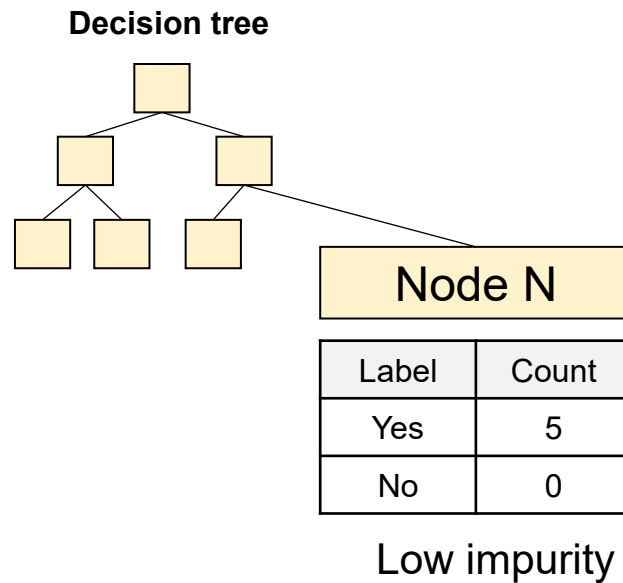
	Binary	Nominal	Ordinal	Numerical	Label
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
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3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes



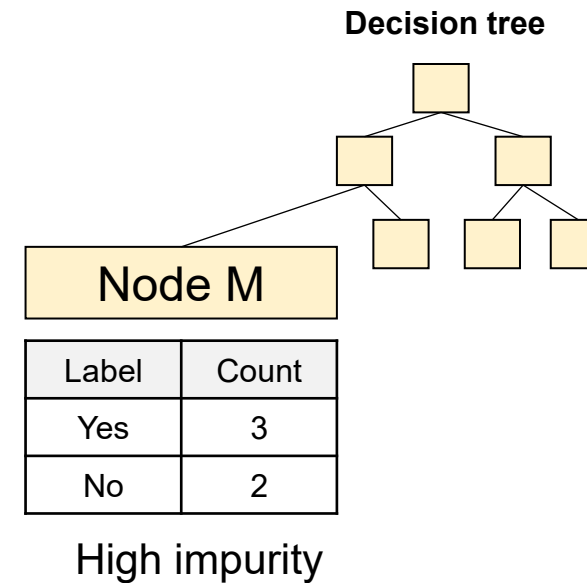
Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)



VS.

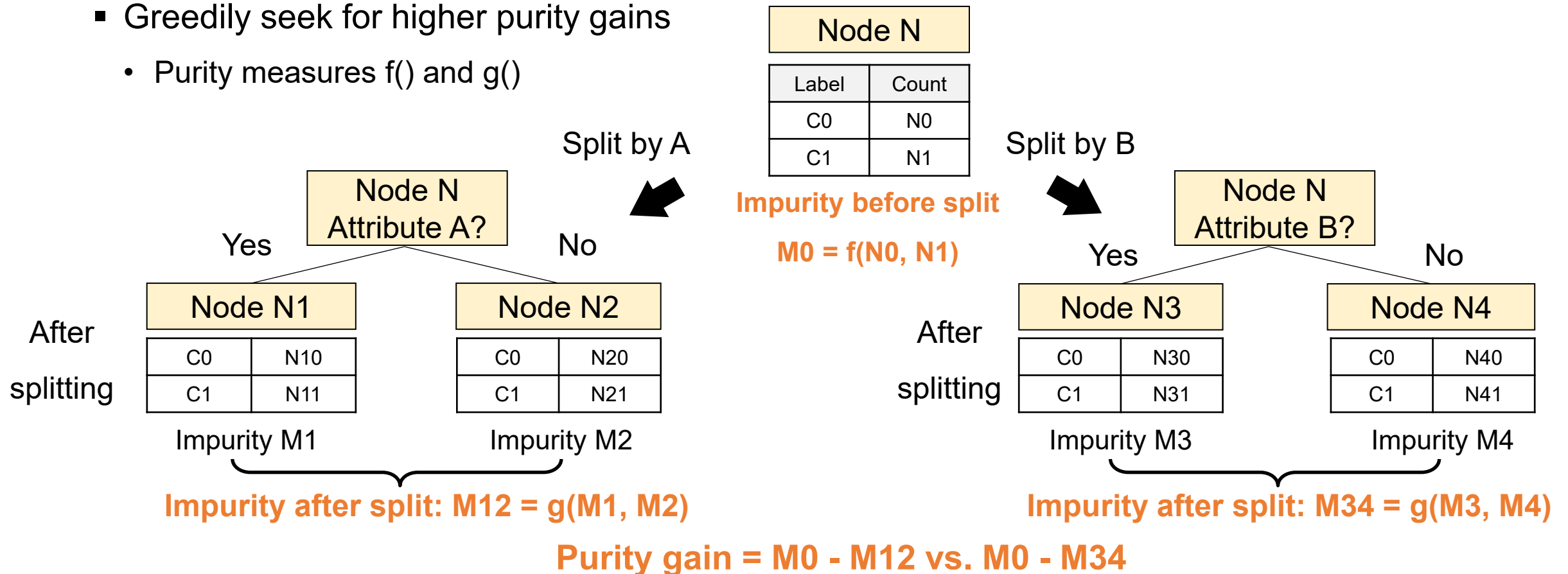


We can say that records falling in node N will have a label “Yes” more likely than those in node M

Classification > Methodologies > Decision Tree

□ How to determine the best split?

- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)
- Greedily seek for higher purity gains
 - Purity measures $f()$ and $g()$



Classification > Methodologies > Decision Tree > GINI

GINI index of a node (i.e., node impurity measure)

- Given a node t , the relative frequency (or probability) $p(j|t)$ of label class j at t

$$p(j|t) = \frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t}$$

- GINI index measures the likelihood of wrong classification, implying node impurity

$$\begin{aligned} GINI(t) &= \sum_j \left(p(j|t) \sum_{k \neq j} p(k|t) \right) = \sum_j p(j|t)(1 - p(j|t)) = 1 - \sum_j p(j|t)^2 \\ &= 1 - \sum_j \left(\frac{\text{\# of records of class } j \text{ in } t}{\text{\# of records in } t} \right)^2 \end{aligned}$$

- Example

Node N	
	
Class label	Count
Yes (blue)	1
No (green)	5

Probability of  = $p(\text{blue} | N) = 1/6$

Probability of  = $p(\text{green} | N) = 5/6$

Miss classify  as  = $1/6 * 5/6 = 5/36$

Misclassify  as  = $5/6 * 1/6 = 5/36$

$GINI(N) = 5/36 + 5/36 = 10/36 = 0.278$

↙

$$\begin{aligned} GINI(N) &= 1 - p(\text{blue} | N)^2 - p(\text{green} | N)^2 \\ &= 1 - (1/6)^2 - (5/6)^2 \\ &= 1 - 1/36 - 25/36 \\ &= 10/36 = 0.278 \end{aligned}$$

■ Classification > Methodologies > Decision Tree > Entropy

□ Entropy (information theory)

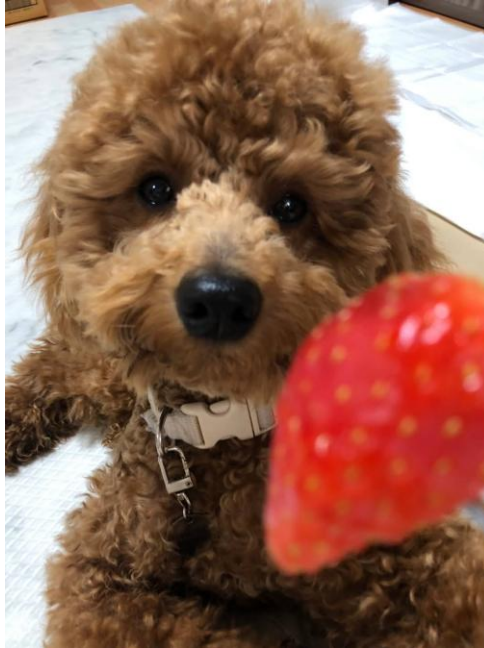
- A measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
 - Higher probability → Lower uncertainty/surprise → Lower information → Lower entropy
 - Lower probability → Higher uncertainty/surprise → Higher information → Higher entropy

Classification > Methodologies > Decision Tree > Entropy

□ Entropy (information theory)

- A measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
 - Higher probability $\rightarrow$ Lower uncertainty/surprise $\rightarrow$ Lower information $\rightarrow$ Lower entropy
 - Lower probability $\rightarrow$ Higher uncertainty/surprise $\rightarrow$ Higher information $\rightarrow$ Higher entropy

Which situation appears to have a higher entropy?



■ Classification > Methodologies > Decision Tree > Entropy

□ Entropy (information theory)

- A measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
- (Shannon) information

$$I(x_i) = \log_2 \left(\frac{1}{P(X = x_i)} \right) = -\log_2(P(X = x_i))$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy (information theory)

- A measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
- (Shannon) information

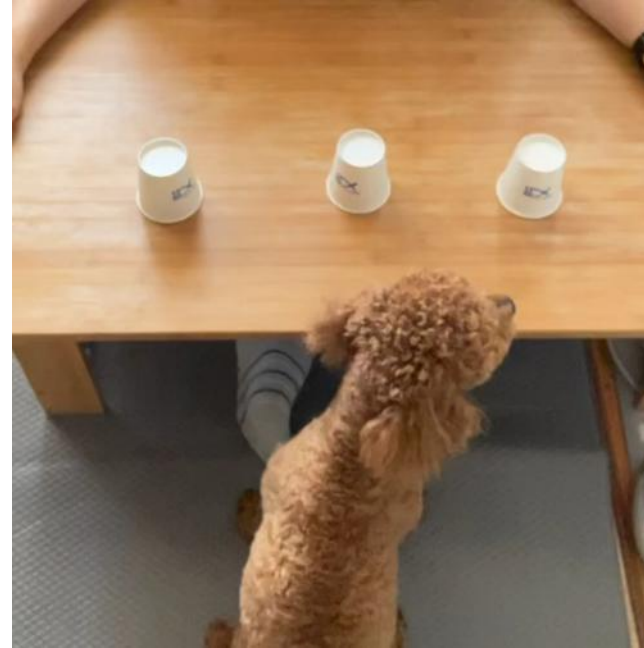
$$I(x_i) = \log_2 \left(\frac{1}{P(X = x_i)} \right) = -\log_2(P(X = x_i))$$



X : a dog snack { something }

$$P(X = \text{strawberry}) = 1$$

$$I(\text{strawberry}) = -\log_2(1) = 0$$



X : a cup with a dog snack

$$P(X = \text{cup}_i) = \frac{1}{3}$$

$$I(\text{cup}_1) = -\log_2 \left(\frac{1}{3} \right) = 1.58$$

$$I(\text{cup}_2) = -\log_2 \left(\frac{1}{3} \right) = 1.58$$

$$I(\text{cup}_3) = -\log_2 \left(\frac{1}{3} \right) = 1.58$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy (information theory)

- A measure of “information”, “surprise”, or “uncertainty” in the variable’s possible outcomes
- (Shannon) information

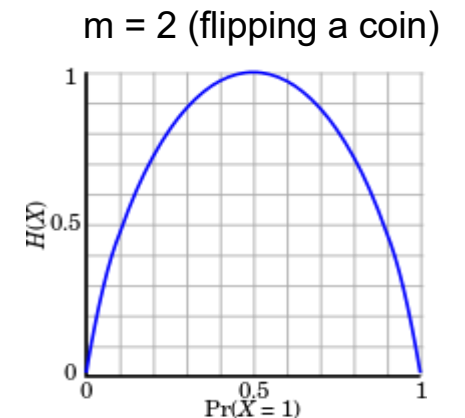
$$I(x_i) = \log_2 \left(\frac{1}{P(X = x_i)} \right) = -\log_2(P(X = x_i))$$

- For a discrete random variable X with m distinct values $\{x_1, x_2, \dots, x_m\}$, entropy of X is

$$H(X) = E[I(X)] = \sum_{i=1}^m p_i I(x_i) = -\sum_{i=1}^m p_i \log_2(p_i), \text{ where } p_i = P(X = x_i)$$

▪ Example

- If probabilities of flipping coins are fair (e.g., 0.5 for head and 0.5 for tail)
 - Highest uncertainty → Highest entropy → highest information
- If probabilities of flipping coins are not fair (e.g., 1.0 for head and 0 for tail)
 - Lowest uncertainty → Lowest entropy → no information



Classification > Methodologies > Decision Tree > Entropy

□ Entropy of a node (i.e., node impurity measure)

- Given a node t , the relative frequency (or probability) $p(j|t)$ of label class j at t

$$p(j|t) = \frac{\# \text{ of records of class } j \text{ in } t}{\# \text{ of records in } t}$$

- Entropy measures the amount of information to identify the label in a node, implying node impurity

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

- Example

Node N	
	
Class label	Count
Yes (blue)	1
No (green)	5

Probability of  = $p(\text{blue}|\text{N}) = 1/6$

Probability of  = $p(\text{green}|\text{N}) = 5/6$

Amount of information for  = $\log_2(1/6) = -2.58$

Amount of information for  = $\log_2(5/6) = -0.26$

$$Entropy(t) = - 1/6 * \log_2(1/6) - 5/6 * \log_2(5/6) = 0.430 + 0.219 = 0.649$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy of a node (i.e., node impurity measure)

- Entropy measures the amount of information to identify the label, implying node impurity

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

- For a node t with n records and c classes,

$$0 \leq Entropy(t) \leq \log_2 c$$

- Upper bound (maximum)

$$- Entropy(t) = - \frac{n/c}{n} \log_2 \frac{n/c}{n} \dots - \frac{n/c}{n} \log_2 \frac{n/c}{n} = - \log_2 \frac{1}{c} = \log_2 c$$

- when records are equally distributed among all classes, implying most information

- Lower bound (minimum)

$$- Entropy(t) = - \frac{n}{n} \log_2 \frac{n}{n} - \frac{0}{n} \log_2 \frac{0}{n} \dots - \frac{0}{n} \log_2 \frac{0}{n} = 0$$

- when all records belong to one class, implying least information

Example with 8 records of 4 classes

Node t1



$$p(\bullet|t1) = p(\bullet|t1) = p(\bullet|t1) = p(\bullet|t1) = 2/8$$

$$Entropy(t1) = 4 \times \left(-\frac{2}{8} \log_2 \frac{2}{8} \right) = \log_2 4 = 2$$

Node t2



$$p(\bullet|t2) = 8/8, p(\bullet|t2) = p(\bullet|t2) = p(\bullet|t2) = 0/8$$

$$Entropy(t2) = 1 \times \left(-\frac{8}{8} \log_2 \frac{8}{8} \right) + 3 \times \left(-\frac{0}{8} \log_2 \frac{0}{8} \right) = 0$$

■ Classification > Methodologies > Decision Tree > Entropy

- When a node t is split into k child nodes, the node information of resulting split is

$$Entropy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

n = the number of records at node t
 n_i = the number of records at t 's child node i

- The quality of split is computed as the purity gain (or information gain)

$$GAIN_{split} = Entropy(t) - Entropy_{split}$$

Classification > Methodologies > Decision Tree > Entropy

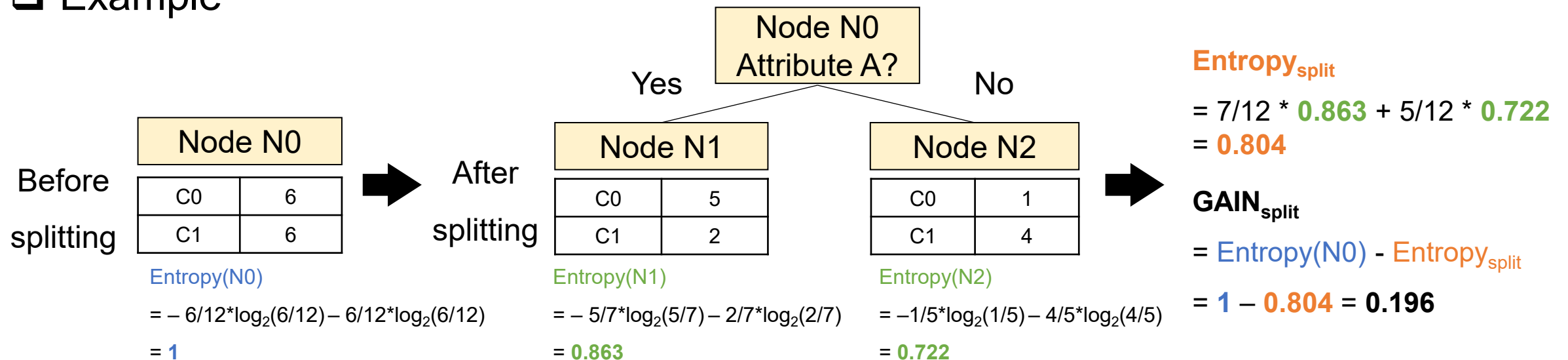
- When a node t is split into k child nodes, the node information of resulting split is

$$Entropy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i) \quad \begin{array}{l} n = \text{the number of records at node } t \\ n_i = \text{the number of records at } t\text{'s child node } i \end{array}$$

- The quality of split is computed as the purity gain (or information gain)

$$GAIN_{split} = Entropy(t) - Entropy_{split}$$

- Example



Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

$$Entorpy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

$$GAIN_{split} = Entorpy(t) - Entorpy_{split}$$

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Root Node	
PlayGolf = Yes	9
PlayGolf = No	5

$$Entropy(\text{Root}) = -\frac{9}{14} \log_2\left(\frac{9}{14}\right) - \frac{5}{14} \log_2\left(\frac{5}{14}\right) = 0.940$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

$$Entorpy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

$$GAIN_{split} = Entorpy(t) - Entorpy_{split}$$

Root Node	
PlayGolf = Yes	9
PlayGolf = No	5

Entropy(Root) = 0.940

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Split option 1

	Outlook		
	Sunny	Overcast	Rainy
Yes	2	4	3
No	3	0	2

Split option 2

	Temp		
	Cold	Mild	Hot
Yes	3	4	2
No	1	2	2

Split option 3

	Humidity	
	Normal	High
Yes	6	3
No	1	4

Split option 4

	Windy	
	True	False
Yes	3	6
No	3	2

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

$$Entropy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

$$GAIN_{split} = Entropy(t) - Entropy_{split}$$

Root Node	
PlayGolf = Yes	9
PlayGolf = No	5

Entropy(Root) = 0.940

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Split option 1

	Outlook		
	Sunny	Overcast	Rainy
Yes	2	4	3
No	3	0	2

$$Entropy(\text{Sunny}) = -\frac{2}{5} \log_2 \left(\frac{2}{5}\right) - \frac{3}{5} \log_2 \left(\frac{3}{5}\right) = 0.971$$

$$Entropy(\text{Overcast}) = -\frac{4}{4} \log_2 \left(\frac{4}{4}\right) - \frac{0}{4} \log_2 \left(\frac{0}{4}\right) = 0$$

$$Entropy(\text{Rainy}) = -\frac{3}{5} \log_2 \left(\frac{3}{5}\right) - \frac{2}{5} \log_2 \left(\frac{2}{5}\right) = 0.971$$

Entropy_{split}

$$= \frac{5}{14} \cdot Entropy(\text{Sunny}) + \frac{4}{14} \cdot Entropy(\text{Overcast}) + \frac{5}{14} \cdot Entropy(\text{Rainy})$$

$$= \frac{5}{14} \cdot 0.971 + \frac{4}{14} \cdot 0 + \frac{5}{14} \cdot 0.971$$

$$= 0.694$$

GAIN_{split}

$$= Entropy(\text{Root}) - Entropy_{split} = 0.940 - 0.694 = 0.246$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t)$$

$$Entorpy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entropy(i)$$

$$GAIN_{split} = Entorpy(t) - Entorpy_{split}$$

Root Node	
PlayGolf = Yes	9
PlayGolf = No	5

Entropy(Root) = 0.940

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
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6	Rainy	Cool	Normal	True	No
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11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

 Split option 1

	Outlook		
	Sunny	Overcast	Rainy
Yes	2	4	3
No	3	0	2

$GAIN_{split} = 0.246$

(Highest GAIN)

 Split option 2

	Temp		
	Cold	Mild	Hot
Yes	3	4	2
No	1	2	2

$GAIN_{split} = 0.029$

 Split option 3

	Humidity	
	Normal	High
Yes	6	3
No	1	4

$GAIN_{split} = 0.151$

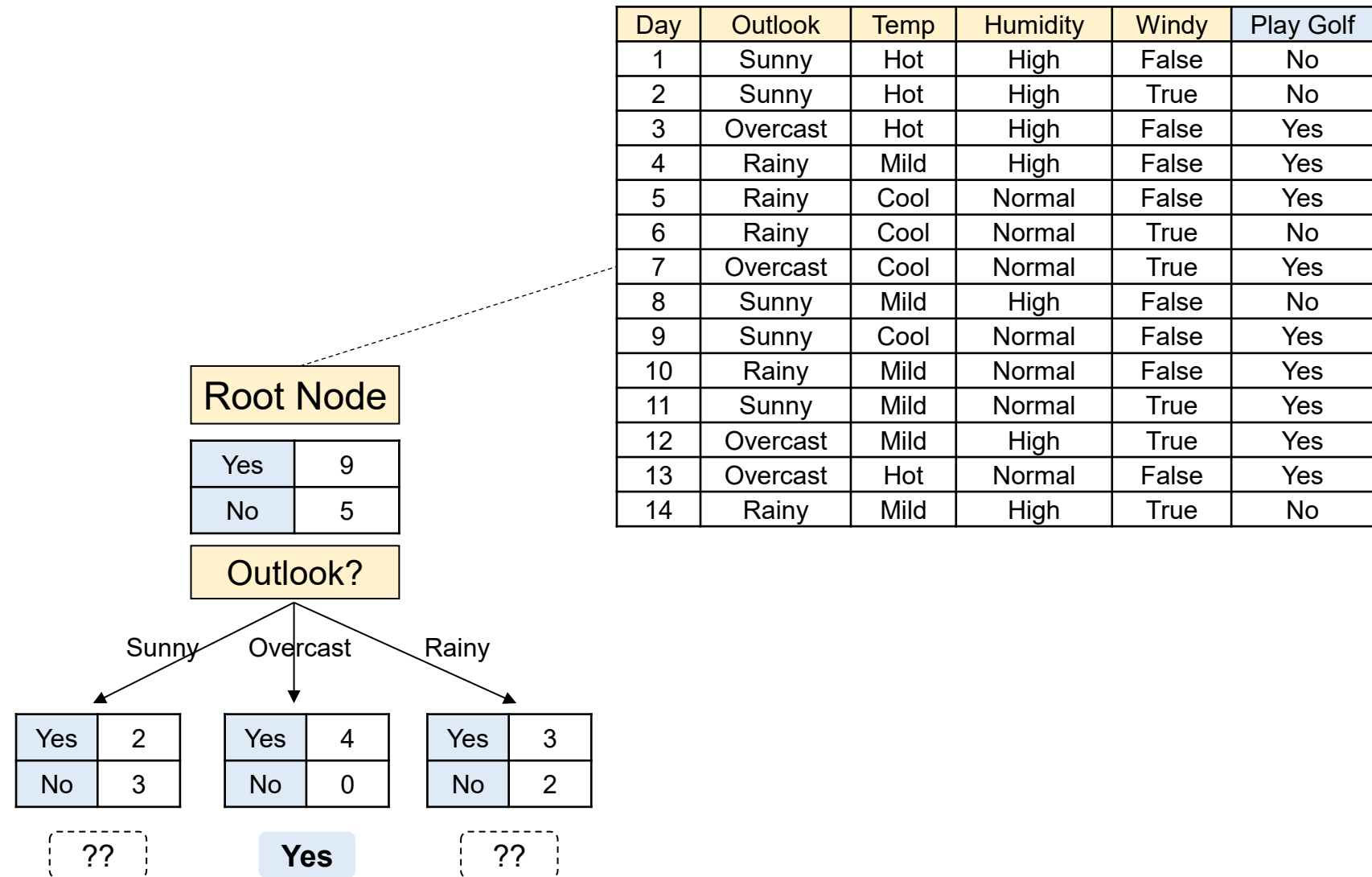
 Split option 4

	Windy	
	True	False
Yes	3	6
No	3	2

$GAIN_{split} = 0.048$

Classification > Methodologies > Decision Tree > Entropy

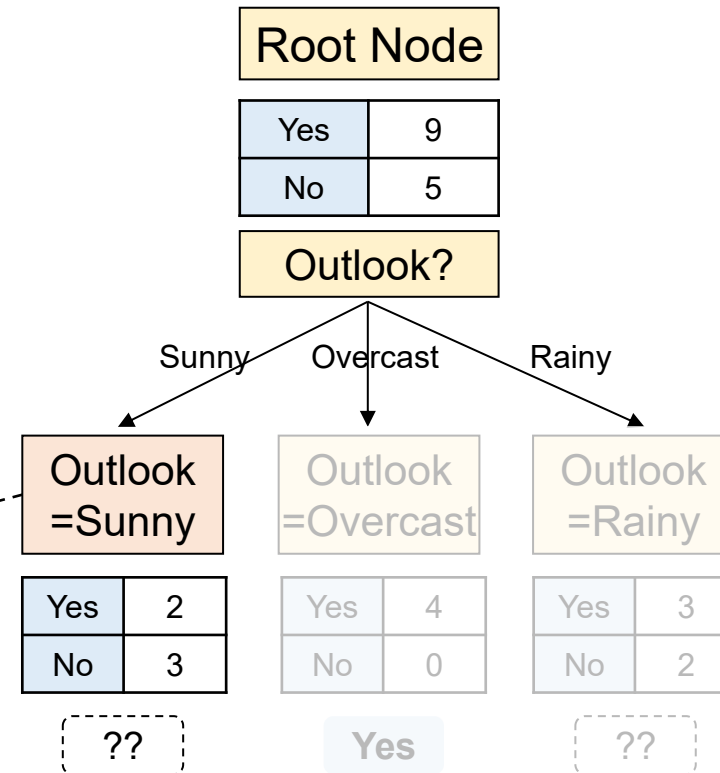
□ Entropy example



Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

Day	Outlook	Temp	Humidity	Windy	Play Golf
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3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
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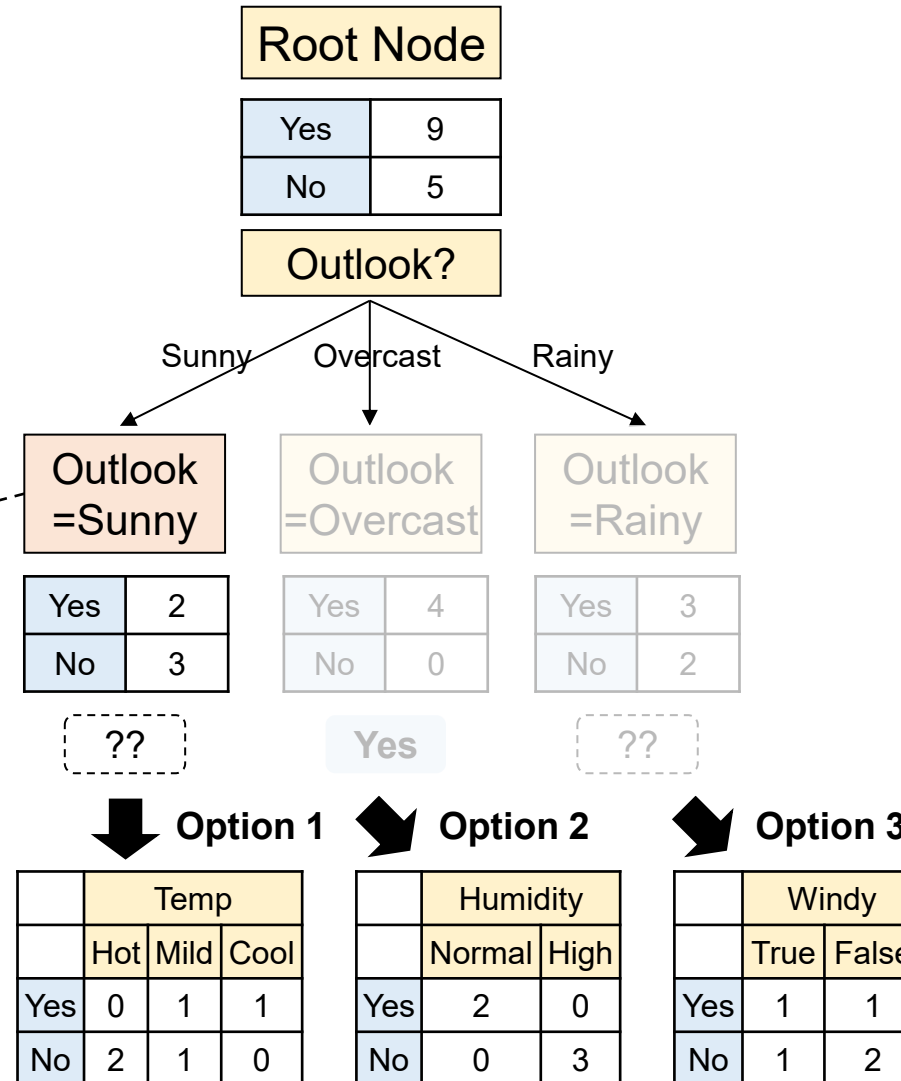


$$\text{Entropy}(\text{Sunny}) = -\frac{2}{5}\log_2\left(\frac{2}{5}\right) - \frac{3}{5}\log_2\left(\frac{3}{5}\right) = 0.971$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

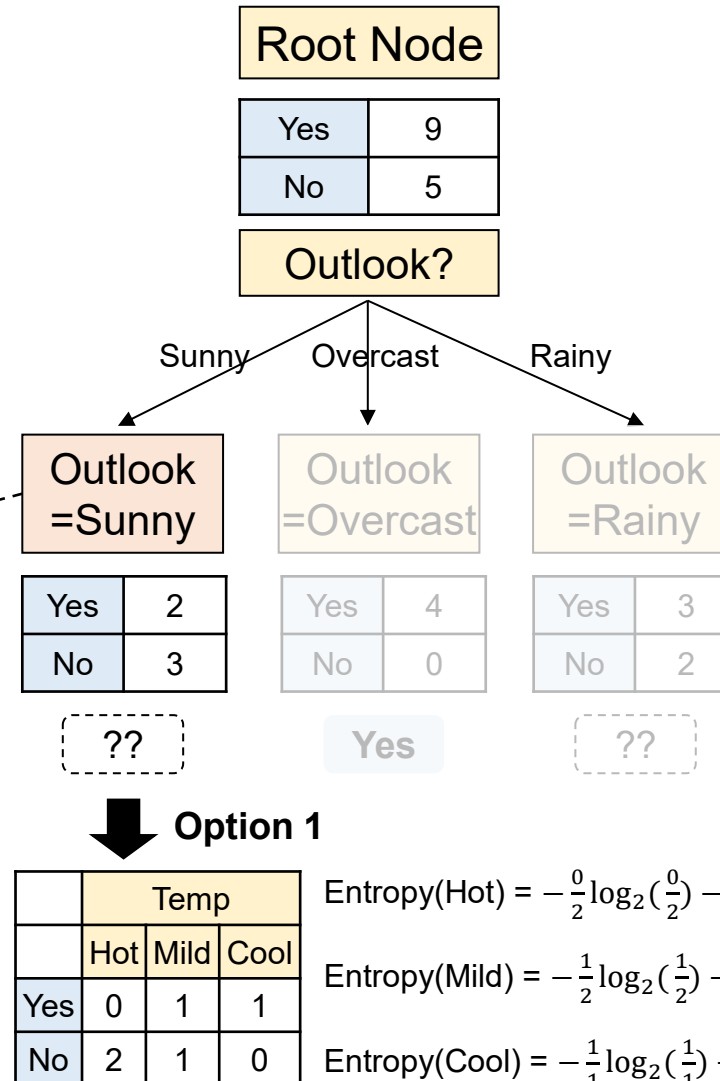
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10	Rainy	Mild	Normal	False	Yes
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12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No



Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
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13	Overcast	Hot	Normal	False	Yes
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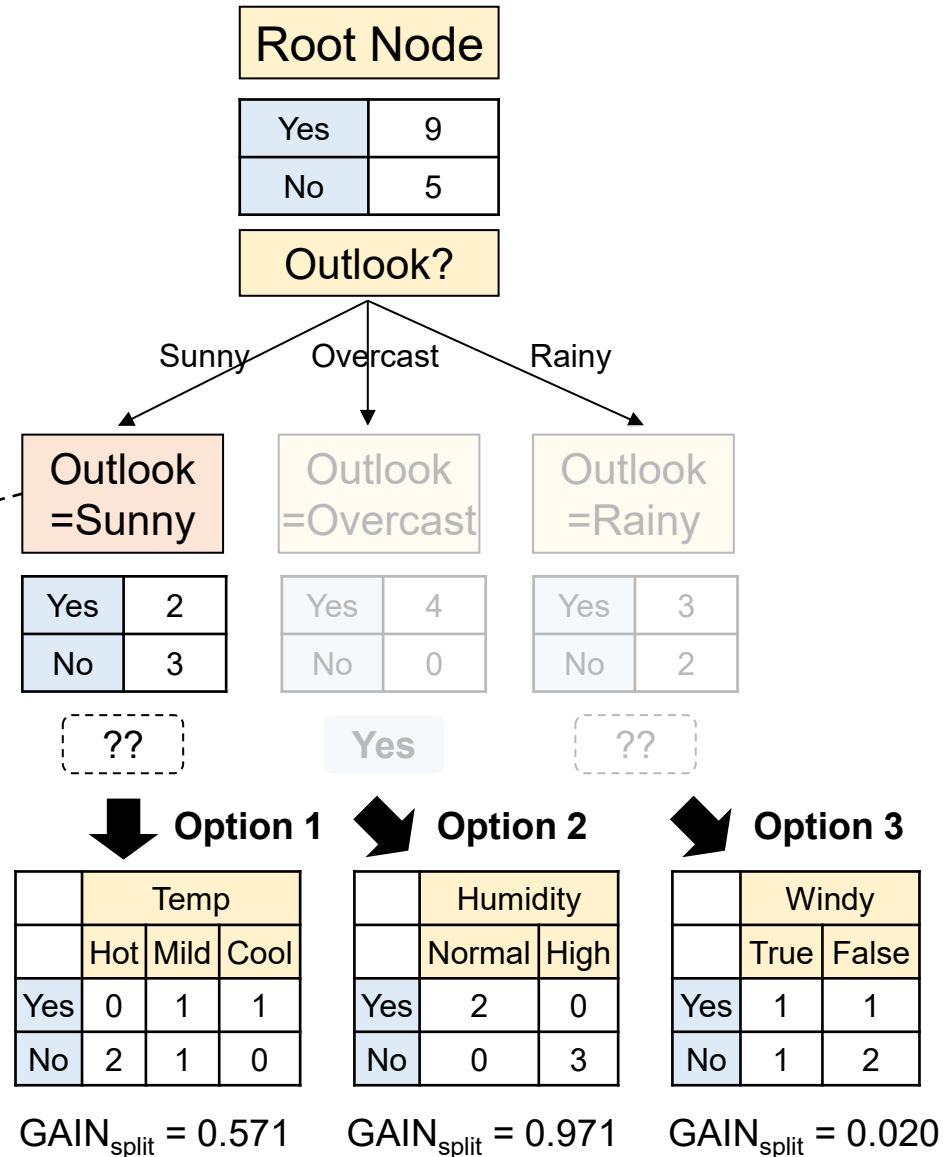
$$\begin{aligned}
 \text{Entropy}_{\text{split}} &= 2/5 * \text{Entropy}(\text{Hot}) \\
 &\quad + 2/5 * \text{Entropy}(\text{Mild}) \\
 &\quad + 1/5 * \text{Entropy}(\text{Cool}) \\
 &= 0.4 \\
 \text{GAIN}_{\text{split}} &= \text{Entropy}(\text{Sunny}) - \text{Entropy}_{\text{split}} \\
 &= 0.971 - 0.4 = 0.571
 \end{aligned}$$

$$\begin{aligned}
 \text{Entropy}(\text{Hot}) &= -\frac{0}{2} \log_2\left(\frac{0}{2}\right) - \frac{2}{2} \log_2\left(\frac{2}{2}\right) = 0 \\
 \text{Entropy}(\text{Mild}) &= -\frac{1}{2} \log_2\left(\frac{1}{2}\right) - \frac{1}{2} \log_2\left(\frac{1}{2}\right) = 1 \\
 \text{Entropy}(\text{Cool}) &= -\frac{1}{1} \log_2\left(\frac{1}{1}\right) - \frac{0}{1} \log_2\left(\frac{0}{1}\right) = 0
 \end{aligned}$$

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

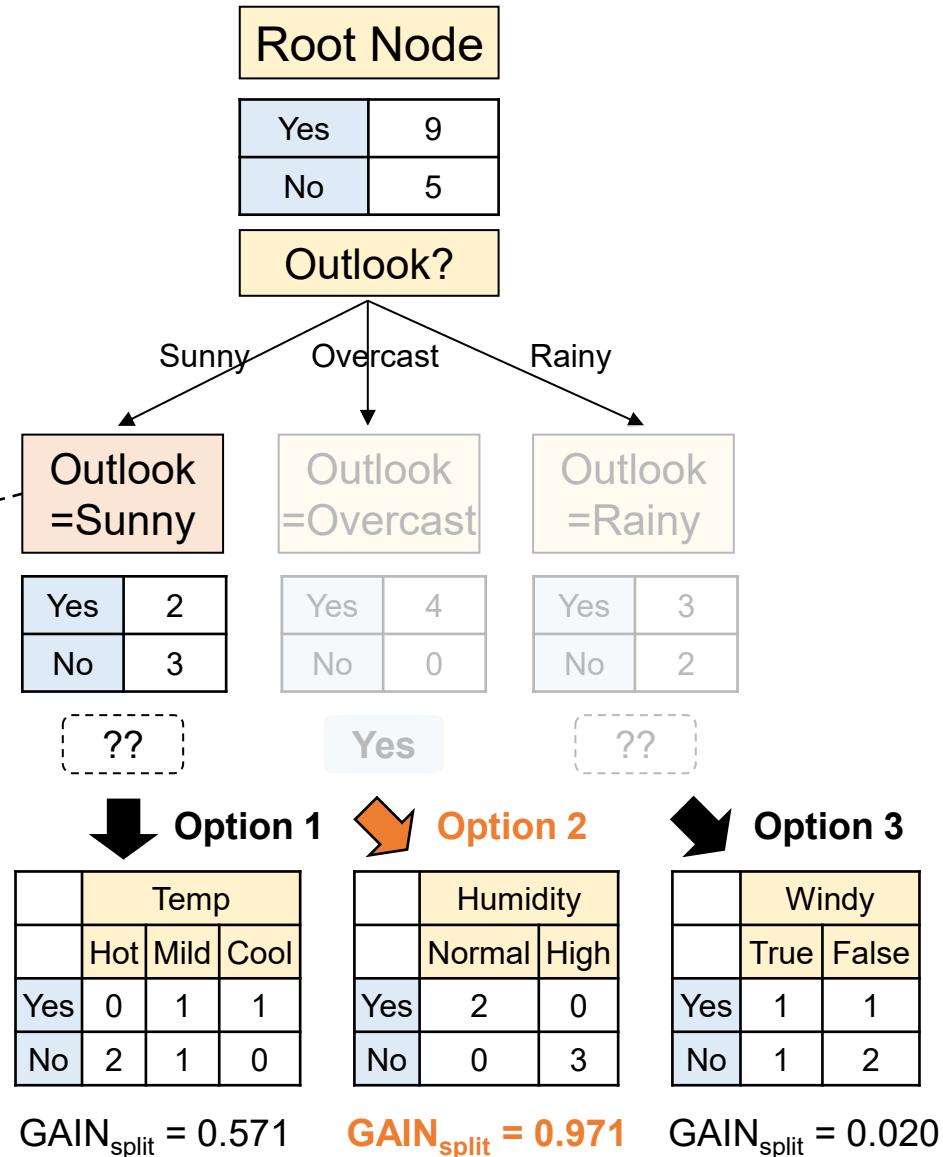
Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No



Classification > Methodologies > Decision Tree > Entropy

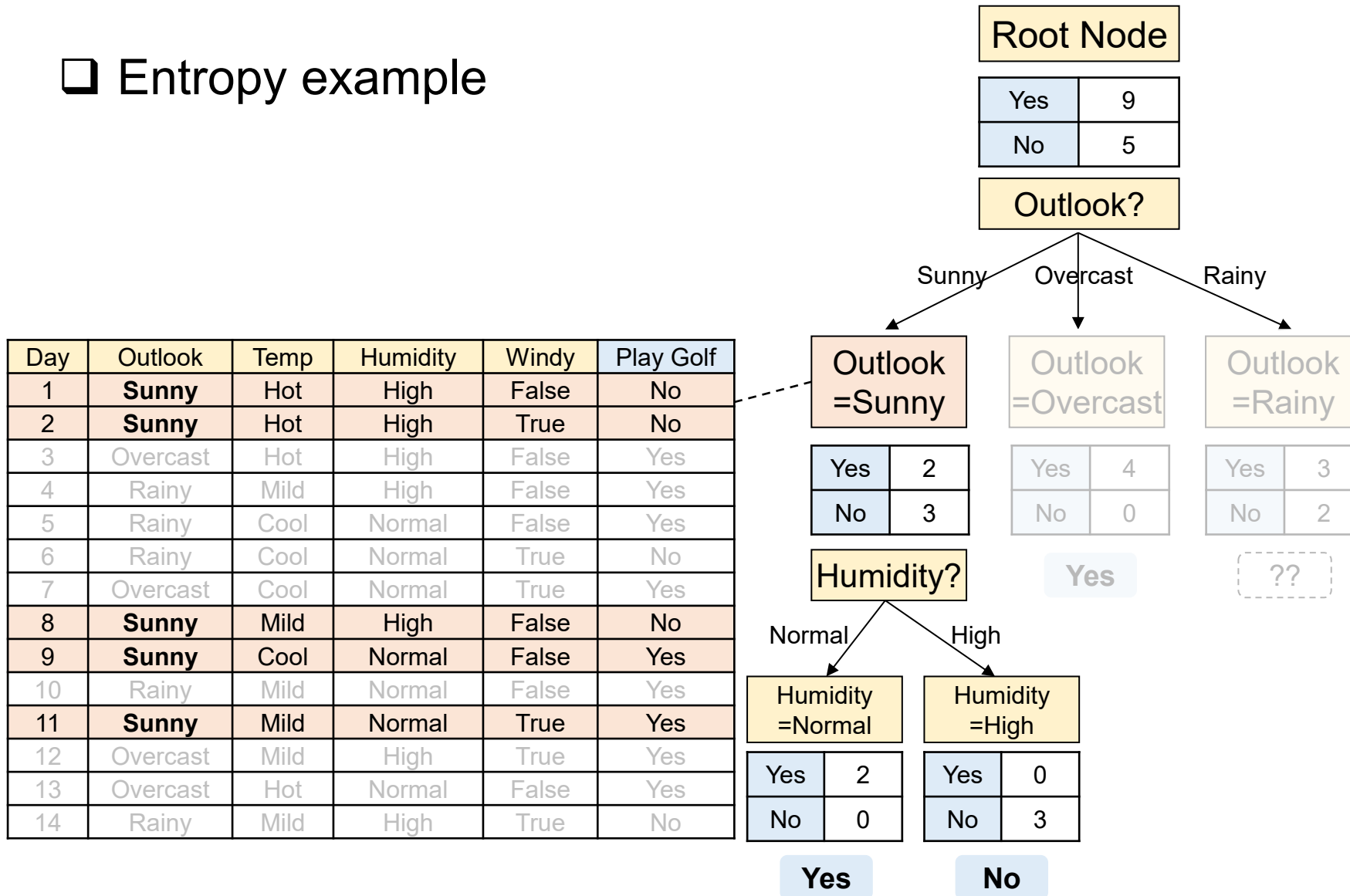
□ Entropy example

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No



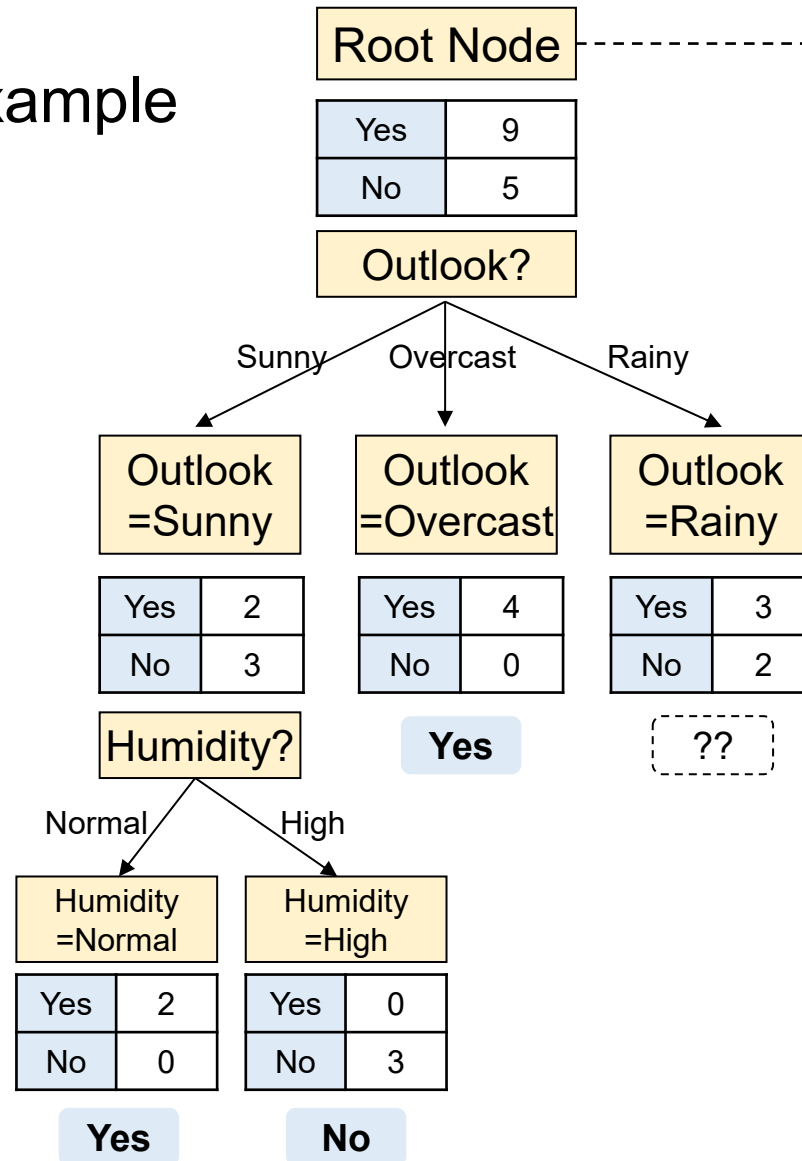
Classification > Methodologies > Decision Tree > Entropy

□ Entropy example



Classification > Methodologies > Decision Tree > Entropy

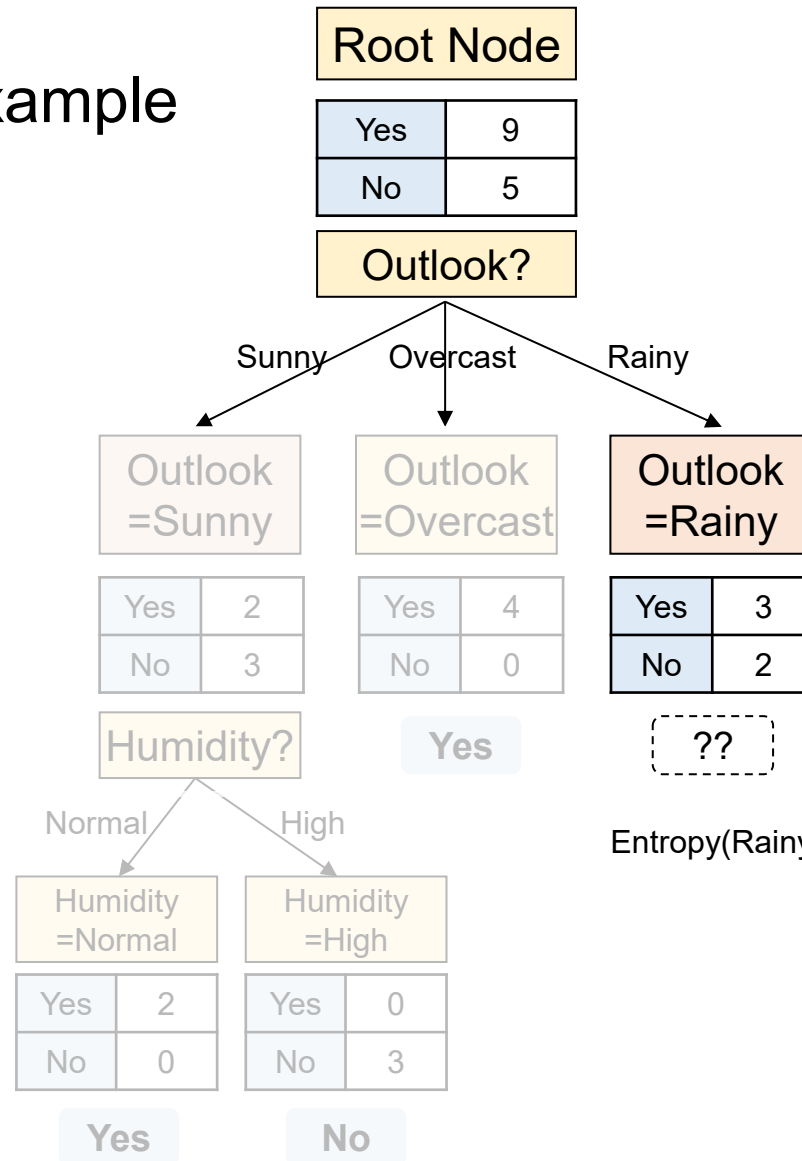
□ Entropy example



Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

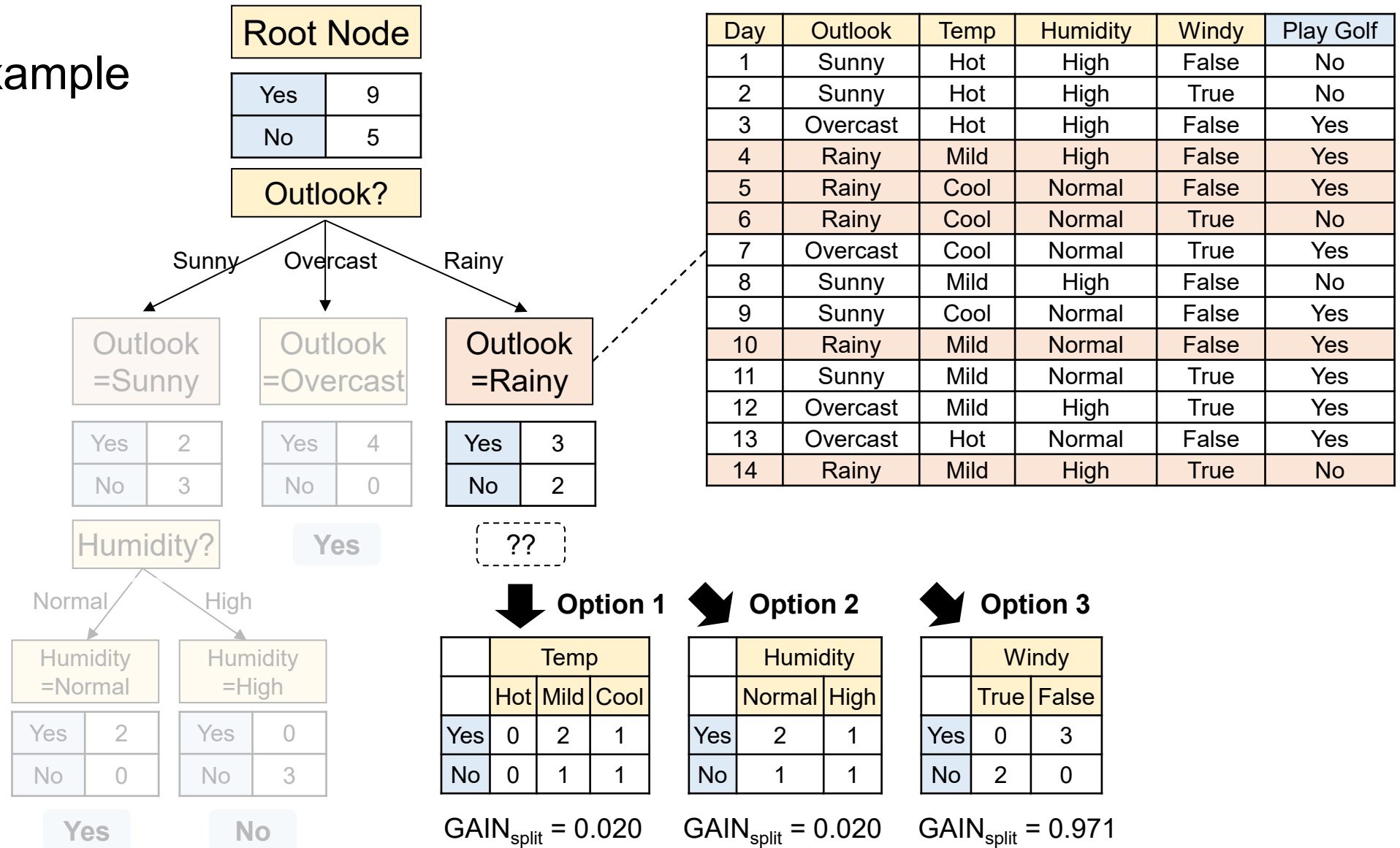


Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

$$\text{Entropy(Rainy)} = -\frac{3}{5}\log_2\left(\frac{3}{5}\right) - \frac{2}{5}\log_2\left(\frac{2}{5}\right) = 0.971$$

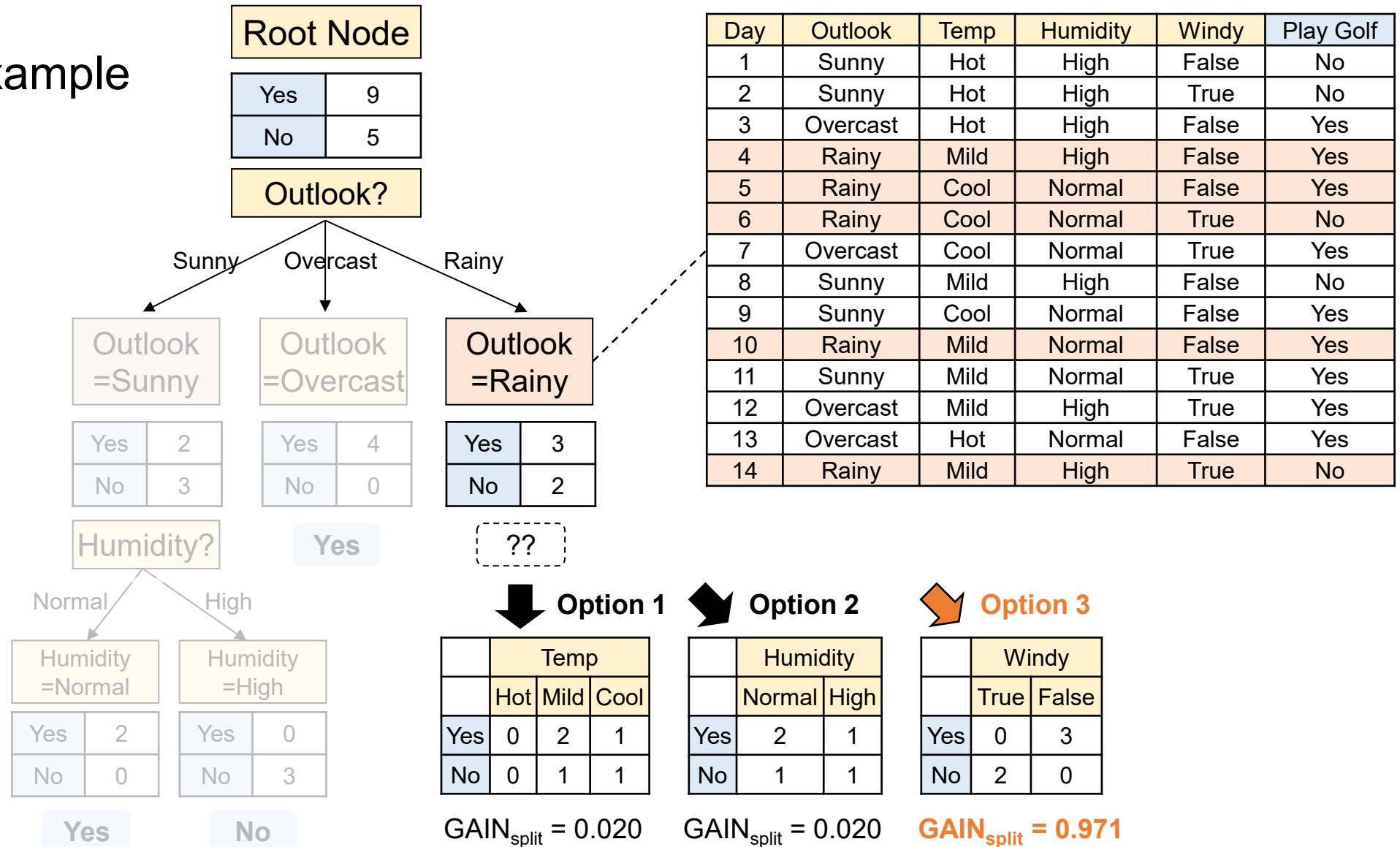
Classification > Methodologies > Decision Tree > Entropy

□ Entropy example



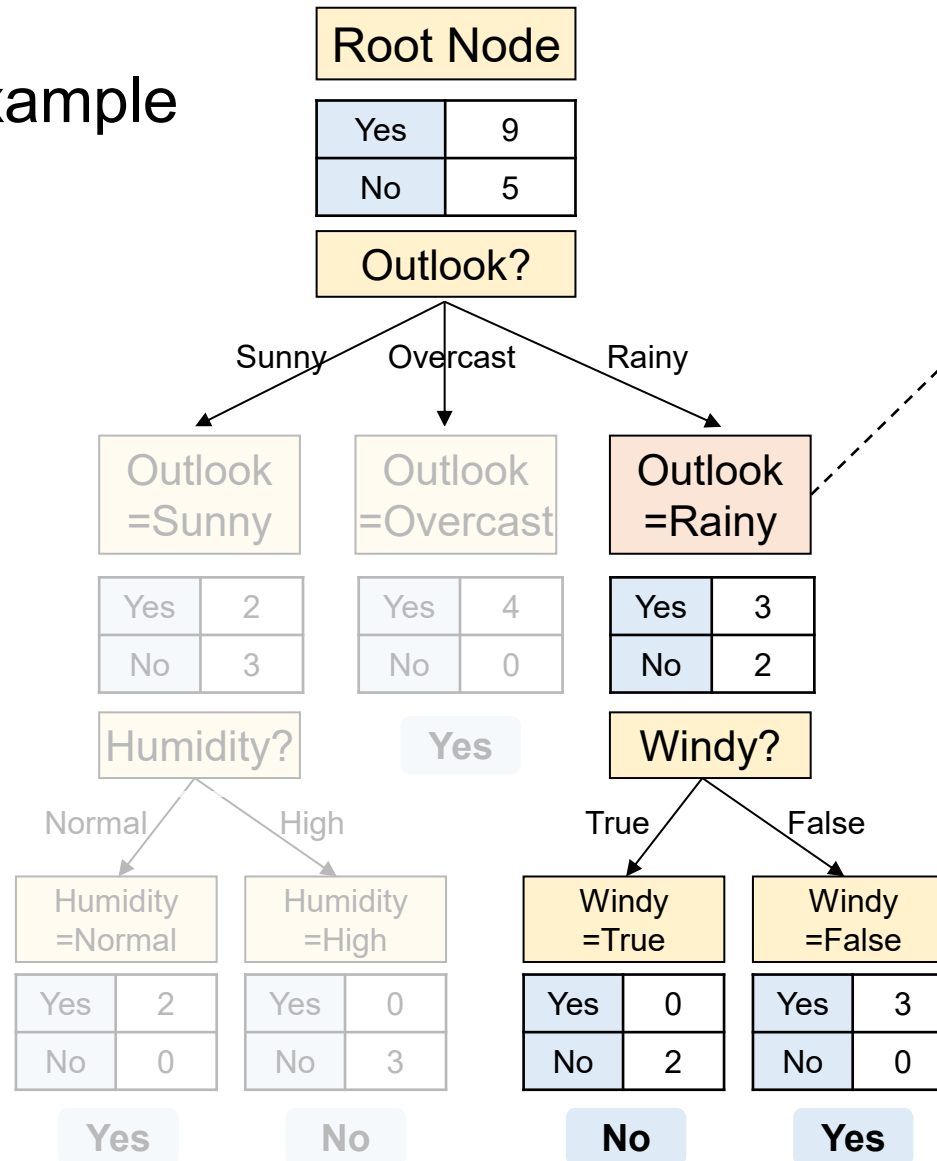
Classification > Methodologies > Decision Tree > Entropy

□ Entropy example



Classification > Methodologies > Decision Tree > Entropy

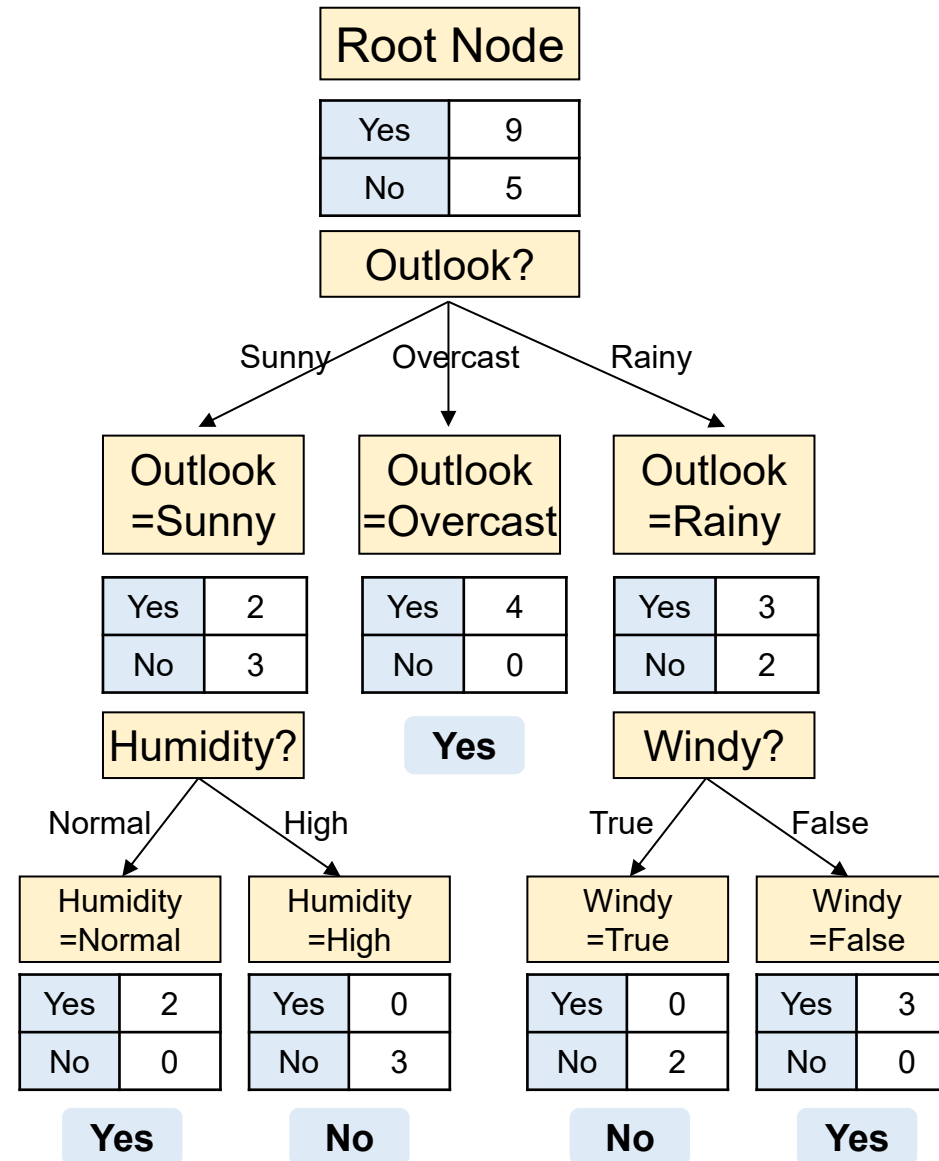
□ Entropy example



Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No

Classification > Methodologies > Decision Tree > Entropy

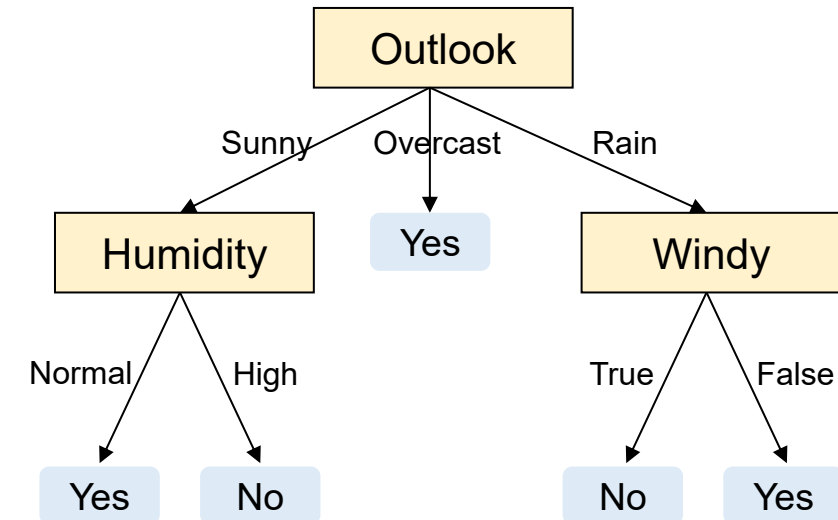
□ Entropy example



Classification > Methodologies > Decision Tree > Entropy

□ Entropy example

Day	Outlook	Temp	Humidity	Windy	Play Golf
1	Sunny	Hot	High	False	No
2	Sunny	Hot	High	True	No
3	Overcast	Hot	High	False	Yes
4	Rainy	Mild	High	False	Yes
5	Rainy	Cool	Normal	False	Yes
6	Rainy	Cool	Normal	True	No
7	Overcast	Cool	Normal	True	Yes
8	Sunny	Mild	High	False	No
9	Sunny	Cool	Normal	False	Yes
10	Rainy	Mild	Normal	False	Yes
11	Sunny	Mild	Normal	True	Yes
12	Overcast	Mild	High	True	Yes
13	Overcast	Hot	Normal	False	Yes
14	Rainy	Mild	High	True	No



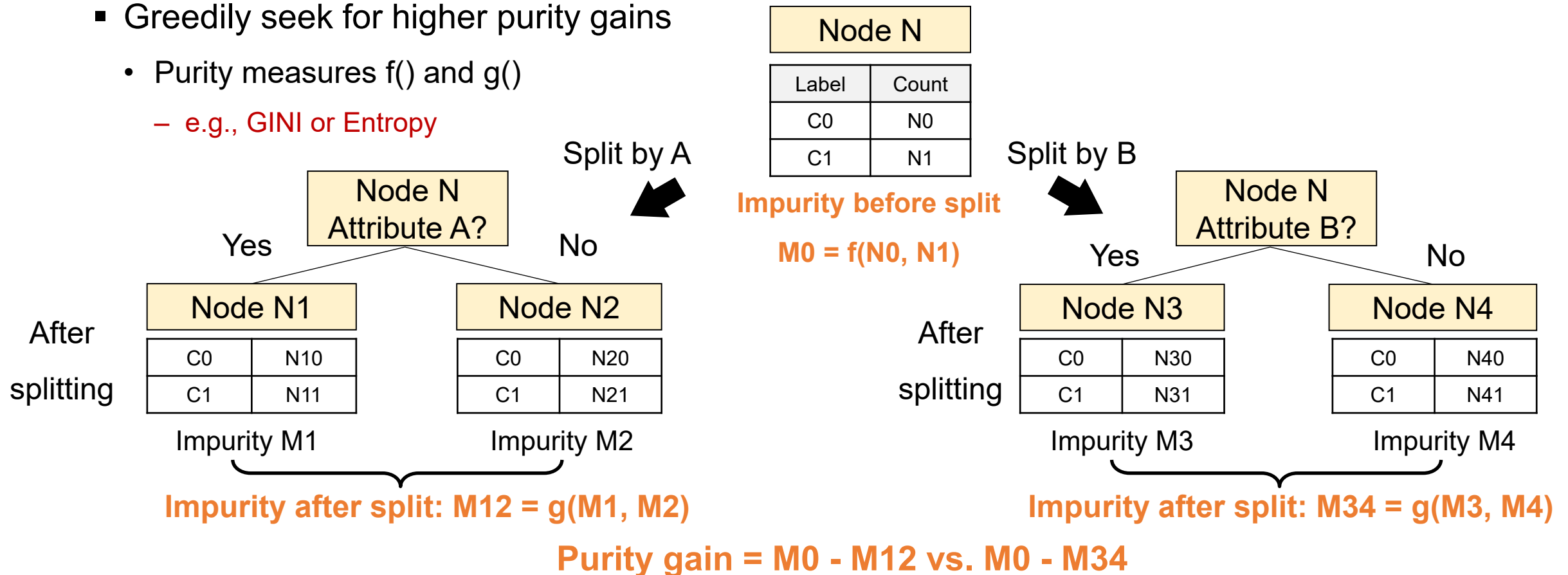
Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - Overview
 - GINI index → CART
 - Entropy → ID3
 - **Gain ratio → C4.5**
- ❑ Classification evaluation
- ❑ Ensemble classifiers

Classification > Methodologies > Decision Tree

□ How to determine the best split?

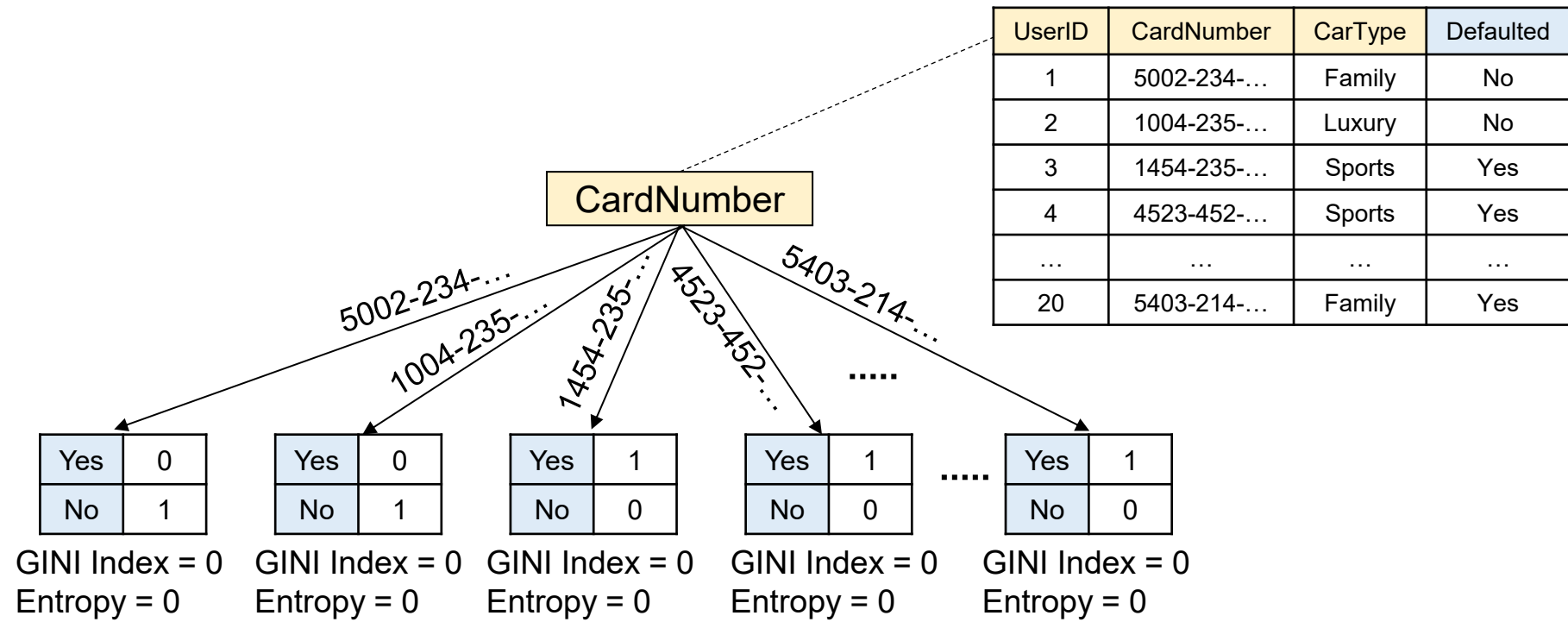
- Intuition: nodes with purer class distribution are preferred (i.e., low impurity)
- Greedily seek for higher purity gains
 - Purity measures $f()$ and $g()$
 - e.g., GINI or Entropy



Classification > Methodologies > Decision Tree > Gain Ratio

❑ Drawback of purity gain (or information gain)

- Tend to prefer splits that result in large number of partitions, each being small but pure
- May lead to less generalizable models (i.e., overfitted to a given data set)
- Example

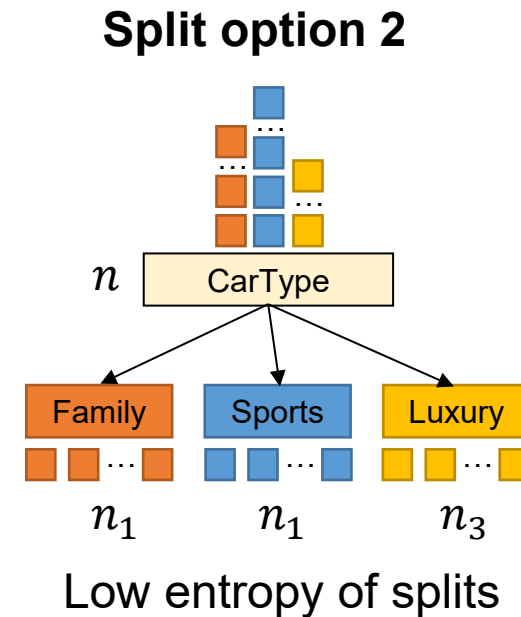
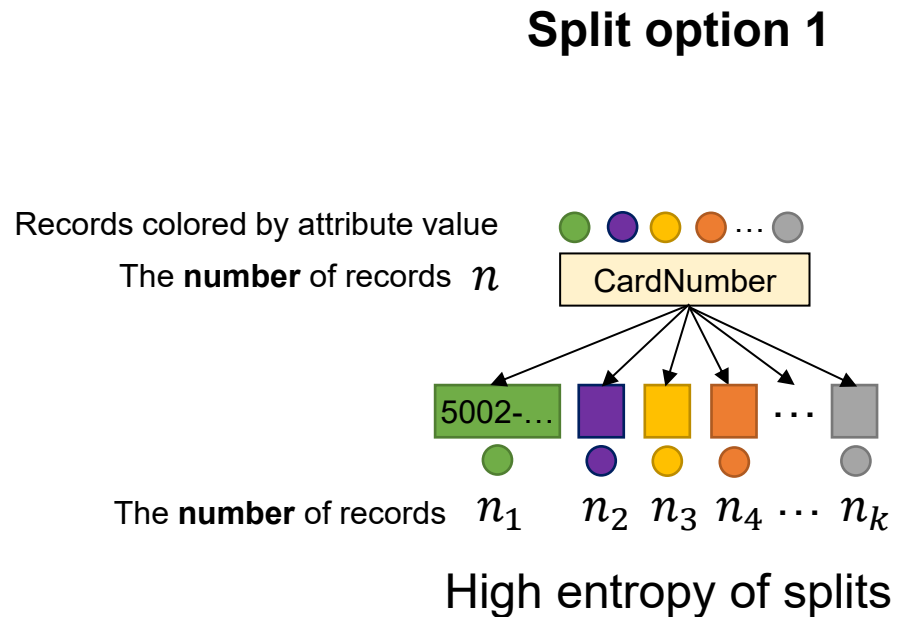


$GINI_{split}$ or $Entropy_{split} = 0 \rightarrow \text{Highest } GAIN_{split}$

Classification > Methodologies > Decision Tree > Gain Ratio

□ Gain ratio

- A normalization to purity gain (or information gain)
 - Adjusts the gain by the entropy of the child node sizes (not the class distribution in each node)
 - Penalize the splitting into too many partitions (i.e., high entropy)



Classification > Methodologies > Decision Tree > Gain Ratio

□ Gain ratio

- A normalization to purity gain (or information gain)
- [Recall] when a node t is split into k child nodes, the purity gain by GINI or Entropy
 - n = the number of records at node t , n_i = the number of records at child node i , $p(j|t)$ = the ratio of records of class j at node t

$$GINI(t) = 1 - \sum_j p(j|t)^2 \quad \rightarrow GINI_{split} = \sum_{i=1}^k \frac{n_i}{n} GINI(i) \quad \rightarrow GAIN_{split} = GINI(t) - GINI_{split}$$

$$Entropy(t) = - \sum_j p(j|t) \log_2 p(j|t) \quad \rightarrow Entorpy_{split} = \sum_{i=1}^k \frac{n_i}{n} Entorpy(i) \quad \rightarrow GAIN_{split} = Entorpy(t) - Entorpy_{split}$$

- Split info (i.e., the entropy of child node sizes) is computed as
 - n = the number of records at node t , n_i = the number of records at child node i

$$SplitInfo = - \sum_{i=1}^k \frac{n_i}{n} \log_2 \frac{n_i}{n}$$

- Gain ratio is computed as

$$GainRatio = \frac{GAIN_{split}}{SplitInfo}$$

Classification > Methodologies > Decision Tree > Gain Ratio

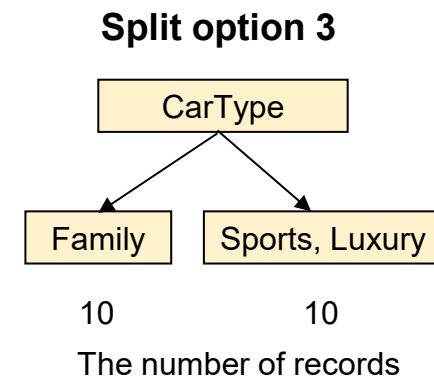
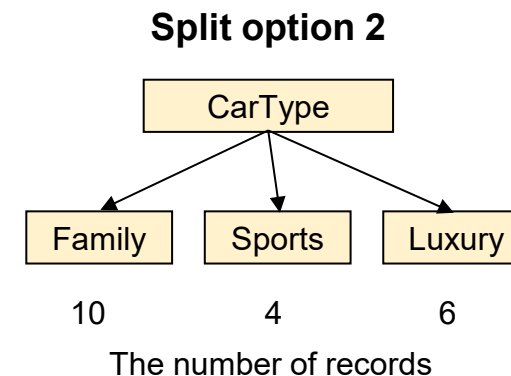
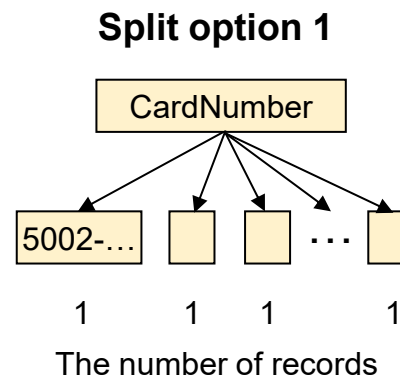
□ Gain ratio

- A normalization to purity gain (or information gain)
- When a node t is split into k child nodes, the gain ratio is computed as:
 - n = the number of records at node t , n_i = the number of records at child i

$$SplitInfo = - \sum_{i=1}^k \frac{n_i}{n} \log_2 \frac{n_i}{n} \rightarrow GainRatio = \frac{GAIN_{split}}{SplitInfo}$$

□ Example

UserID	CardNumber	CarType	Defaulted
1	5002-234-...	Family	No
2	1004-235-...	Luxury	No
3	1454-235-...	Sports	Yes
4	4523-452-...	Sports	Yes
...	...	...	...
20	5403-214-...	Family	Yes



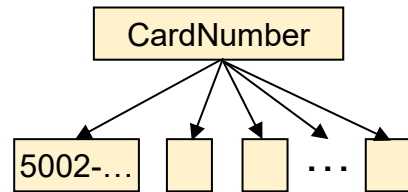
Classification > Methodologies > Decision Tree > Gain Ratio

□ Example (GINI index)

$$SplitInfo = - \sum_{i=1}^k \frac{n_i}{n} \log_2 \frac{n_i}{n} \rightarrow GainRatio = \frac{GAIN_{split}}{SplitInfo}$$

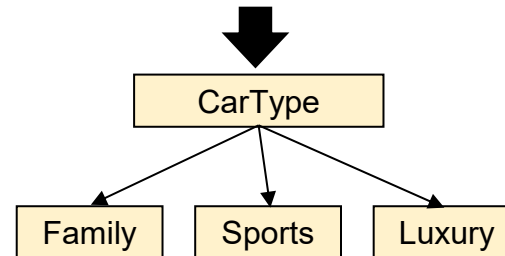
Root node: 9 Yes, 11 No → GINI index = 0.495

UserID	CardNumber	CarType	Defaulted
1	5002-234-...	Family	No
2	1004-235-...	Luxury	No
3	1454-235-...	Sports	Yes
4	4523-452-...	Sports	Yes
...	...	...	...
20	5403-214-...	Family	Yes

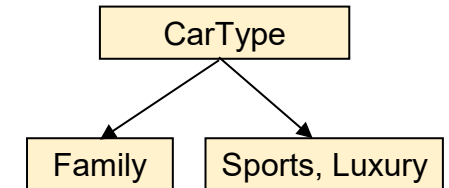


	CardNumber			
	5002-...	1004-...	...	5403-...
Yes	0	0	...	1
No	1	1	...	0
GINI	0	0	0	0
GINI _{split}	0			
GAIN_{split}	0.495 - 0 = 0.495			

Best split?



	CarType		
	Family	Sports	Luxury
Yes	8	0	1
No	2	4	5
GINI	0.32	0	0.28
GINI _{split}	0.243		
GAIN_{split}	0.495 - 0.243 = 0.252		



	CarType	
	Family	Sports, Luxury
Yes	8	1
No	2	9
GINI	0.32	0.18
GINI _{split}	0.25	
GAIN_{split}	0.495 - 0.25 = 0.245	

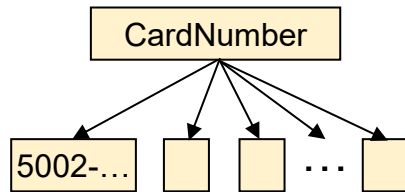
Classification > Methodologies > Decision Tree > Gain Ratio

□ Example (GINI index)

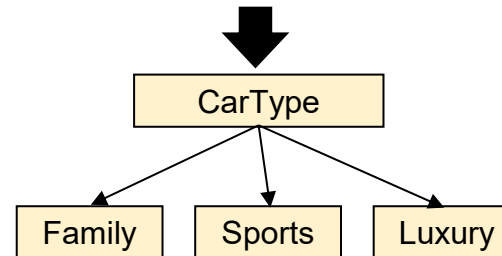
Root node: 9 Yes, 11 No → GINI index = 0.495

UserID	CardNumber	CarType	Defaulted
1	5002-234-...	Family	No
2	1004-235-...	Luxury	No
3	1454-235-...	Sports	Yes
4	4523-452-...	Sports	Yes
...	...	...	...
20	5403-214-...	Family	Yes

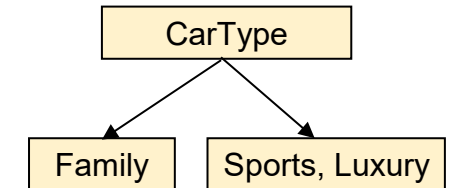
$$SplitInfo = - \sum_{i=1}^k \frac{n_i}{n} \log_2 \frac{n_i}{n} \rightarrow GainRatio = \frac{GAIN_{split}}{SplitInfo}$$



	CardNumber			
	5002-...	1004-...	...	5403-...
Yes	0	0	...	1
No	1	1	...	0
GINI	0	0	0	0
GINI _{split}	0			
GAIN _{split}	0.495 - 0 = 0.495			
SplitInfo	$-\frac{1}{20} \log_2 \frac{1}{20} - \frac{1}{20} \log_2 \frac{1}{20} - \dots - \frac{1}{20} \log_2 \frac{1}{20}$ = 4.32			
GainRatio	0.495/4.32 = 0.114			



	CarType		
	Family	Sports	Luxury
Yes	8	0	1
No	2	4	5
GINI	0.32	0	0.28
GINI _{split}	0.243		
GAIN _{split}	0.495 - 0.243 = 0.252		
SplitInfo	$-\frac{10}{20} \log_2 \frac{10}{20} - \frac{4}{20} \log_2 \frac{4}{20} - \frac{6}{20} \log_2 \frac{6}{20}$ = 1.485		
GainRatio	0.252/1.485 = 0.169		



	CarType	
	Family	Sports, Luxury
Yes	8	1
No	2	9
GINI	0.32	0.18
GINI _{split}	0.25	
GAIN _{split}	0.495 - 0.25 = 0.245	
SplitInfo	$-\frac{10}{20} \log_2 \frac{10}{20} - \frac{10}{20} \log_2 \frac{10}{20}$ = 1	
GainRatio	0.245/1 = 0.245	

Best split by GainRatio

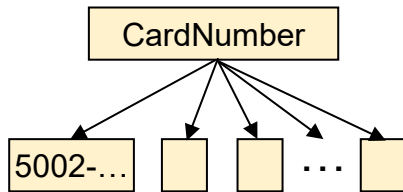
Classification > Methodologies > Decision Tree > Gain Ratio

□ Example (Entropy)

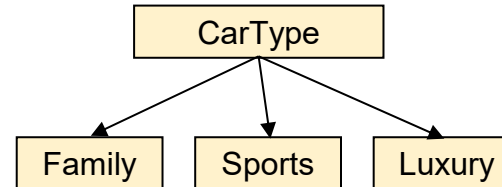
$$SplitInfo = - \sum_{i=1}^k \frac{n_i}{n} \log_2 \frac{n_i}{n} \rightarrow GainRatio = \frac{GAIN_{split}}{SplitInfo}$$

Root node: 9 Yes, 11 No → Entropy = 0.992

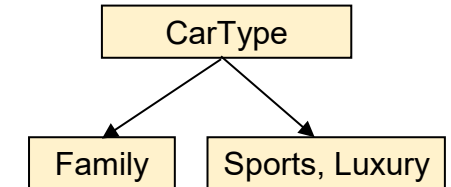
UserID	CardNumber	CarType	Defaulted
1	5002-234-...	Family	No
2	1004-235-...	Luxury	No
3	1454-235-...	Sports	Yes
4	4523-452-...	Sports	Yes
...	...	...	...
20	5403-214-...	Family	Yes



	CardNumber			
	5002-...	1004-...	...	5403-...
Yes	0	0	...	1
No	1	1	...	0
Entropy	0	0	0	0
Entropy _{split}	0			
GAIN_{split}	0.992 - 0 = 0.992			
SplitInfo	$-\frac{1}{20} \log_2 \frac{1}{20} - \frac{1}{20} \log_2 \frac{1}{20} - \dots - \frac{1}{20} \log_2 \frac{1}{20}$ = 4.32			
GainRatio	0.992/4.32 = 0.229			



	CarType		
	Family	Sports	Luxury
Yes	8	0	1
No	2	4	5
Entropy	0.721	0	0.650
Entropy _{split}	0.556		
GAIN_{split}	0.992 - 0.556 = 0.437		
SplitInfo	$-\frac{10}{20} \log_2 \frac{10}{20} - \frac{4}{20} \log_2 \frac{4}{20} - \frac{6}{20} \log_2 \frac{6}{20}$ = 1.485		
GainRatio	0.437/1.485 = 0.294		



	CarType	
	Family	Sports, Luxury
Yes	8	1
No	2	9
Entropy	0.721	0.469
Entropy _{split}	0.595	
GAIN_{split}	0.992 - 0.595 = 0.397	
SplitInfo	$-\frac{10}{20} \log_2 \frac{10}{20} - \frac{10}{20} \log_2 \frac{10}{20}$ = 1	
GainRatio	0.397/1 = 0.397	

Best split by GainRatio

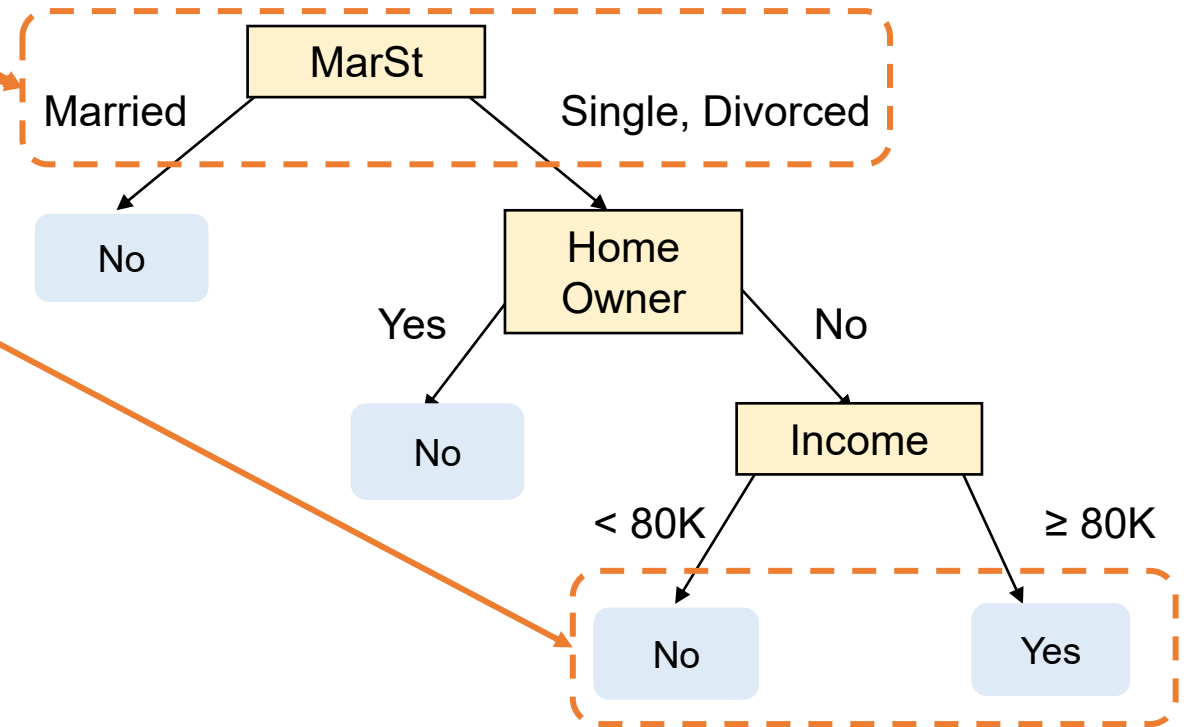
Outline

- ❑ Basic concepts of classification
- ❑ **Classification methodologies**
 - **Decision tree**
 - Overview
 - GINI index → CART
 - Entropy → ID3
 - Gain ratio → C4.5
- ❑ Classification evaluation
- ❑ Ensemble classifiers

Classification > Methodologies > Decision Tree

□ Two important questions for decision tree induction

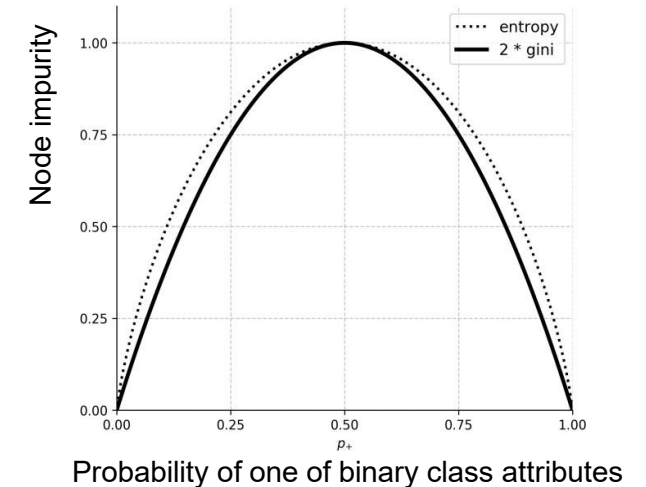
- How to split?
 - How to specify the split criterion?
 - How to determine the best split?
- When to stop splitting?



Classification > Methodologies > Decision Tree

□ Split criteria

- Gini index (CART) and Entropy (ID3)
 - Measures of node impurity $\rightarrow$ purity gain (or information gain)
 - Both give similar results in practice (e.g., only disagree in 2% [Răileanu & Stoffel 2004])
 - Entropy might be slower to compute because of the log
- Gain ratio (C4.5)
 - An adjusted purity gain (normalized by split info)
 - Penalize the splitting into too many partitions



□ Stopping criteria

- Node/tree size conditions
 - Stop splitting a node when it reaches a minimum node size threshold
 - Stop splitting a node when it reaches a maximum depth threshold
- Node impurity conditions
 - Stop splitting a node when all the records belong to the same class (i.e., node impurity = 0)
 - Stop splitting a node when its purity gain is less than a minimum purity gain threshold

Outline

□ Today

- Basic concepts of classification
- Classification methodologies
- **Classification evaluation**
 - Holdout
 - Trade-off
 - Cross validation
 - Metrics
- Ensemble classifiers
 - Bagging
 - Boosting

Classification > Evaluation

□ How to evaluate a classifier?

Existing dataset

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

New dataset

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
11	No	Married	Small	80K	??
12	Yes	Married	Large	60K	??
13	No	Single	Extra Large	105K	??
14	No	Married	Large	60K	??
15	Yes	Married	Large	120K	??

Induction



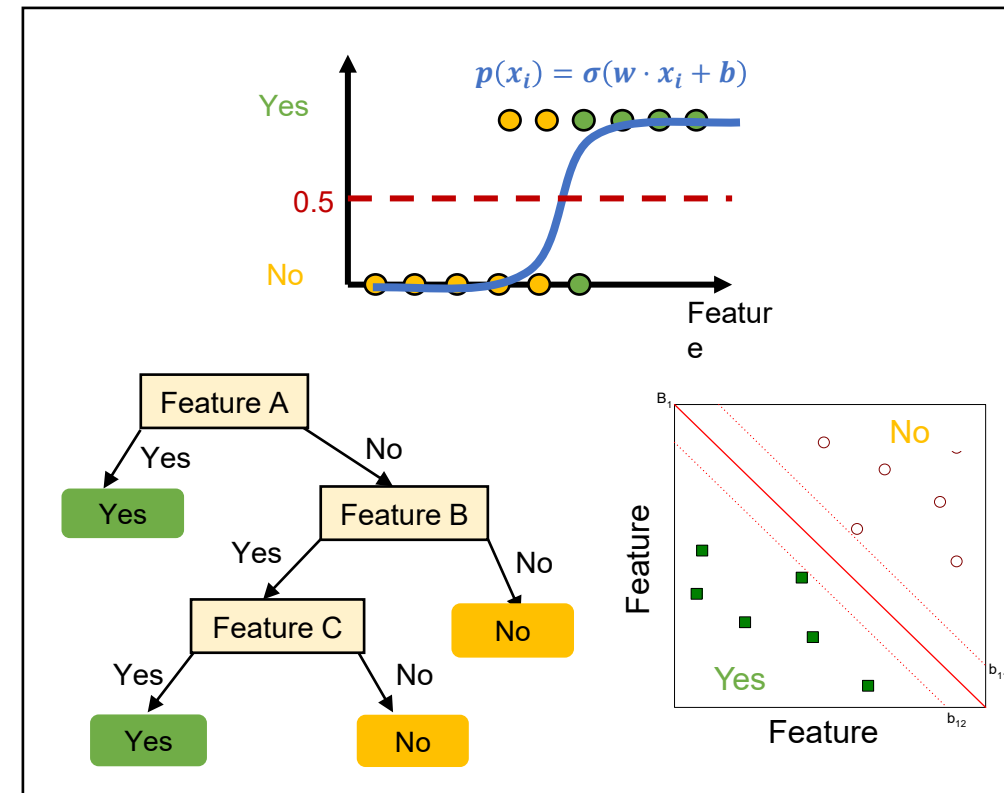
Learning
a model

Deduction



Applying
a model

A classifier model



Classification > Evaluation

□ How to evaluate a classifier?

Existing dataset → **Seen data**

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

New dataset → **Unseen data**

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
11	No	Married	Small	80K	??
12	Yes	Married	Large	60K	??
13	No	Single	Extra Large	105K	??
14	No	Married	Large	60K	??
15	Yes	Married	Large	120K	??

Induction



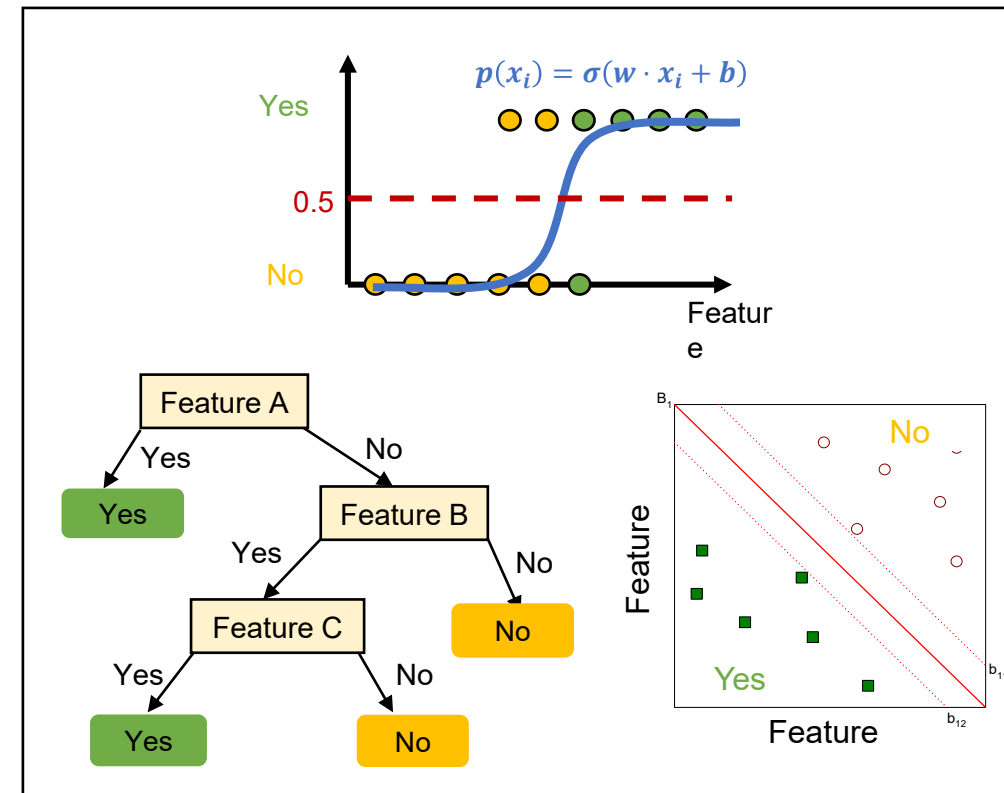
Learning
a model

Deduction



Applying
a model

A classifier model



Classification > Evaluation

□ How to evaluate a classifier?

Existing dataset → Seen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

New dataset → Unseen data

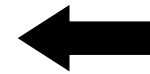
ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
11	No	Married	Small	80K	??
12	Yes	Married	Large	60K	??
13	No	Single	Extra Large	105K	??
14	No	Married	Large	60K	??
15	Yes	Married	Large	120K	??

Induction



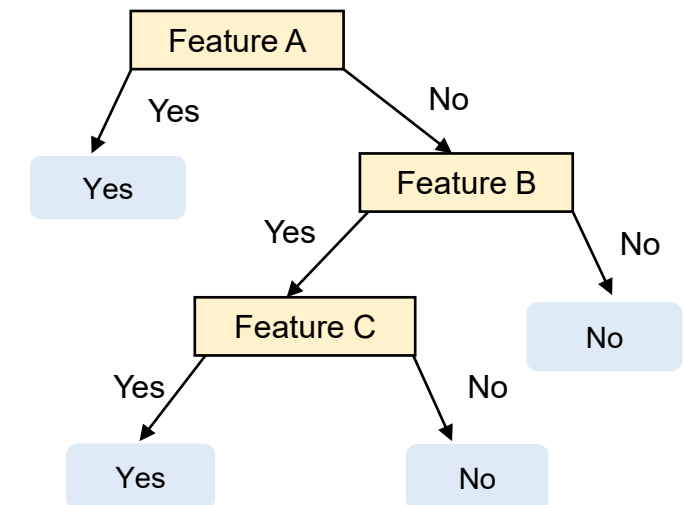
Learning
a model

Deduction



Applying
a model

**An example decision tree
built from the existing dataset**



Classification > Evaluation

□ How to evaluate a classifier?

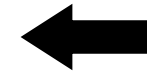
Existing dataset → Seen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes

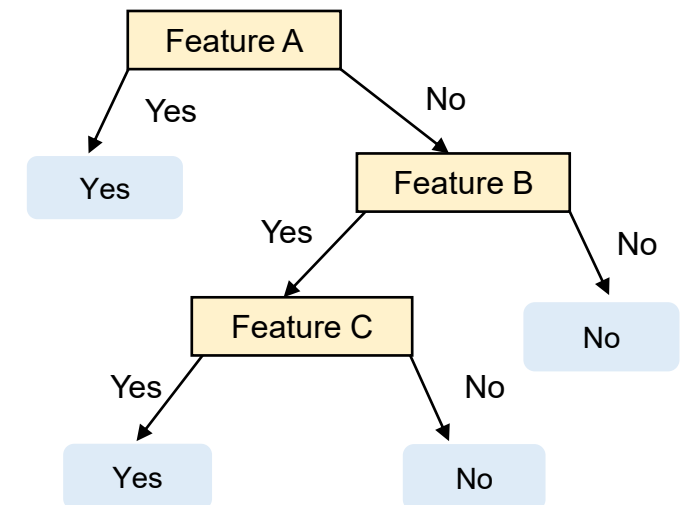
New dataset → Unseen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
11	No	Married	Small	80K	??
12	Yes	Married	Large	60K	??
13	No	Single	Extra Large	105K	??
14	No	Married	Large	60K	??
15	Yes	Married	Large	120K	??

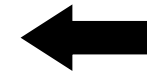
Applying
a model



**An example decision tree
built from the existing dataset**



Applying
a model



Classification > Evaluation

□ How to evaluate a classifier?

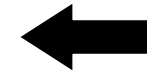
Existing dataset → Seen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	No
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	Yes
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes

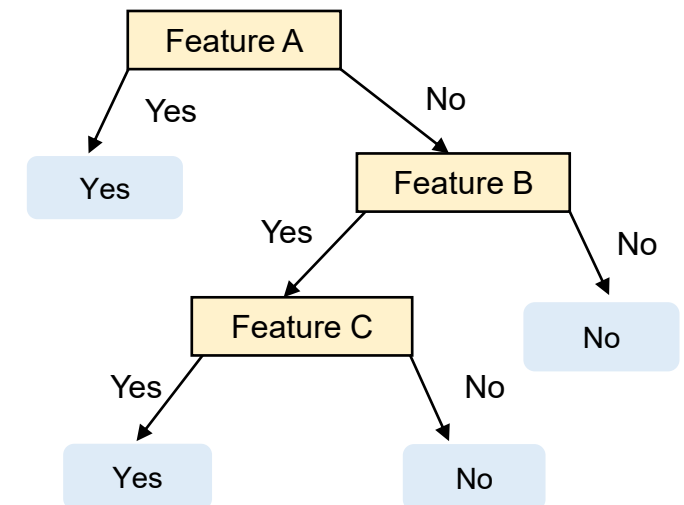
New dataset → Unseen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
11	No	Married	Small	80K	??	Yes
12	Yes	Married	Large	60K	??	No
13	No	Single	Extra Large	105K	??	Yes
14	No	Married	Large	60K	??	No
15	Yes	Married	Large	120K	??	Yes

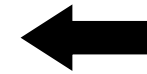
Applying
a model



**An example decision tree
built from the existing dataset**



Applying
a model



Classification > Evaluation

□ How to evaluate a classifier?

Existing dataset → Seen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	No
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	Yes
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes

1/10 incorrect
Error = 10%

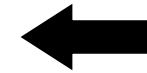
New dataset → Unseen data

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
11	No	Married	Small	80K	Yes	Yes
12	Yes	Married	Large	60K	No	No
13	No	Single	Extra Large	105K	Yes	Yes
14	No	Married	Large	60K	No	No
15	Yes	Married	Large	120K	No	Yes

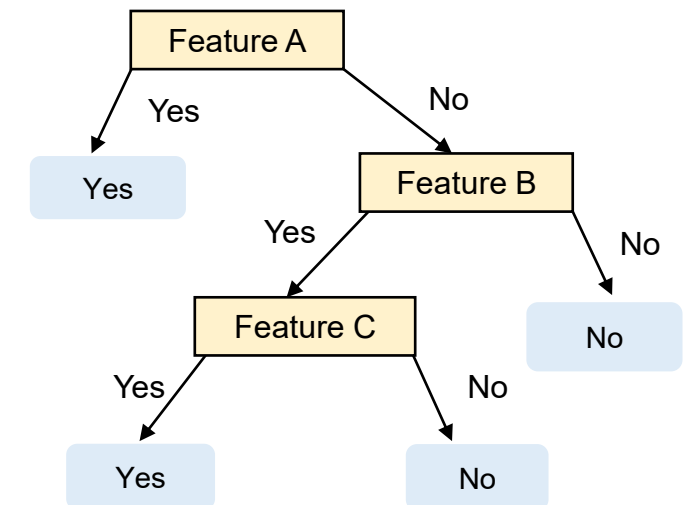
1/5 incorrect
Error = 20%

*Assume that
we know the answers*

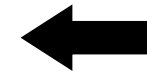
Applying
a model



**An example decision tree
built from the existing dataset**



Applying
a model



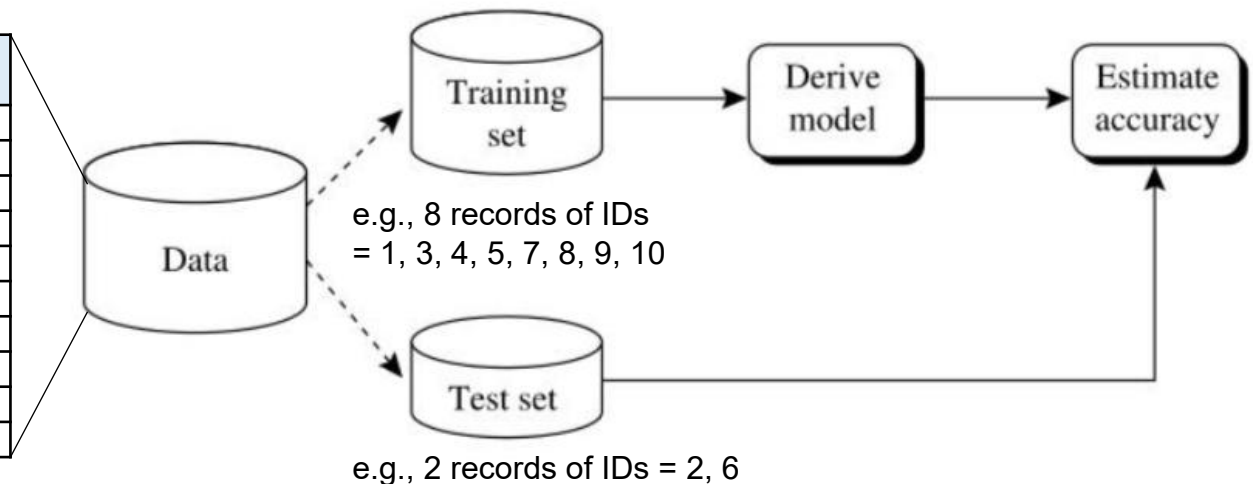
Classification > Evaluation > Holdout

❑ In practice, we do not know the ground-truth labels for the new data

❑ Holdout evaluation

- The existing dataset is randomly partitioned into training set and test set (e.g., typically, 8:2 ratio)
- Training set is used to derive a model → true labels are used to learn the model
 - e.g., maximize likelihood for logistic regression, margin for SVM, or purity for decision tree
- Test set is used to evaluate the model → true labels are used only for evaluation

ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower
1	Yes	Single	Small	125K	No
2	No	Married	Medium	100K	No
3	No	Single	Small	70K	No
4	Yes	Married	Large	120K	No
5	No	Divorced	Extra Large	95K	Yes
6	No	Married	Large	60K	No
7	Yes	Divorced	Medium	220K	Yes
8	No	Single	Extra Large	85K	Yes
9	No	Married	Small	75K	No
10	No	Single	Medium	90K	Yes



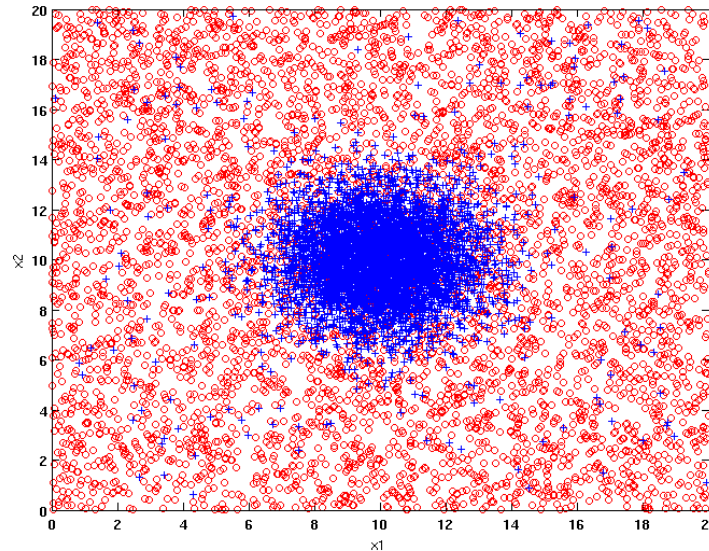
Classification > Evaluation > Trade-off

□ Model complexity and error

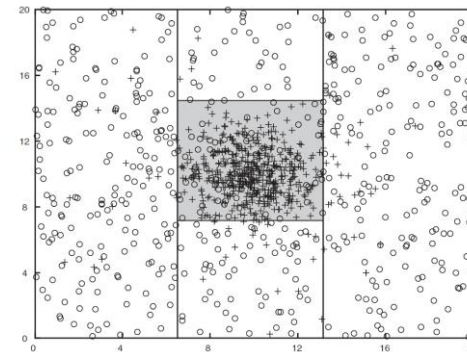
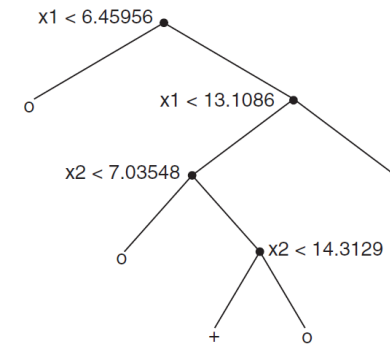
- Underfitting: when a model is too simple, both training and test errors are large
- Overfitting: when a model is too complex, training error is small but test error is large

□ Motivating example

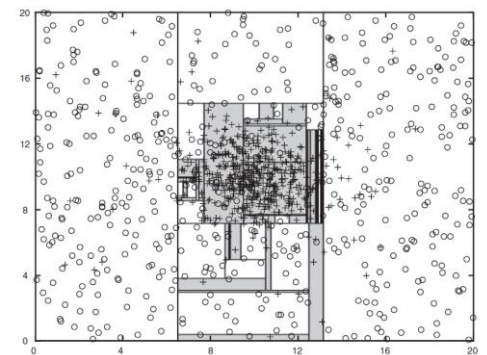
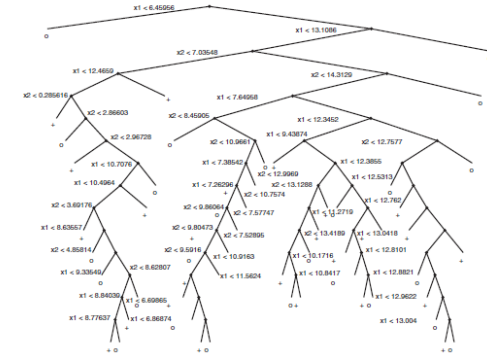
- A simple classification problem with two numerical attributes
- Classification of data points into Blue or Red



Decision Tree with 4 nodes



Decision Tree with 50 nodes

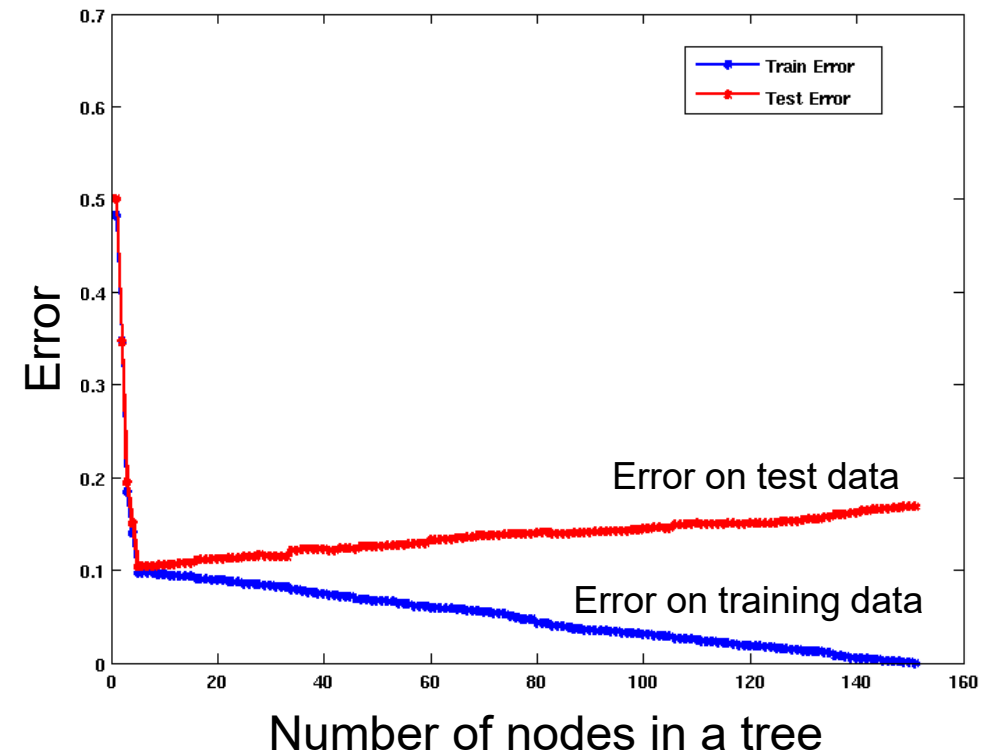
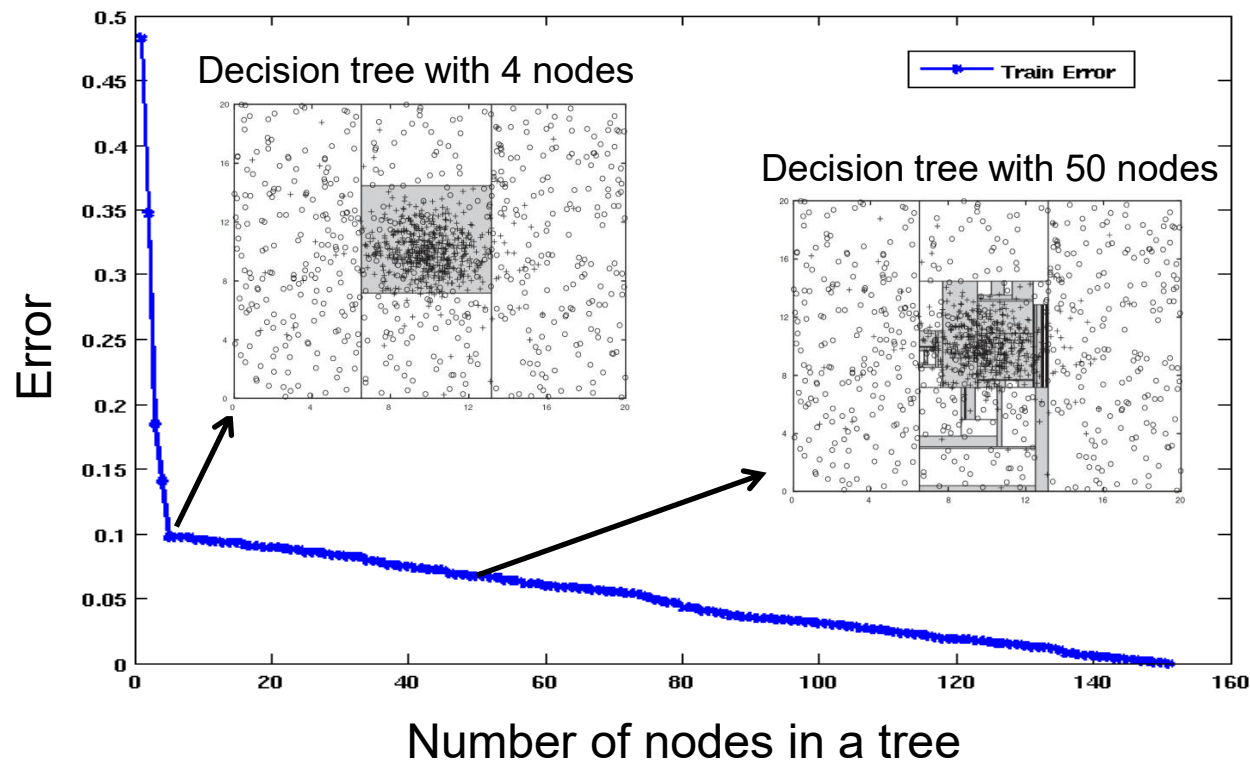


Classification > Evaluation > Trade-off

❑ Model complexity and error

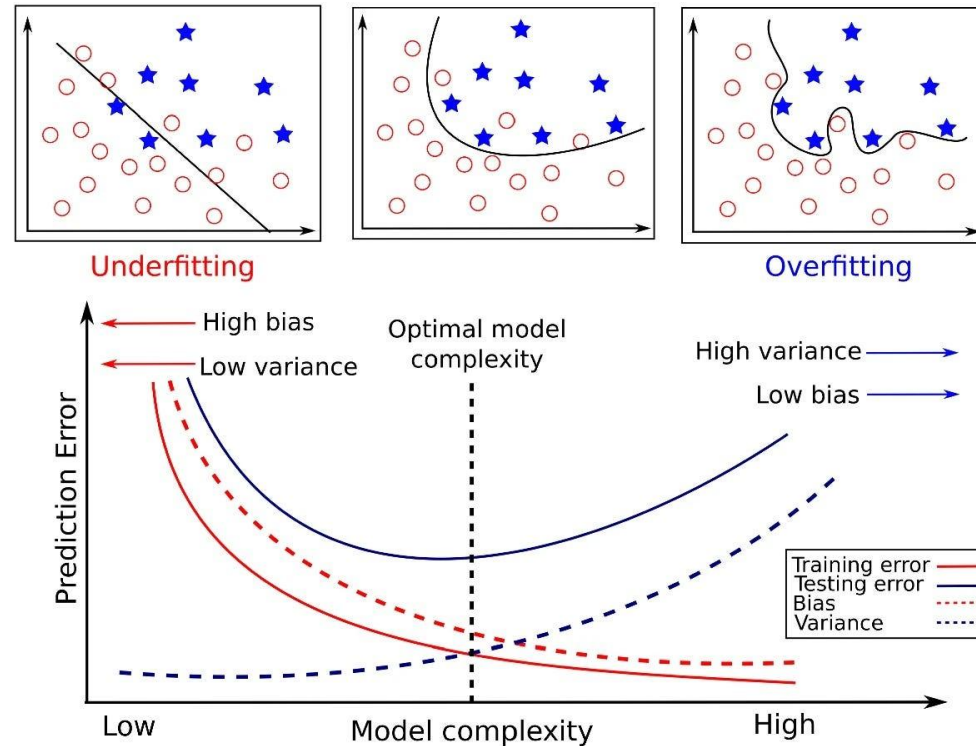
- Underfitting: when a model is too simple, both training and test errors are large
- Overfitting: when a model is too complex, training error is small but test error is large

❑ Motivating example



Classification > Evaluation > Trade-off

- The underfitting (bias) and overfitting (variance) trade-off in learning classifiers

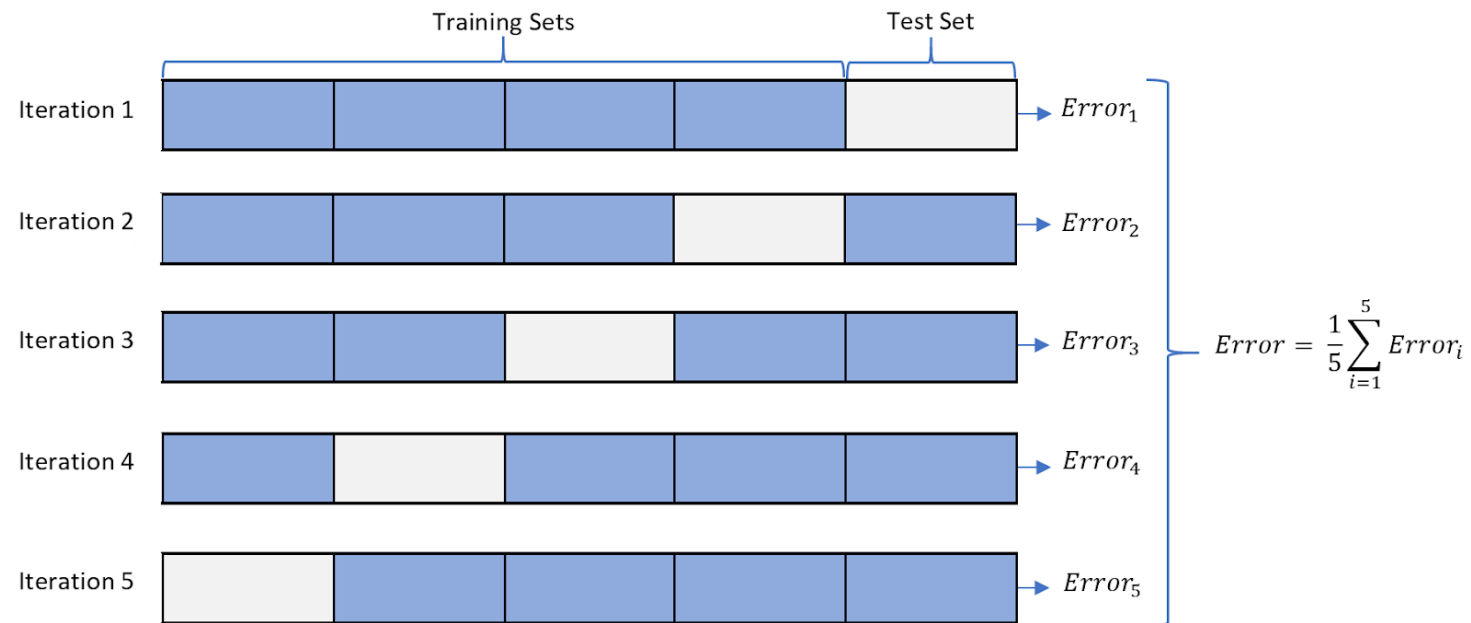


- Finding the optimal model is the goal of model evaluation
 - Possible strategies: cross validation, pruning, regularization, early stopping, ensemble, etc.

Classification > Evaluation > Cross Validation

□ k-fold cross validation

- Randomly partition data into k disjoint subsets
 - Iteratively train on k-1 subsets and test on the remaining 1 subset
- Stratified cross validation
 - Stratified sampling → preserve the class distribution for each partition



Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j
- Terminology
 - Positive/Negative → the class label (i.e., Yes/No or 1/0)
 - True/False → the prediction correctness (i.e., is the predicted label the same with the true label?)
- Components
 - True Positives (TP)
 - # of actual positive cases correctly predicted as positive
 - True Negatives (TN)
 - # of actual negative cases correctly predicted as negative
 - False Positives (FP), a.k.a. Type I error
 - # of actual negative cases incorrectly predicted as positive
 - False Negatives (FN), a.k.a. Type II error
 - # of actual positive cases incorrectly predicted as negative

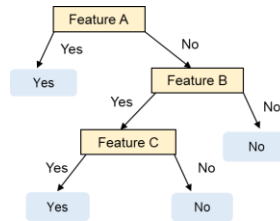
		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP (True Positive)	FN (False Negative) (Type II error)
	Class=No (Negative)	FP (False Positive) (Type I error)	TN (True Negative)

Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j

❑ Example



ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	Yes
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes



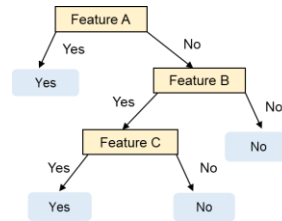
		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP (True Positive)	FN (False Negative)
	Class=No (Negative)	FP (False Positive)	TN (True Negative)

Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j

❑ Example



ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
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5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes



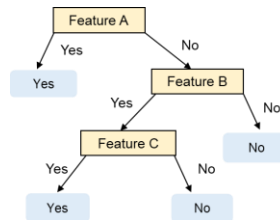
		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP = 3 (True Positive)	FN (False Negative)
	Class=No (Negative)	FP (False Positive)	TN (True Negative)

Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j

❑ Example



ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	Yes
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes



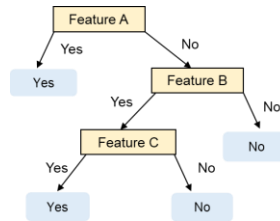
		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP = 3 (True Positive)	FN = 1 (False Negative)
	Class=No (Negative)	FP (False Positive)	TN (True Negative)

Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j

❑ Example



ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	Yes
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes



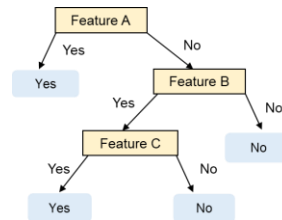
		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP = 3 (True Positive)	FN = 1 (False Negative)
	Class=No (Negative)	FP = 2 (False Positive)	TN (True Negative)

Classification > Evaluation > Metrics

❑ Confusion matrix

- Shows the number of data in class i that was predicted by the classifier as class j

❑ Example



ID	Home Owner	Marital Status	Car Size	Annual Income	Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	Yes
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes



		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP = 3 (True Positive)	FN = 1 (False Negative)
	Class=No (Negative)	FP = 2 (False Positive)	TN = 4 (True Negative)

Classification > Evaluation > Metrics

Major metrics

- **Accuracy** = $\frac{True}{All} = \frac{TP+TN}{TP+TN+FP+FN}$
 - How many data were correctly predicted?
 - Error rate = 1- Accuracy = $\frac{False}{All} = \frac{FP+FN}{TP+TN+FP+FN}$
 - Class imbalance problem (e.g., positive:negative = 10:90)
- **Precision** = $\frac{TP}{TP+FP}$
 - How many data predicted as positive were correct?
- **Recall** = $\frac{TP}{TP+FN}$
 - How many data of the actual positive class were correctly predicted?
 - a.k.a. True positive rate (TPR)
- **F1 score** = $\frac{2 \times precision \times recall}{precision + recall}$
 - How well did it predict the positive class in general?
 - Harmonic mean of precision and recall

		Predicted class	
		Class=Yes (Positive)	Class=No (Negative)
Actual class	Class=Yes (Positive)	TP (True Positive)	FN (False Negative)
	Class=No (Negative)	FP (False Positive)	TN (True Negative)

Others metrics

- True positive rate (TPR) = $\frac{TP}{(Actual) P} = \frac{TP}{TP+FN}$
 - a.k.a. sensitivity, recall
- True negative rate (TNR) = $\frac{TN}{(Actual) N} = \frac{TN}{TN+FP}$
 - a.k.a., specificity
- False positive rate (FPR) = $\frac{FP}{(Actual) N} = \frac{FP}{TN+FP}$
- False negative rate (FNR) = $\frac{FN}{(Actual) P} = \frac{FN}{TP+FN}$

Classification > Evaluation > Metrics

□ Example

▪ Evaluation metrics results

- Accuracy = $\frac{True}{All} = \frac{TP+TN}{TP+TN+FP+FN}$

- How many data were correctly predicted? $\frac{90+9560}{10000} = 96.5\%$

- Precision = $\frac{TP}{TP+FP}$

- How many data predicted as positive were correct? $\frac{90}{230} = 39.13\%$

- Recall = $\frac{TP}{TP+FN}$

- How many data of the actual positive class were correctly predicted? $\frac{90}{300} = 30.00\%$

- F1 score = $\frac{2 \times \text{precision} \times \text{recall}}{\text{precision} + \text{recall}}$

- How well did it predict the positive class in general? $\frac{2 \times 39.13 \times 30.00}{39.13 + 30.00} = 33.96\%$

Actual\Predicted	Cancer = yes	Cancer = no	Total
Cancer = yes	TP = 90	FN = 210	300
Cancer = no	FP = 140	TN = 9560	9700
Total	230	9770	10000

▪ Implication

- Cancer patients are very rare (3%) → a large number of True Negative results → misleading high Accuracy
- High accuracy but low Recall, Precision, F1 scores → the classifier is actually not very good

Classification > Evaluation > Metrics

Quiz

- Accuracy = $\frac{True}{All} = \frac{TP+TN}{TP+TN+FP+FN}$
 - How many data were correctly predicted?
 - **Answer:**
- Precision = $\frac{TP}{TP+FP}$
 - How many data predicted as positive were correct?
 - **Answer:**
- Recall = $\frac{TP}{TP+FN}$
 - How many data of the actual positive class were correctly predicted?
 - **Answer:**
- F1 score = $\frac{2 \times precision \times recall}{precision + recall}$
 - How well did it predict the positive class in general?
 - (optional, not a quiz) Answer:

ID	Home Owner	Marital Status	Car Size	Annual Income	Target attribute	Predicted label
					Defaulted Borrower	Predicted by the model
1	Yes	Single	Small	125K	No	Yes
2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
5	No	Divorced	Extra Large	95K	Yes	Yes
6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes

Actual\Predicted	Defaulted = yes	Defaulted = no	Total
Defaulted = yes			
Defaulted = no			
Total			

Classification > Evaluation > Metrics

Quiz

- Accuracy = $\frac{True}{All} = \frac{TP+TN}{TP+TN+FP+FN}$
 - How many data were correctly predicted?
 - **Answer:** $\frac{3+4}{10} = 0.7$
- Precision = $\frac{TP}{TP+FP}$
 - How many data predicted as positive were correct?
 - **Answer:** $\frac{3}{3+2} = 0.6$
- Recall = $\frac{TP}{TP+FN}$
 - How many data of positive class were correctly predicted?
 - **Answer:** $\frac{3}{3+1} = 0.75$
- F1 score = $\frac{2 \times precision \times recall}{precision + recall}$
 - How well did it predict the positive class in general?
 - **Answer:** $\frac{2 \times 0.6 \times 0.75}{0.6 + 0.75} = 0.667$

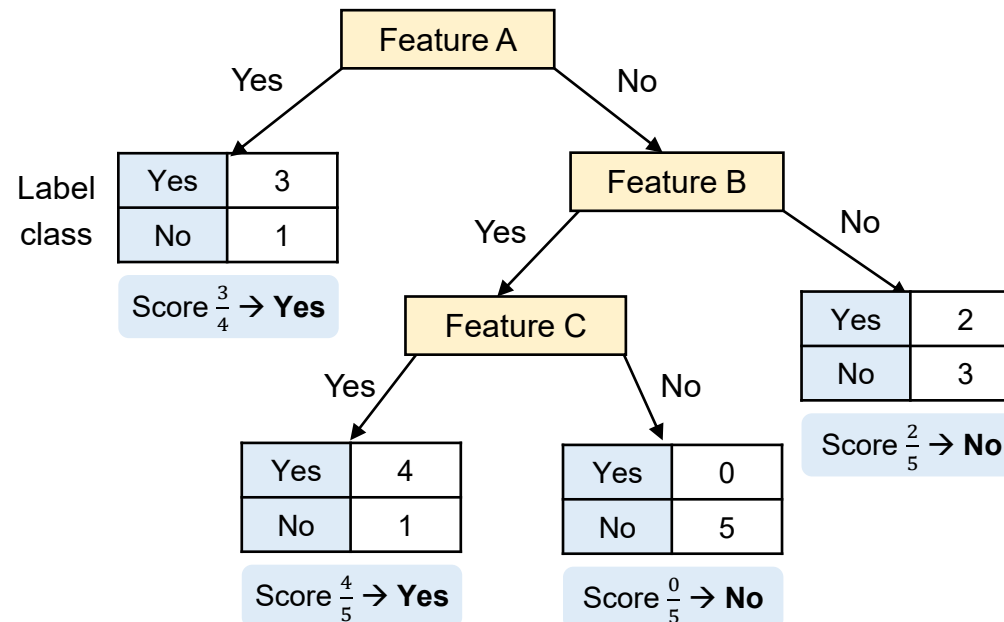
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					Defaulted Borrower	Predicted by the model
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2	No	Married	Medium	100K	No	No
3	No	Single	Small	70K	No	No
4	Yes	Married	Large	120K	No	Yes
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6	No	Married	Large	60K	No	No
7	Yes	Divorced	Medium	220K	Yes	Yes
8	No	Single	Extra Large	85K	Yes	No
9	No	Married	Small	75K	No	No
10	No	Single	Medium	90K	Yes	Yes

Actual\Predicted	Defaulted = yes	Defaulted = no	Total
Defaulted = yes	TP = 3	FN = 1	4
Defaulted = no	FP = 2	TN = 4	6
Total	5	5	10

Classification > Evaluation > Metrics > ROC

□ Motivation

- We implicitly assume a majority vote for determining the prediction class
 - e.g., for leaf nodes in a decision tree, a prediction score (= a proportion of a certain class) > 0.5?



- Can we consider all possible thresholds to evaluate classifiers' performance?

Classification > Evaluation > Metrics > ROC

❑ ROC (Receiver Operating Characteristics) curve

▪ Visual comparison of classification models

- Shows the trade-off between TPR and FPR
- The closer to the diagonal line, the less accurate the model

$$\text{True positive rate (TPR)} = \frac{TP}{P} = \frac{TP}{TP+FN}$$

$$\text{False positive rate (FPR)} = \frac{FP}{N} = \frac{FP}{TN+FP}$$

▪ Construction procedure

- Rank the test data by its predicted score
 - e.g., for a decision tree, the class proportions in leaf nodes
- Change the threshold for class prediction and compute TPR and FPR

▪ Example

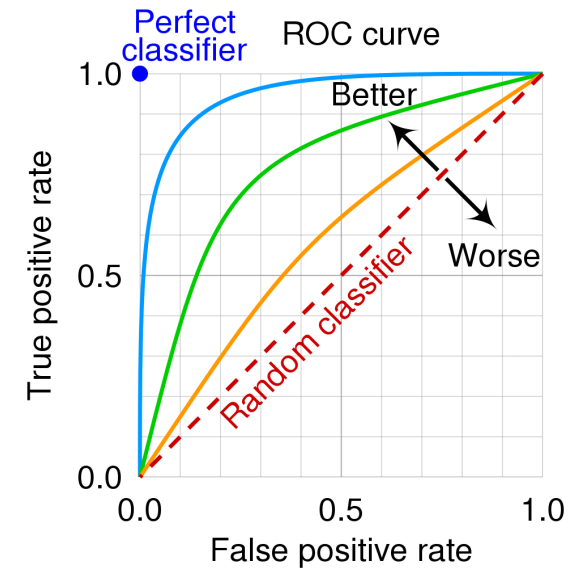
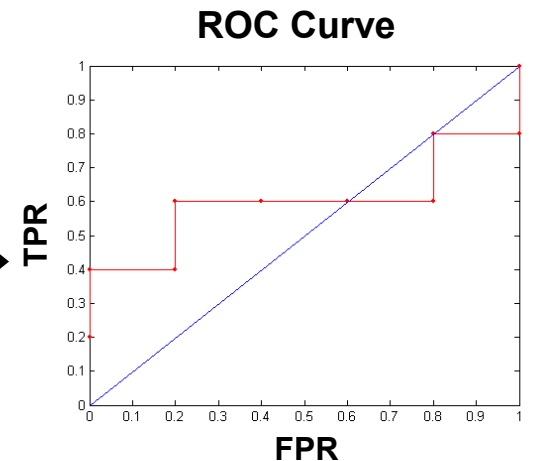
Example prediction results

ID	Actual class	Predicted score for Yes
1	Yes	0.95
2	No	0.85
3	No	0.88
4	Yes	0.93
5	Yes	0.53
6	Yes	0.25
7	No	0.43
8	No	0.76
9	No	0.86
10	Yes	0.87



e.g., score ≥ 0.76 ?

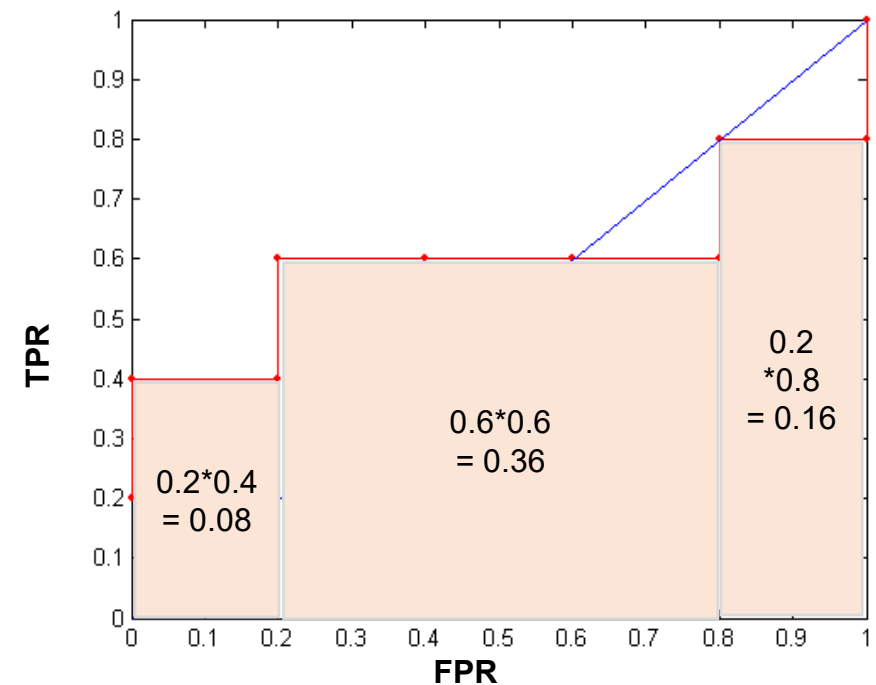
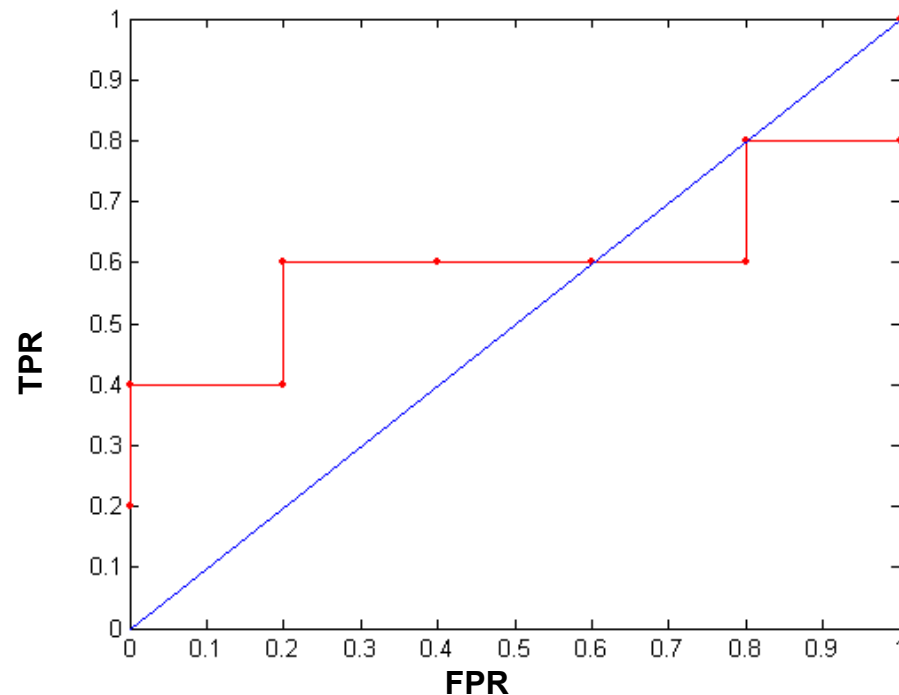
	Predict as Neg ←						Pos →					
ID	6	7	5	8	2	9	10	3	4	1		
Actual class	P	N	P	N	N	N	P	N	P	P		
Pred. score	0.25	0.43	0.53	0.76	0.85	0.86	0.87	0.88	0.93	0.95	1.00	
TP	5	4	4	3	3	3	3	2	2	1	0	
FP	5	5	4	4	3	2	1	1	0	0	0	
TN	0	0	1	1	2	3	4	4	5	5	5	
FN	0	1	1	2	2	2	2	3	3	4	5	
TPR	1	0.8	0.8	0.6	0.6	0.6	0.6	0.4	0.4	0.2	0	
FPR	1	1	0.8	0.8	0.6	0.4	0.2	0.2	0	0	0	



Classification > Evaluation > Metrics > AUC

□ AUC (Area Under Curve) score

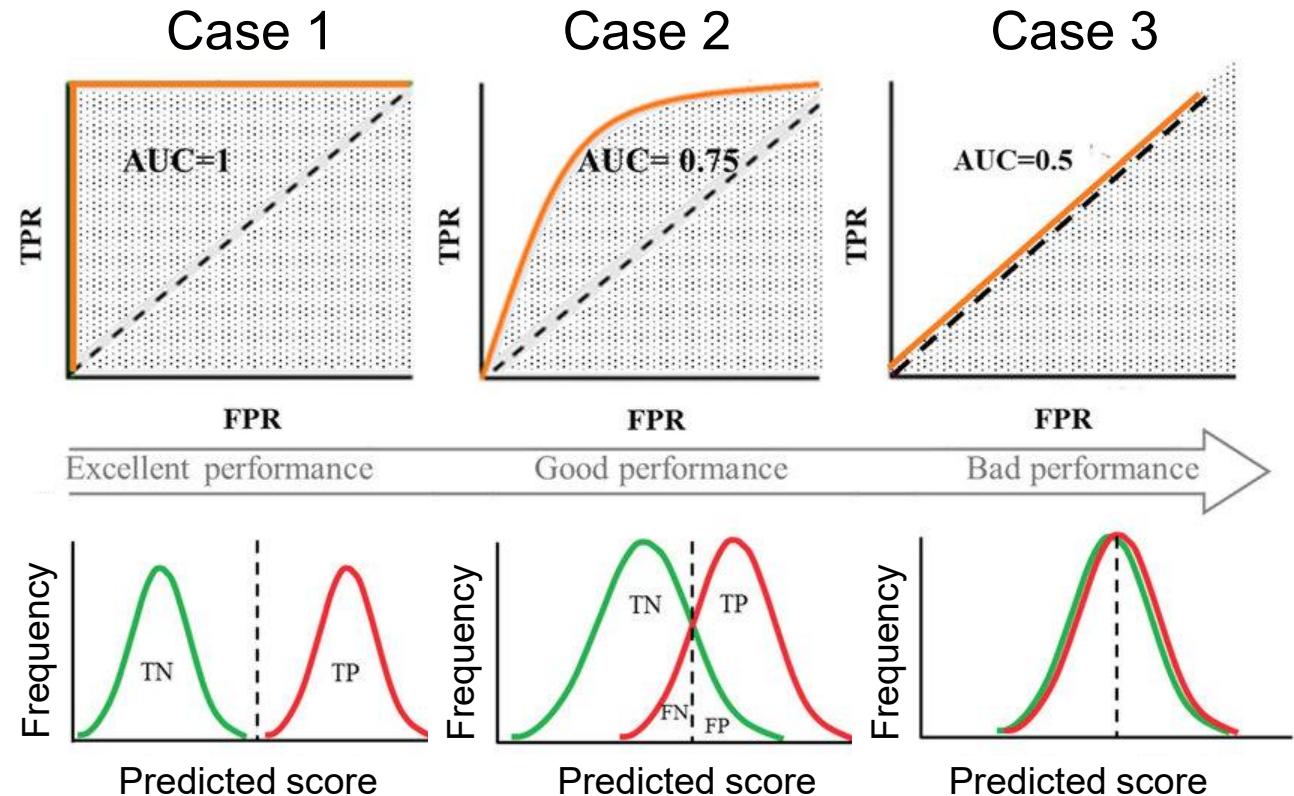
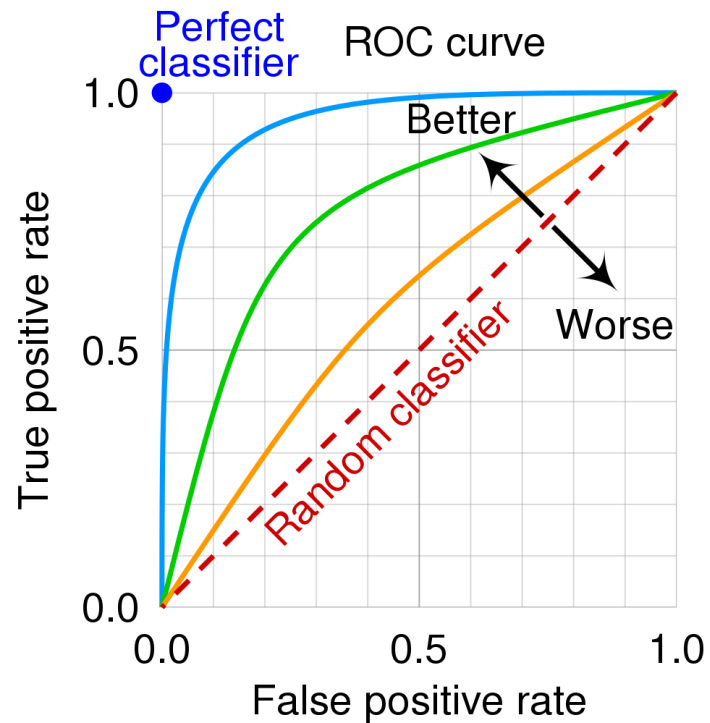
- The area under the ROC curve to quantify the overall accuracy of the model
- Example: $AUC = 0.08 + 0.36 + 0.16 = 0.6$



Classification > Evaluation > Metrics > AUC

□ AUC (Area Under Curve) score

- The area under the ROC curve to quantify the overall accuracy of the model



Outline

- ❑ Classification evaluation
- ❑ **REVISIT: Cluster/outlier evaluation**
 - Supervised measure for clustering evaluation
 - Outlier evaluation when labels are present
- ❑ Ensemble classifiers

REVISIT: Clustering > Evaluation > Clustering Quality > Comparison-based

□ Supervised measure (i.e., extrinsic index)

- Dataset $D = \{o_1, \dots, o_n\}$, Ground truth $G = \{G_1, \dots, G_l\}$, Clustering result $C = \{C_1, \dots, C_m\}$,
- Pairwise comparison-based
 - Treat each cluster in the ground truth as a class
 - For each pair of objects (o_i, o_j) in D ,
if they are assigned to the same cluster, the assignment is regarded as positive; otherwise, negative
 - Depending on assignments, we have four possible cases for the total number of pairs (i.e., $\binom{n}{2}$)
 - Basic metrics
 - Jaccard index = $TP / (TP + FP + FN)$
 - Rand index = $(TP + TN) / (TP + TN + FP + FN)$
- Cf. advanced metrics: adjusted rand index (ARI), B-cubed score

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

REVISIT: Clustering > Evaluation > Clustering Quality > Comparison-based

□ Example

- Dataset $D = \{a, b, c, d, e\}$
- Ground truth $G = [G1=\{a,b,c\}, G2=\{d,e\}]$
- Clustering result $C = [C1= \{a,b\}, C2 = \{c,d,e\}]$

- All possible pairs $= \binom{5}{2} = 10$
 - $(a,b), (a,c), (a,d), (a,e), (b,c), (b,d), (b,e), (c,d), (c,e), (d,e)$
- Counting each case
 - TP = 2: $(a,b), (d,e)$
 - FP = 2: $(c,d), (c,e)$
 - FN = 2: $(a,c), (b,c)$
 - TN = 4: $(a,d), (a,e), (b,d), (b,e)$

	$C(o_i) = C(o_j)$	$C(o_i) \neq C(o_j)$
$G(o_i) = G(o_j)$	true positive	false negative
$G(o_i) \neq G(o_j)$	false positive	true negative

- Rand index $= (TP + TN) / (TP + TN + FP + FN) = 2 + 4 / (2 + 2 + 2 + 4) = 0.6$

REVISIT: Outlier Detection > Evaluation

□ When labels are present (i.e., binary classification?)

- Standard evaluation measures
 - ROC (TPR – FPR) curve → AUC score
 - Precision-Recall(=TPR) curve → Average Precision score
 - More useful when the classes are imbalanced ($P \ll N$)
 - FPR ($FP/(FP+TN)$) in ROC → Precision ($TP/(FP+TP)$)
- Top-N evaluation measures (N: # of true outliers)
 - Precision@N

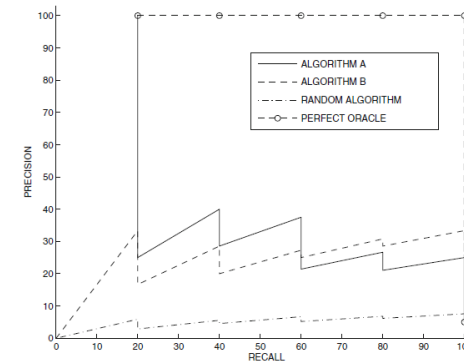
$$P@n = \frac{|\{o \in O \mid \text{rank}(o) \leq n\}|}{n}$$

- Average Precision@N

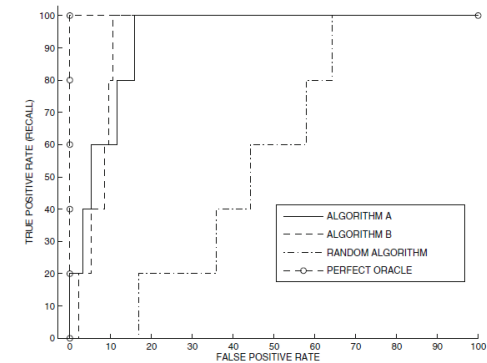
$$AP = \frac{1}{|O|} \sum_{o \in O} P@ \text{rank}(o)$$

Ranks of true outliers out of 100 objects

Algorithm	Rank of Ground-truth Outliers
Algorithm A	1, 5, 8, 15, 20
Algorithm B	3, 7, 11, 13, 15
Random Algorithm	17, 36, 45, 59, 66
Perfect Oracle	1, 2, 3, 4, 5



(a) Precision-recall



(b) Receiver operating characteristic

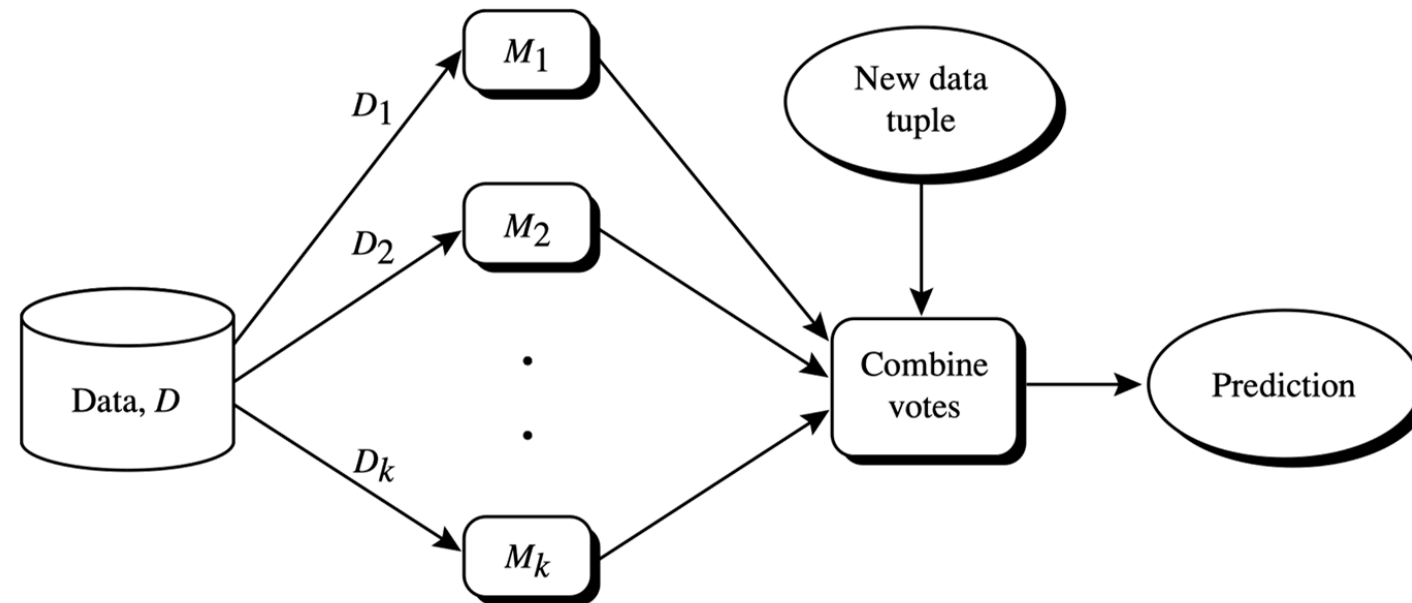
Outline

- ❑ Classification evaluation
- ❑ REVISIT: Cluster/outlier evaluation
- ❑ **Ensemble classifiers**
 - Bagging
 - Boosting

Classification > Ensemble

□ Ensemble methods

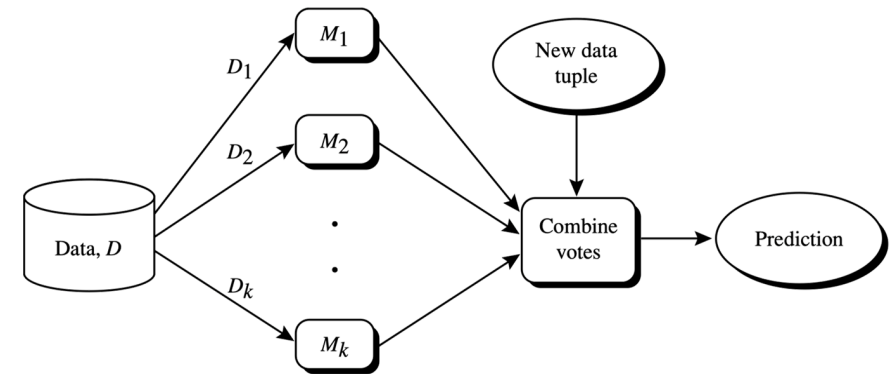
- Use a combination of models to increase performance
- Combine a set of k models, $M_1, M_2, \dots, M_k$, to create an improved model M^*
- Final decision will be made by aggregating the outputs of models



Classification > Ensemble

❑ Is an ensemble method always good?

- Example of an ensemble method with majority voting
 - Three base models M_1 , M_2 , M_3
 - Three test data instances x_1 , x_2 , x_3



✓: correct prediction

✗: wrong prediction

Case 1

		x_1	x_2	x_3
Base model performance	M_1	✓	✓	✗
	M_2	✗	✓	✓
	M_3	✓	✗	✓
Ensemble performance	Voting	✓	✓	✓

Ensemble has positive effect

Case 2

		x_1	x_2	x_3
M_1	✓	✓	✓	✗
M_2	✓	✓	✓	✗
M_3	✓	✓	✓	✗
Voting	✓	✓	✓	✗

Ensemble has no effect

Case 3

	x_1	x_2	x_3
M_1	✓	✗	✗
M_2	✗	✓	✗
M_3	✗	✗	✓
Voting	✗	✗	✗

Ensemble has negative effect

Classification > Ensemble

□ Is an ensemble method always good?

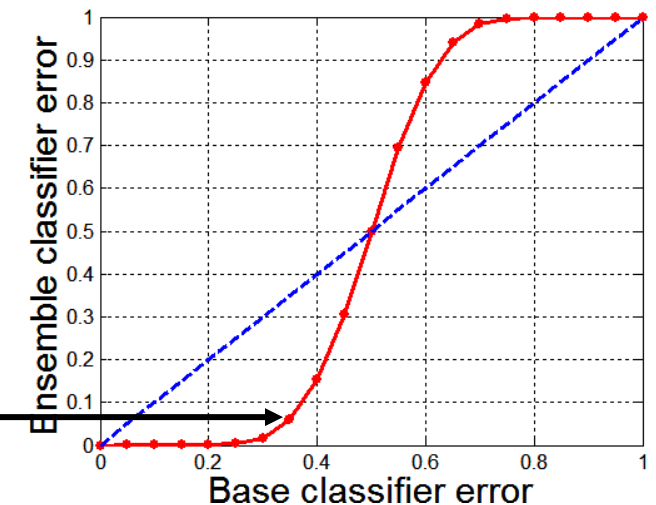
- Given n independent classifiers with an error rate ε , the probability that k classifiers are wrong is

$$P(k \text{ wrong classifiers}) = \binom{n}{k} \varepsilon^k (1 - \varepsilon)^{n-k}$$

- Example with 25 base classifiers

- Assume that the classifiers are not correlated and their error rate $\varepsilon = 0.35$
- The probability of having more than half of the classifiers being wrong is:

$$P(\# \text{ of wrong classifiers} \geq 13) = \sum_{i=13}^{25} \binom{25}{i} \varepsilon^i (1 - \varepsilon)^{25-i} = 0.06$$



- Theoretically, the error rate has been decreased from 0.35 to 0.06 by adopting the ensemble approach

Classification > Ensemble

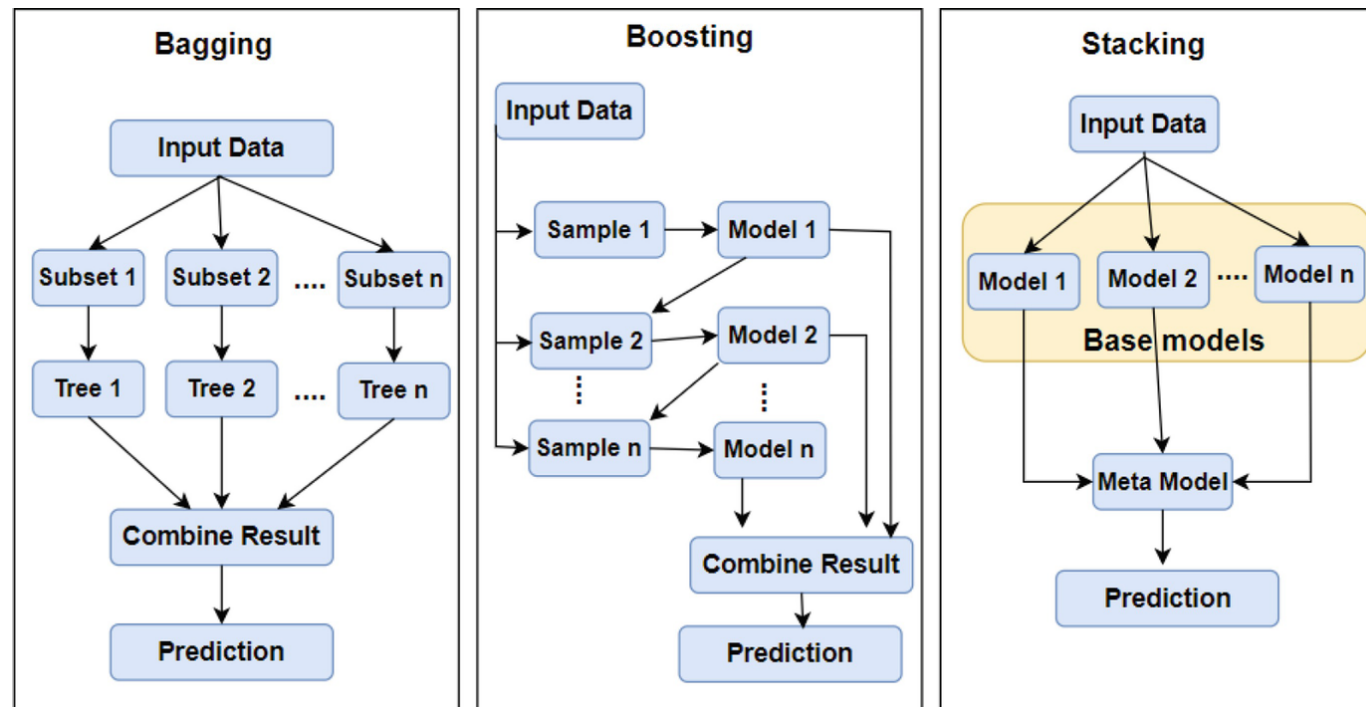
❑ Is an ensemble method always good?

- Ensemble methods work better than a single base model if:
 - All base models are independent of each other (i.e., diverse)
 - All base models perform better than random guessing (i.e., accurate)
 - e.g., error rate < 0.5 for binary classification
- Ensemble methods work best with unstable base models
 - Unstable models: small variations in the training data cause big changes in the predictions
 - Usually due to high model complexity (e.g., overfitting)
 - Examples: decision tree, artificial neural networks

Classification > Ensemble

□ Three representative approaches of the ensemble method

- **Bagging**: learn base models parallelly over subsets/subspace of data
- **Boosting**: learn base models sequentially over weighted data
- **Stacking**: learn different types of base models, whose results are ultimately learned by a meta model



Outline

- ❑ Classification evaluation
- ❑ REVISIT: Cluster/outlier evaluation
- ❑ **Ensemble classifiers**
 - **Bagging**
 - Boosting

Classification > Ensemble > Bagging

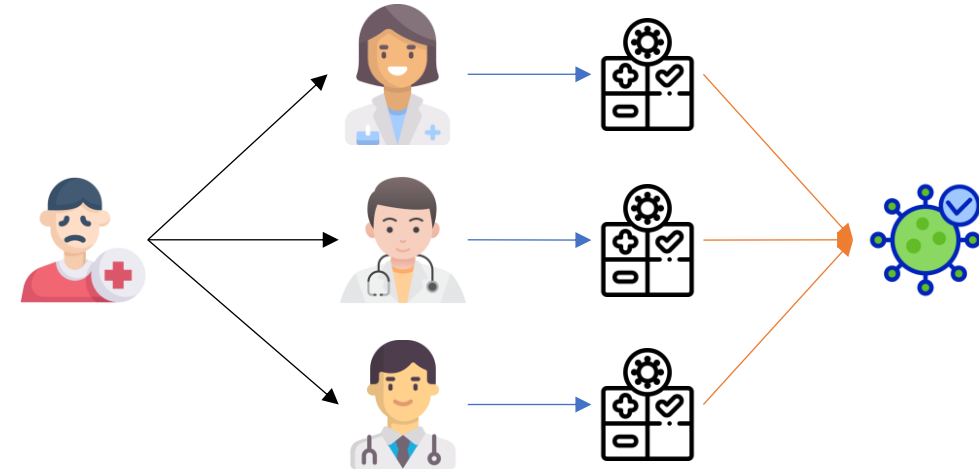
□ Bootstrap AGGREGatING

▪ Analogy

- *Diagnosis based on the parallel discussion by multiple doctors*

▪ Procedure

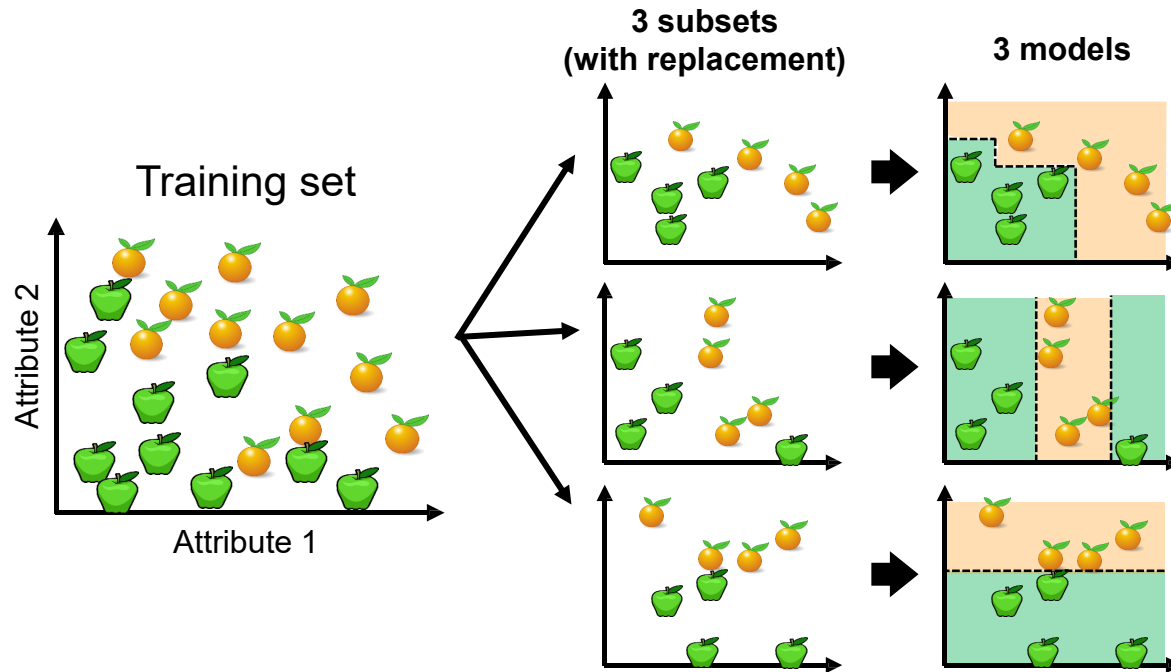
- Training for each model $i = 1$ to k
 - Data bagging
 - > Choose a bootstrap sample D_i by sampling with replacement from all data D and learn a model M_i
 - Feature bagging
 - > Choose a random subspace F_i from a set of attributes F and learn a model M_i
- Testing for an unknown sample
 - Test with each of k models and return the majority vote (or average value) on the predicted class labels (or values)



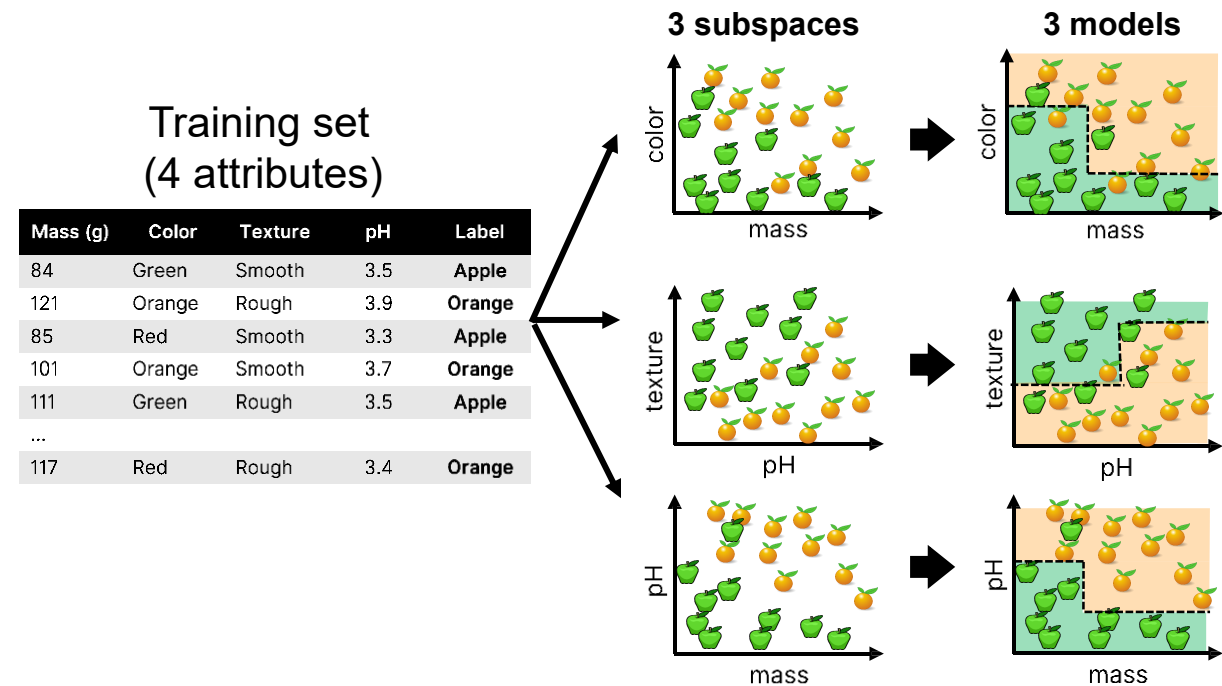
Classification > Ensemble > Bagging

□ Examples of bagging

Subset bagging



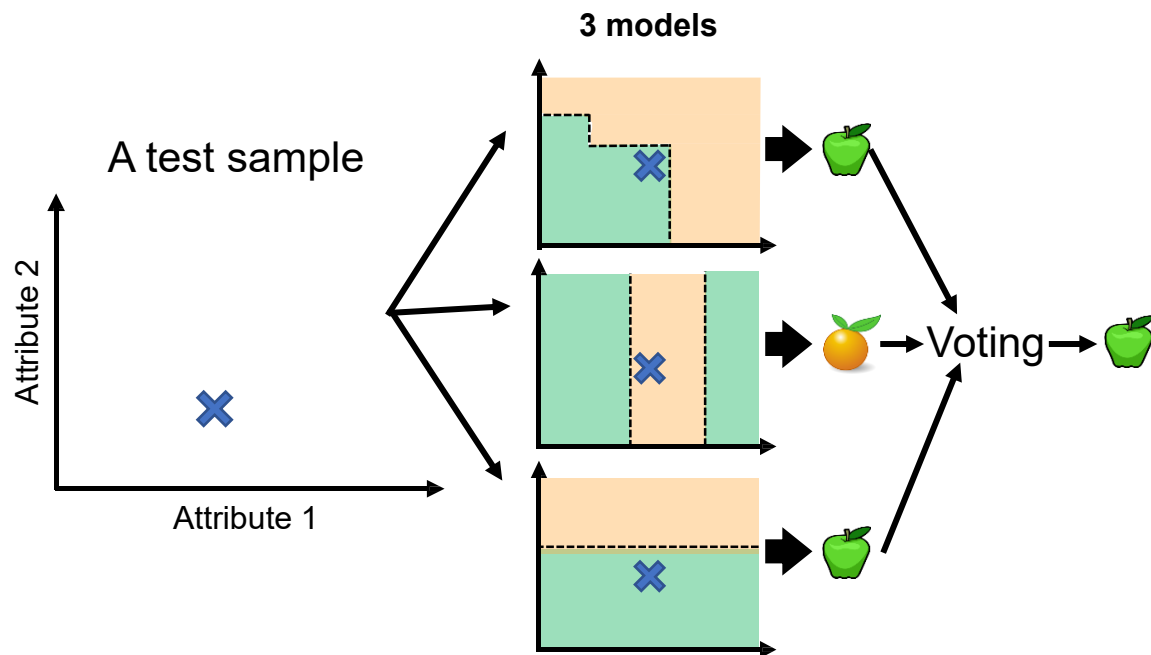
Feature bagging



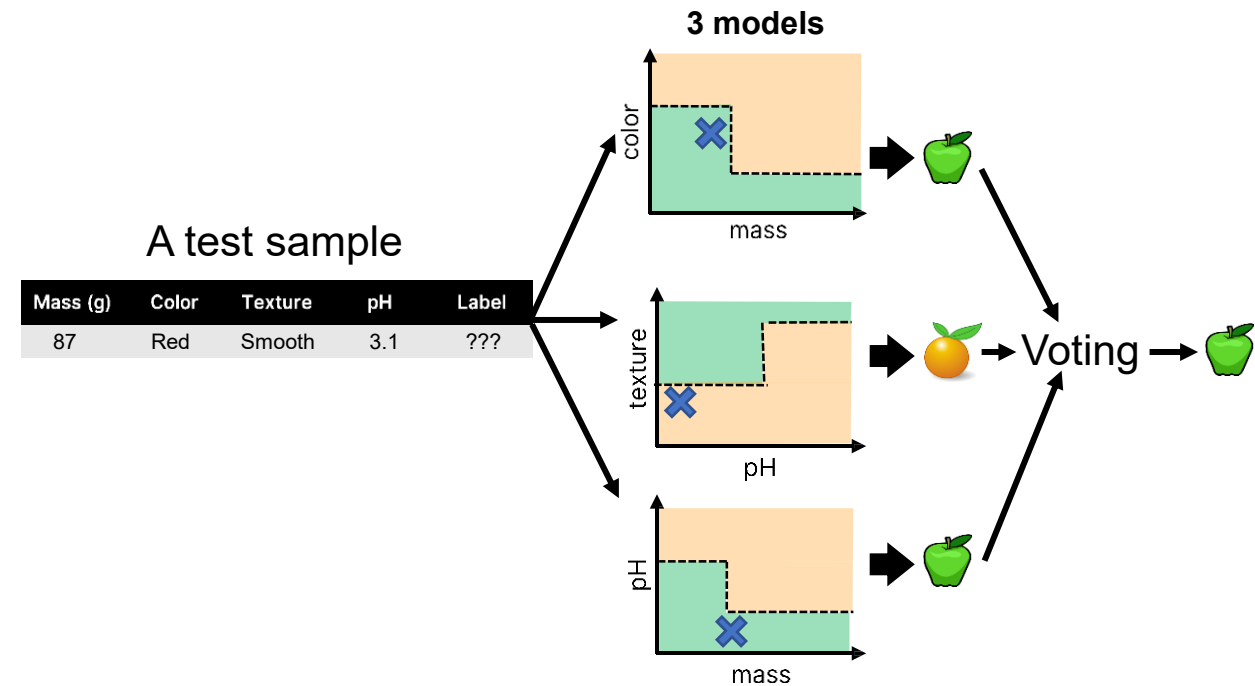
Classification > Ensemble > Bagging

□ Examples of bagging

Subset bagging



Feature bagging



Classification > Ensemble > Bagging

□ Random Forest

- Data bagging + Feature bagging + Decision Tree

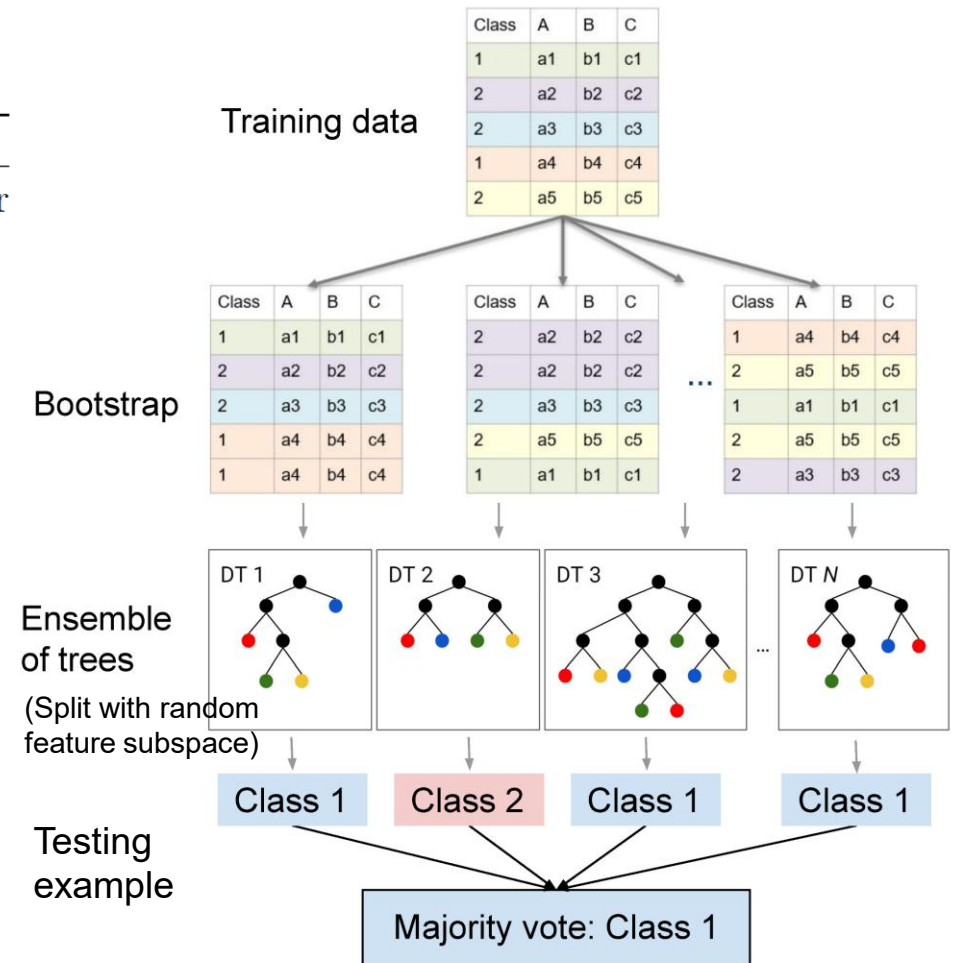
Algorithm 1 Random Forest

Precondition: A training set $S := (x_1, y_1), \dots, (x_n, y_n)$, features F , and number of trees in forest B .

```

1 function RANDOMFOREST( $S, F$ )
2    $H \leftarrow \emptyset$ 
3   for  $i \in 1, \dots, B$  do
4      $S^{(i)} \leftarrow$  A bootstrap sample from  $S$ 
5      $h_i \leftarrow$  RANDOMIZEDTREELEARN( $S^{(i)}, F$ )
6      $H \leftarrow H \cup \{h_i\}$ 
7   end for
8   return  $H$ 
9 end function
10 function RANDOMIZEDTREELEARN( $S, F$ )
11   At each node:
12      $f \leftarrow$  very small subset of  $F$ 
13     Split on best feature in  $f$ 
14   return The learned tree
15 end function

```



Outline

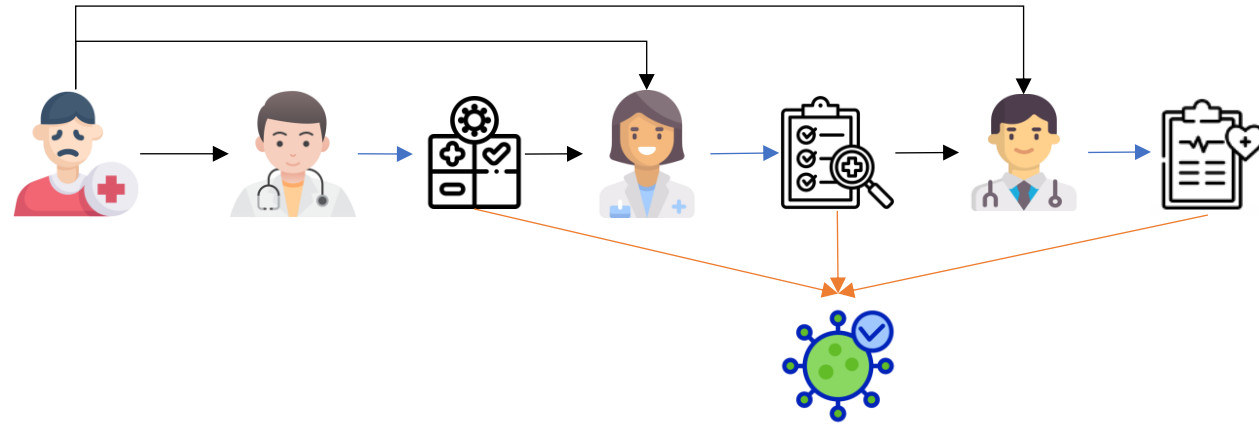
- ❑ Classification evaluation
- ❑ REVISIT: Cluster/outlier evaluation
- ❑ **Ensemble classifiers**
 - Bagging
 - **Boosting**

Classification > Ensemble > Boosting

□ Boosting

▪ Analogy

- *Diagnosis based on the sequential discussion by multiple doctors*

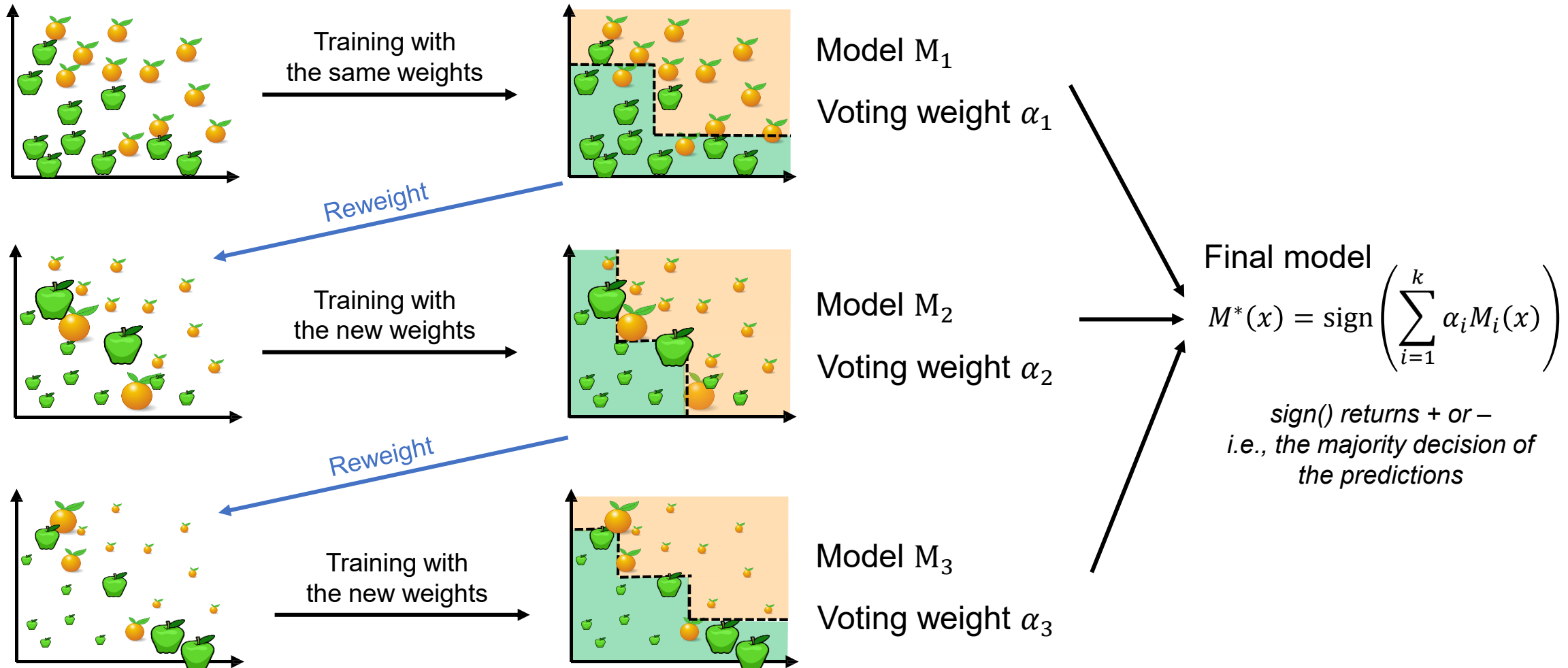


▪ Procedure

- Set the same initial weight to all data D (i.e., D_0)
- Training for each model $i = 1$ to k
 - Build a new classifier M_i with the weighted data D_i
 - Identify the data misclassified by M_i and give more weight to the data (i.e., D_{i+1})
- Testing for an unknown sample
 - Classifiers are weighted by their accuracy
 - Test with each of k models and return the weighted vote (or average value) on the predicted class labels (or values)

Classification > Ensemble > Boosting

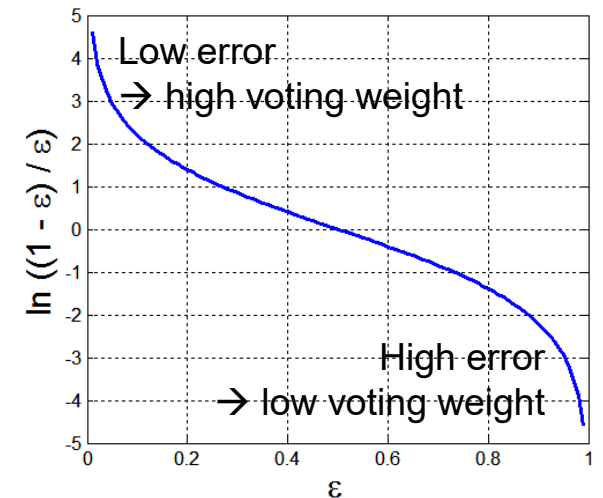
□ Examples of boosting



Classification > Ensemble > Boosting

□ AdaBoost (Adaptive Boosting)

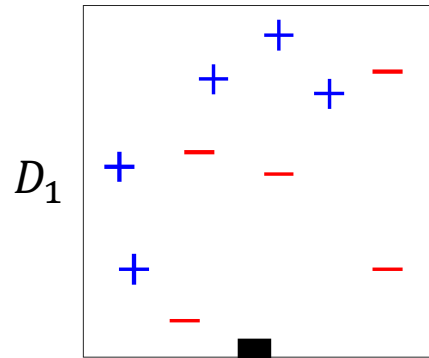
- Given a training set $D = \{(x_j, y_j)\}$ for $j = 1 \dots n$, initialize all weights $\{w_j^{(1)}\}$ of data to $1/n$
- For round $i = 1$ to k ,
 - Fit a classifier M_i to a training subset D_i sampled according to $\{w_n^{(i)}\}$
 - Evaluate the model M_i 's **error rate** $\epsilon_i = \frac{\sum_{j=1}^n w_j^{(i)} I(M_i(x_j) \neq y_j)}{\sum_{j=1}^n w_j^{(i)}}$
 - $I()$: indicator function [True, False] $\rightarrow$ [1, 0]
 - Stop iteration if $\epsilon_i < \text{threshold}$
 - Evaluate the model M_i 's **voting weight** $\alpha_i = \frac{1}{2} \ln \left(\frac{1-\epsilon_i}{\epsilon_i} \right)$
 - **Update weights** of data and normalize (e.g., to be in $[0,1]$)
 - $w_j^{(i+1)} = w_j^{(i)} \exp\{\alpha_i \cdot \text{sign}(M_i(x_j) \neq y_j)\} / z$ (e.g., z is the normalization factor for $\sum_j w_j^{(i)} = 1$)
- The final model is given by weighted voting of models with α_i
 - $M^*(x) = \text{sign}(\sum_{i=1}^k \alpha_i M_i(x))$



Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers

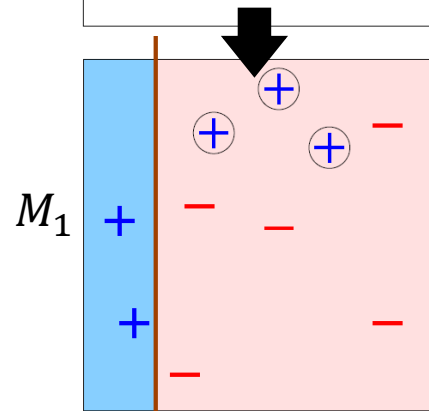
Weighted data at iteration 1



Weights of all samples

$$w_j^{(1)} = \frac{1}{10} = 0.1$$

Initial data weight $\{w_j^{(1)}\} = \frac{1}{n}$

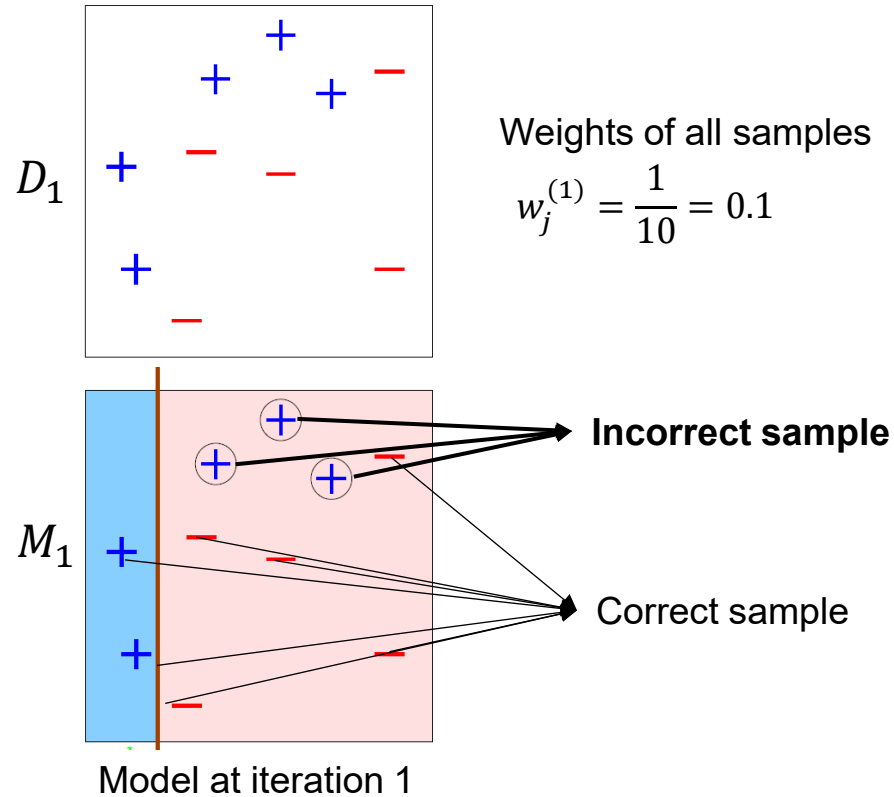


Model at iteration 1

Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers

Weighted data at iteration 1

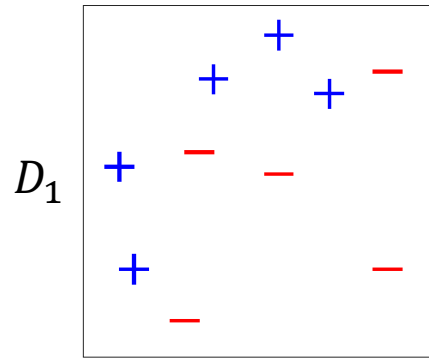


Initial data weight $\{w_j^{(1)}\} = \frac{1}{n}$

Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers

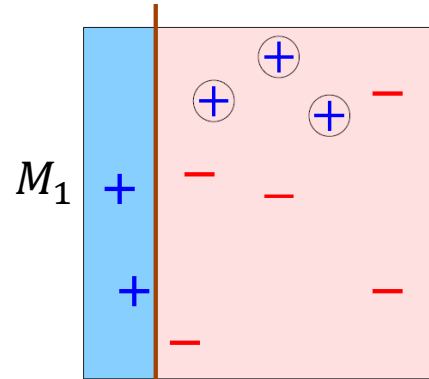
Weighted data at iteration 1



Weights of all samples

$$w_j^{(1)} = \frac{1}{10} = 0.1$$

Initial data weight $\{w_j^{(1)}\} = \frac{1}{n}$



Error rate of the model M_1

$$\epsilon_1 = \frac{3 \times 0.1}{10 \times 0.1} = 0.3$$

$$\text{Error rate } \epsilon_i = \frac{\sum_{j=1}^n w_j^{(i)} I(M_i(x_j) \neq y_j)}{\sum_{j=1}^n w_j^{(i)}}$$

Voting weight of the model M_1

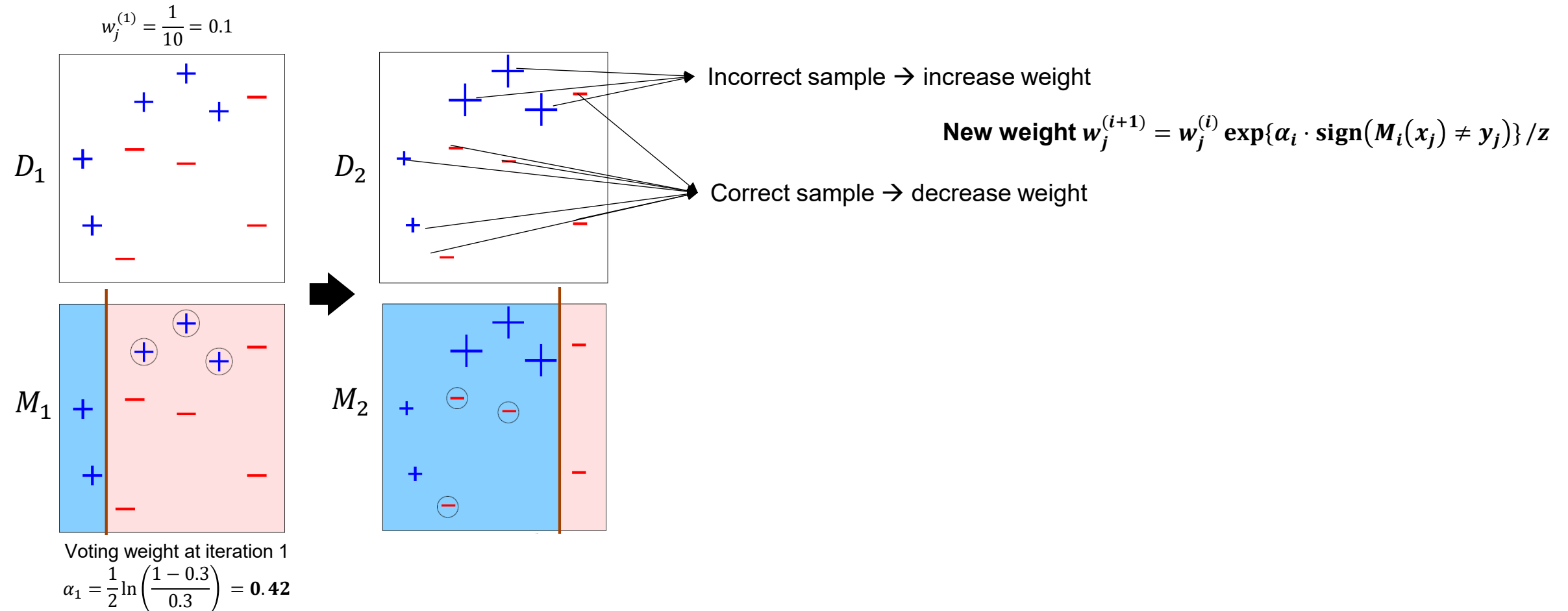
$$\alpha_1 = \frac{1}{2} \ln \left(\frac{1 - 0.3}{0.3} \right) = 0.42$$

$$\text{Voting weight } \alpha_i = \frac{1}{2} \ln \left(\frac{1 - \epsilon_i}{\epsilon_i} \right)$$

Model at iteration 1

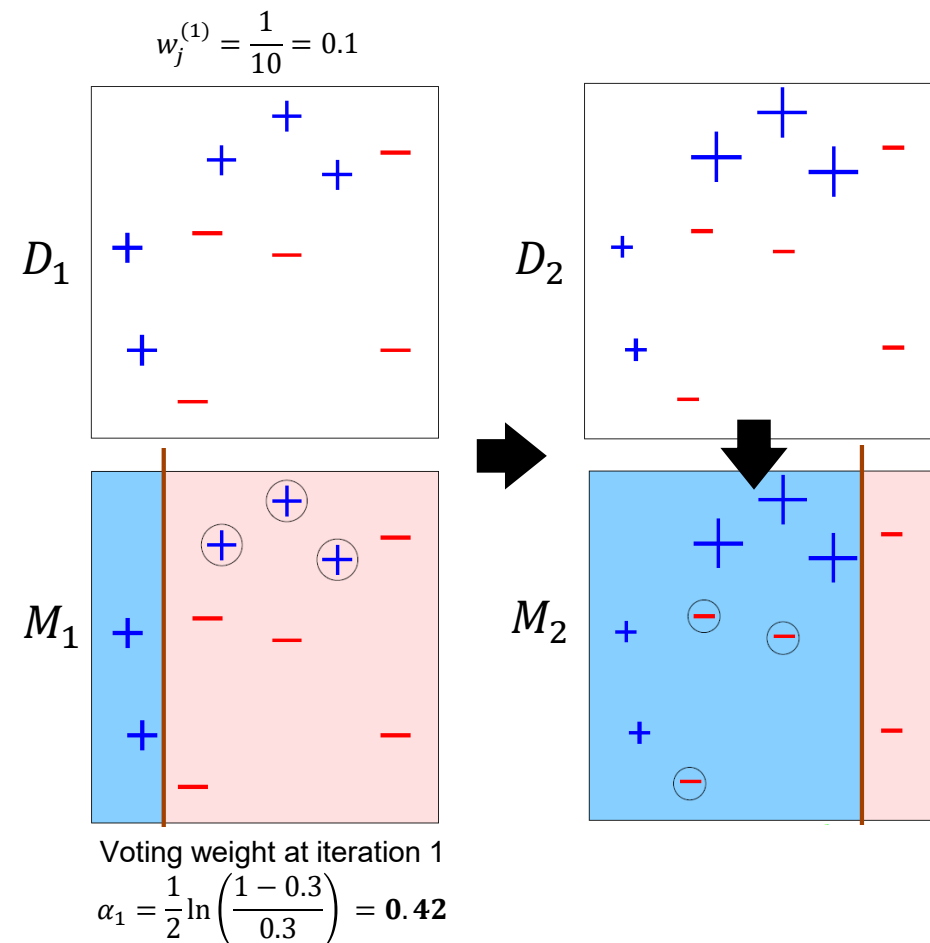
Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers



Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers



New weight $w_j^{(i+1)} = w_j^{(i)} \exp\{\alpha_i \cdot \text{sign}(M_i(x_j) \neq y_j)\} / z$

Unnormalized new weights for

- previously incorrect samples: $w_j^{(2)'} = w_j^{(1)} \exp(0.42) = 0.1522$
- previously correct samples: $w_j^{(2)'} = w_j^{(1)} \exp(-0.42) = 0.0657$

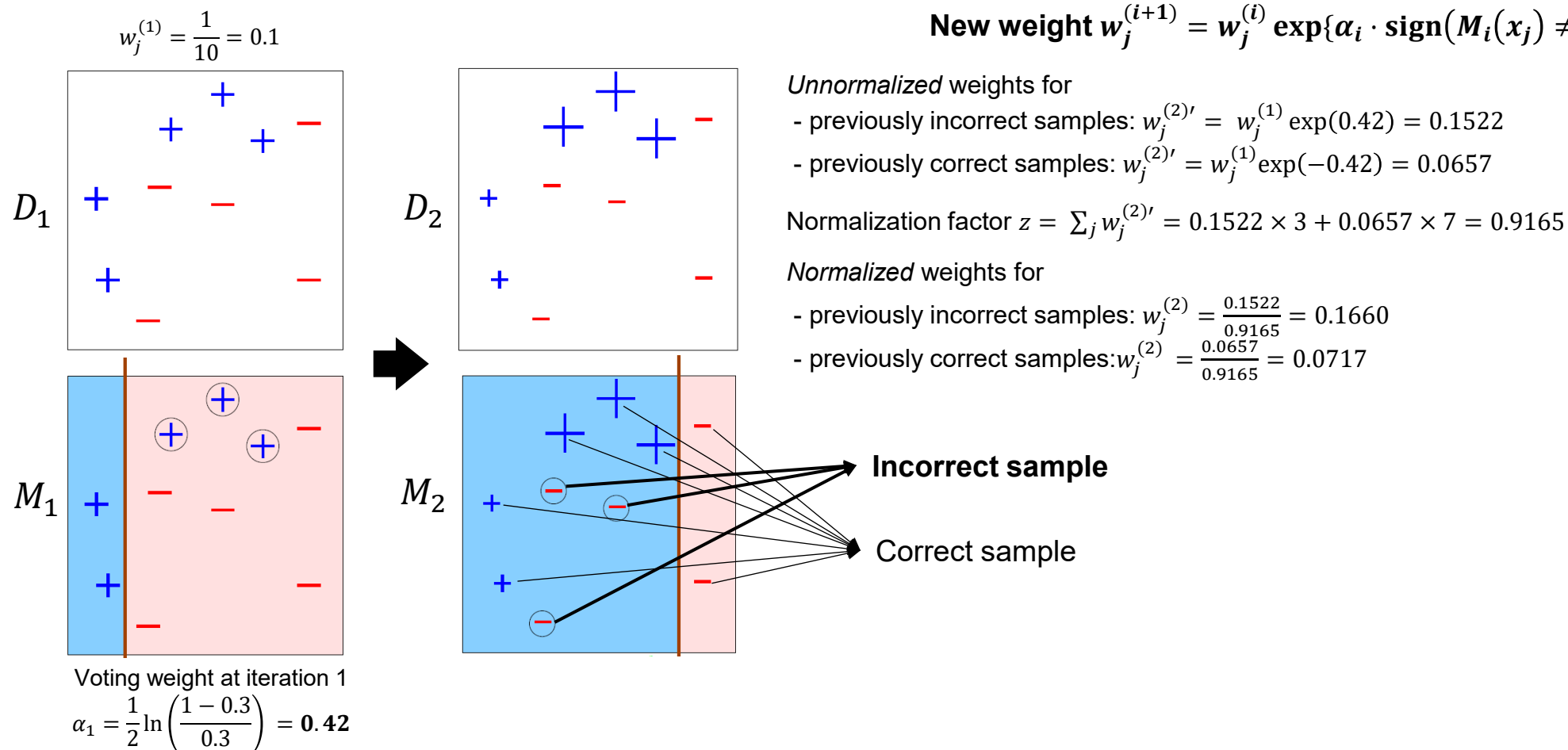
Normalization factor $z = \sum_j w_j^{(2)'} = 0.1522 \times 3 + 0.0657 \times 7 = 0.9165$

Normalized new weights for

- previously incorrect samples: $w_j^{(2)} = \frac{0.1522}{0.9165} = 0.1660$
- previously correct samples: $w_j^{(2)} = \frac{0.0657}{0.9165} = 0.0717$

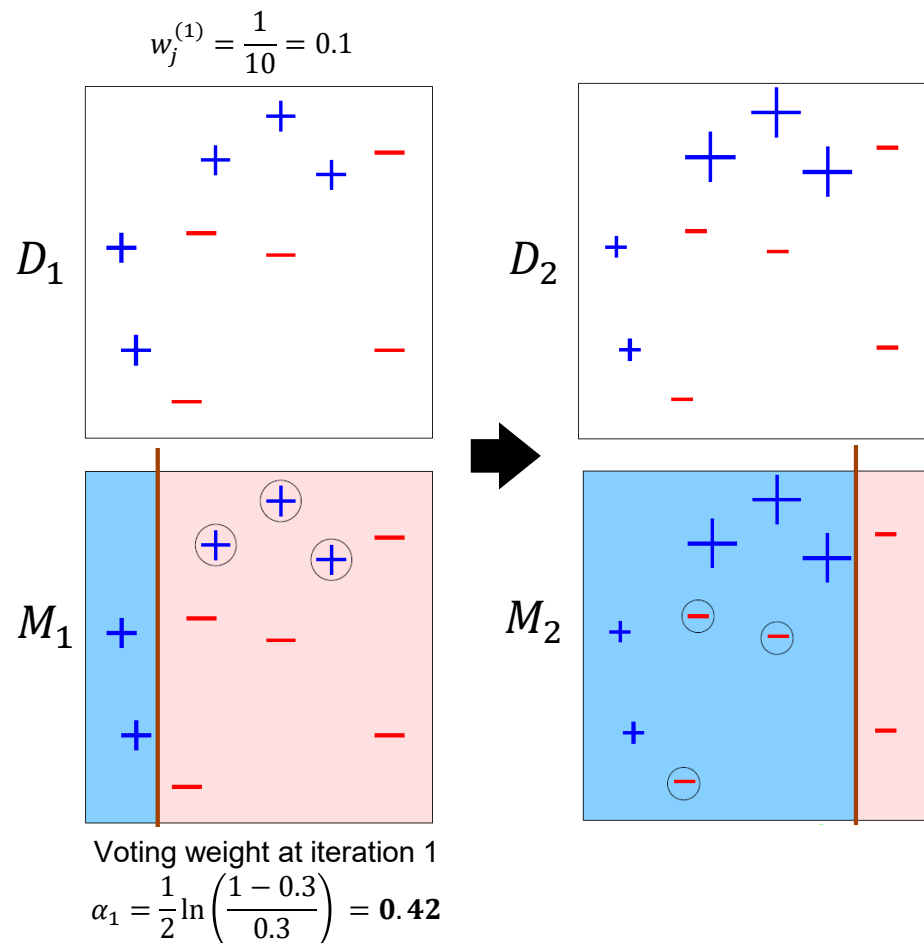
Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers



Classification > Ensemble > Boosting

□ Example of AdaBoost with 10 samples and 3 base classifiers



$$\text{New weight } w_j^{(i+1)} = w_j^{(i)} \exp\{\alpha_i \cdot \text{sign}(M_i(x_j) \neq y_j)\} / z$$

Unnormalized weights for

- previously incorrect samples: $w_j^{(2)'} = w_j^{(1)} \exp(0.42) = 0.1522$
- previously correct samples: $w_j^{(2)'} = w_j^{(1)} \exp(-0.42) = 0.0657$

Normalization factor $z = \sum_j w_j^{(2)'} = 0.1522 \times 3 + 0.0657 \times 7 = 0.9165$

Normalized weights for

- previously incorrect samples: $w_j^{(2)} = \frac{0.1522}{0.9165} = 0.1660$
- previously correct samples: $w_j^{(2)} = \frac{0.0657}{0.9165} = 0.0717$

Error rate of the model M_2

$$\epsilon_2 = \frac{3 \times 0.0717}{3 \times 0.1660 + 7 \times 0.0717} = 0.213$$

$$\text{Error rate } \epsilon_i = \frac{\sum_{j=1}^n w_j^{(i)} I(M_i(x_j) \neq y_j)}{\sum_{j=1}^n w_j^{(i)}}$$

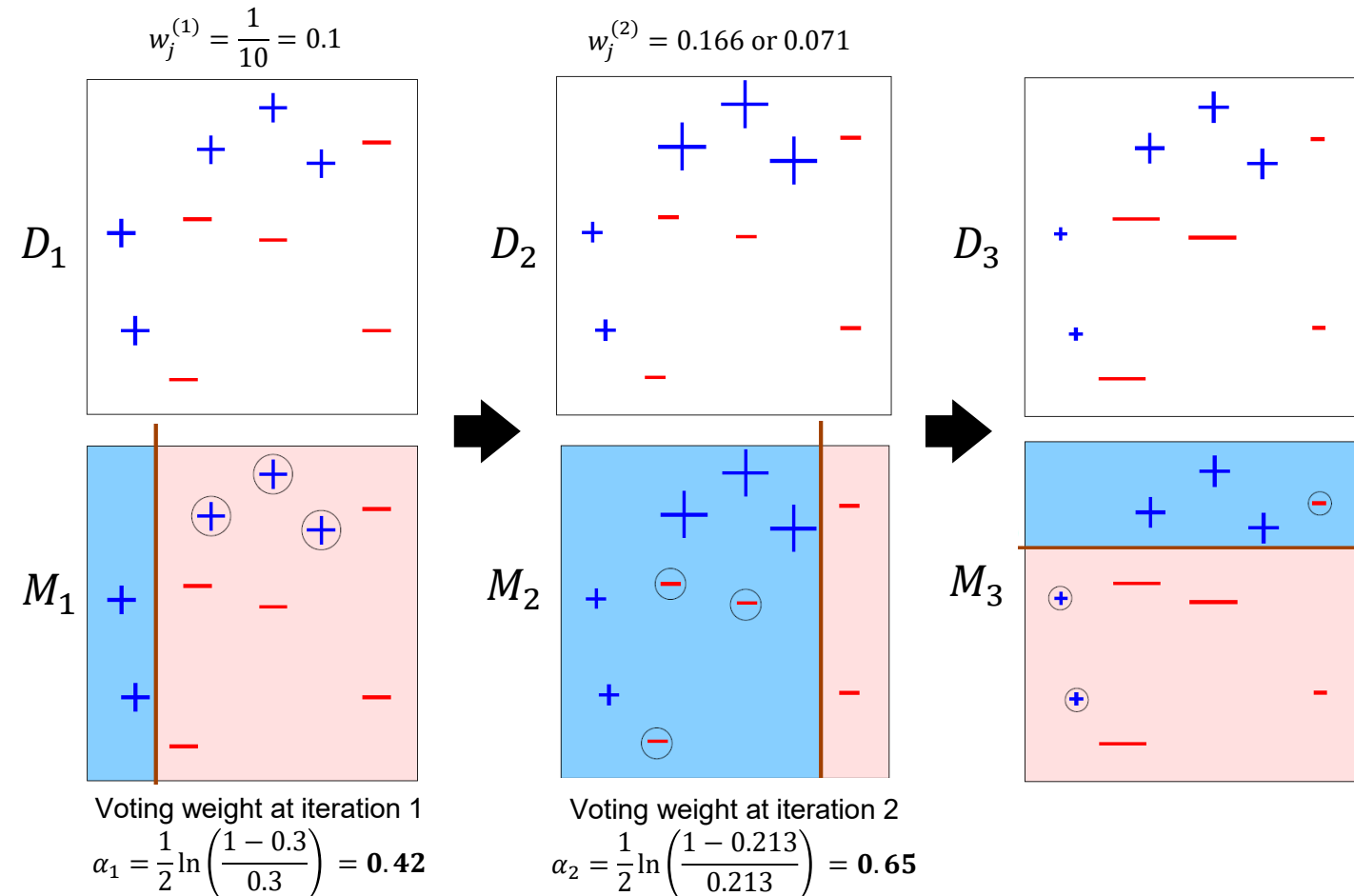
Voting weight of the model M_2

$$\alpha_2 = \frac{1}{2} \ln \left(\frac{1 - 0.213}{0.213} \right) = 0.65$$

$$\text{Voting weight } \alpha_i = \frac{1}{2} \ln \left(\frac{1 - \epsilon_i}{\epsilon_i} \right)$$

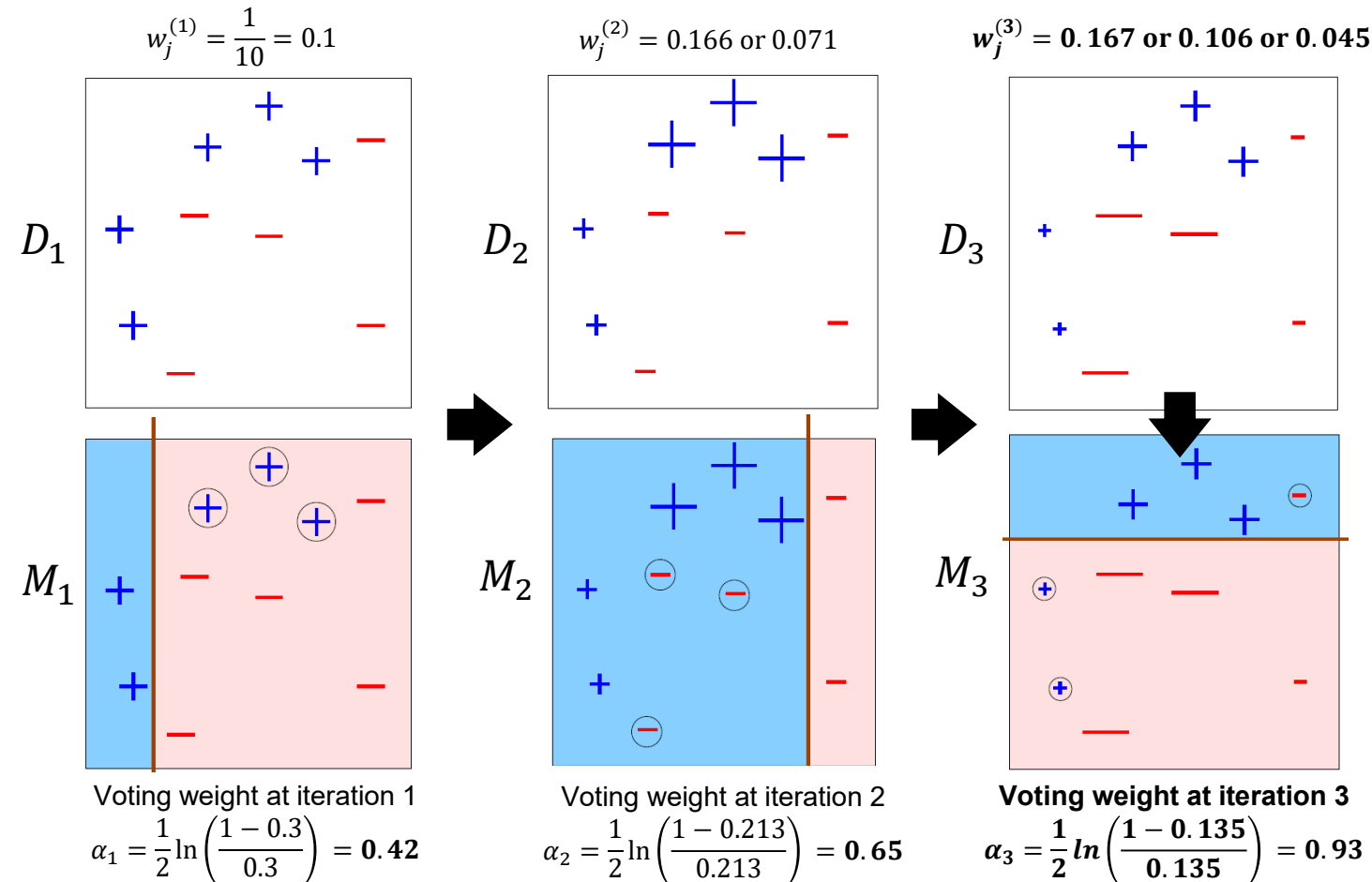
Classification > Ensemble > Boosting

□ AdaBoost – example of 10 samples and 3 base classifiers



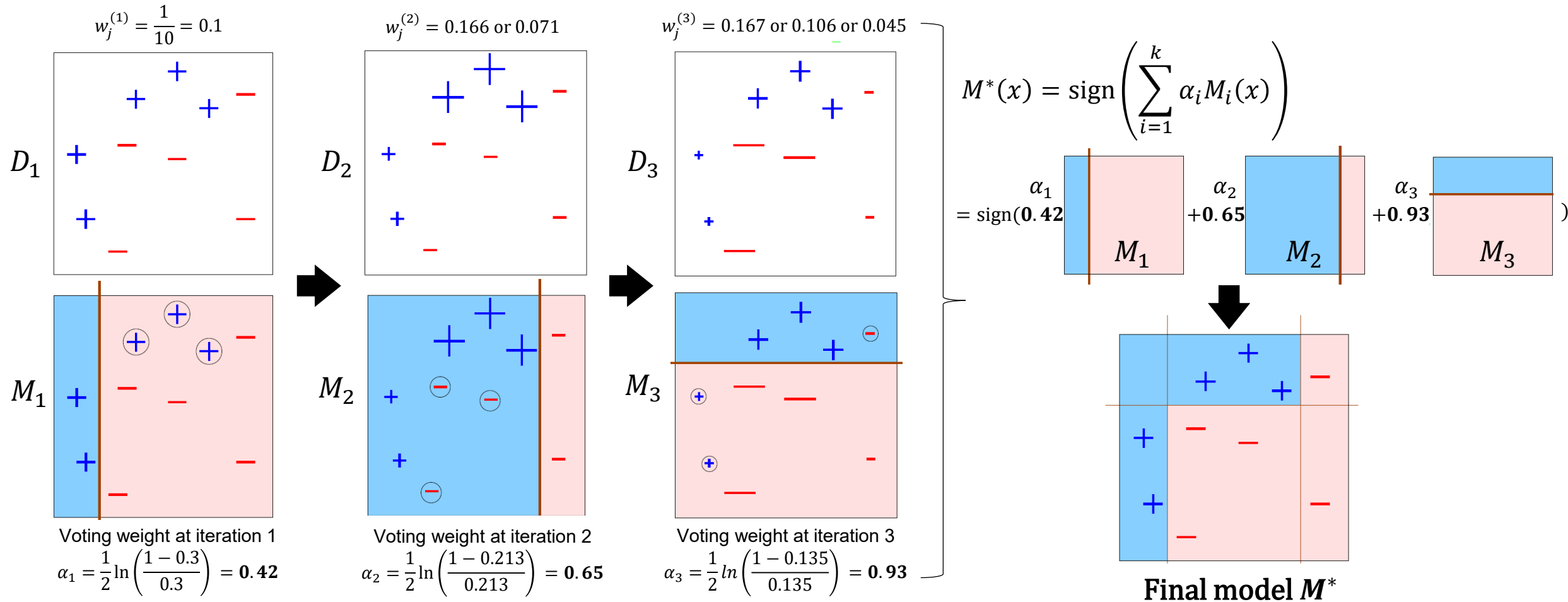
Classification > Ensemble > Boosting

□ AdaBoost – example of 10 samples and 3 base classifiers



Classification > Ensemble > Boosting

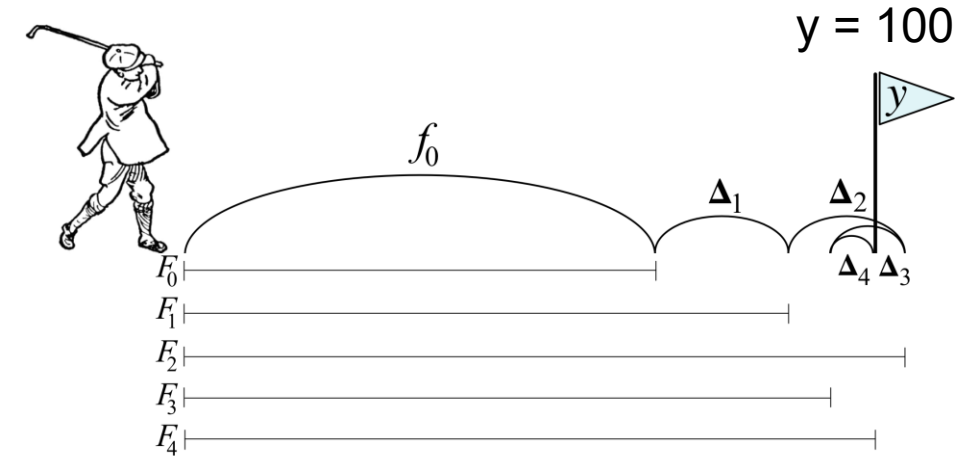
AdaBoost – example of 10 samples and 3 base classifiers



Classification > Ensemble > Gradient Boosting

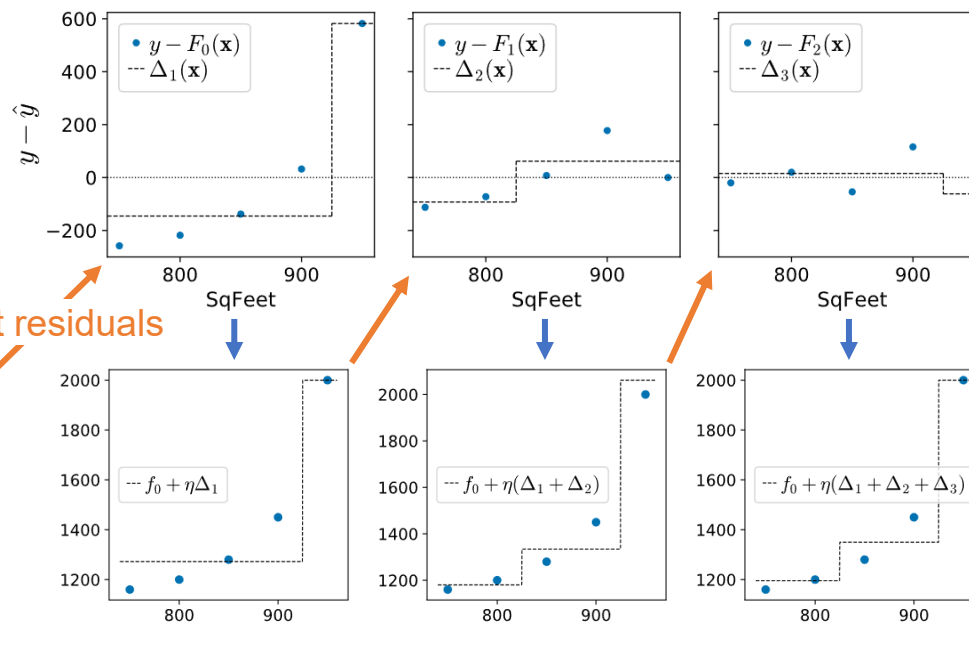
□ Gradient Boosting Model (GBM)

- Learn models sequentially to predict the error residuals
 - Analogy: *a golfer repeatedly tapping a ball towards the hole*
- Scalable variants: **XGBoost**, LightGBM
- Example



Initial model (average)

sqfeet	rent	F_0	$y - F_0$
750	1160	1418	-258
800	1200	1418	-218
850	1280	1418	-138
900	1450	1418	32
950	2000	1418	582



Stage m	Boosted Model	Model Output $\hat{y}$	Train Δ_m on $y - F_{m-1}$	Noisy Prediction Δ_m
0	F_0	70		
1	$F_1 = F_0 + \Delta_1$	$70 + 15 = 85$	$100 - 70 = 30$	$\Delta_1 = 15$
2	$F_2 = F_1 + \Delta_2$	$85 + 20 = 105$	$100 - 85 = 15$	$\Delta_2 = 20$
3	$F_3 = F_2 + \Delta_3$	$105 - 10 = 95$	$100 - 105 = -5$	$\Delta_3 = -10$
4	$F_4 = F_3 + \Delta_4$	$95 + 5 = 100$	$100 - 95 = 5$	$\Delta_4 = 5$

Algorithm: $l2boost(X, y, M, \eta)$ returns model F_M

Let $F_0(X) = \frac{1}{N} \sum_{i=1}^N y_i$, mean of target y across all observations

for $m = 1$ **to** M **do**

Let $\mathbf{r}_{m-1} = \mathbf{y} - F_{m-1}(X)$ be the residual direction vector

Train regression tree Δ_m on $\mathbf{r}_{m-1}$, minimizing squared error

$F_m(X) = F_{m-1}(X) + \eta \Delta_m(X)$

end

return F_M

Hands-on Tutorials

❑ Examples

- [kNN, SVM, and Logistic regression](#)
- [Decision tree](#)
- [Ensemble \(Random forest, Adaboost\)](#)
- [Evaluation \(confusion matrix, ROC, AUC\)](#)
- [XGBoost](#)

❑ Library

- Scikit-learn
 - [k nearest neighbors](#)
 - [Logistic regression](#)
 - [Support vector machines](#)
 - [Decision tree](#)
 - [Ensemble](#)
 - [Cross validation](#)
 - [Evaluation metric](#)
- [XGBoost](#)

Cookie Slides – Multi-class Classification Evaluation

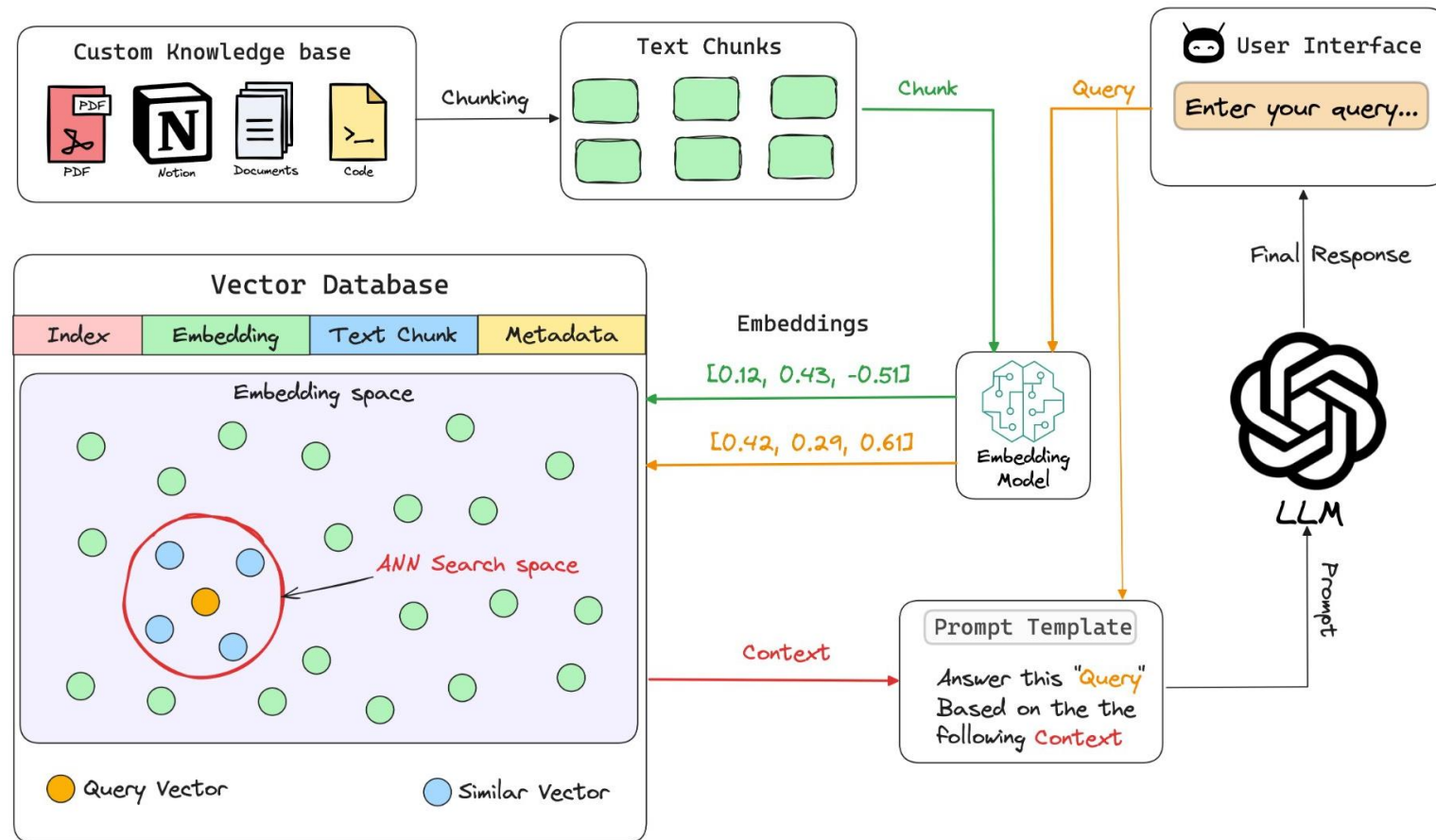
❑ Confusion matrix for multi-class classification

- Generalization of confusion matrix to multiple classes
- Example with three classes: A, B, C

		Predicted					Positive class = A					Positive class = B					Positive class = C				
		A	B	C			A	B	C			A	B	C			A	B	C		
Actual	A	N_{AA}	N_{AB}	N_{AC}	➔	A	TP_A	FN_A		A <th>A</th> <td>TN_B</td> <td>FP_B</td> <td>TN_B</td> <td data-kind="parent" data-rs="3">B<th>A</th><td colspan="2">TN_C</td><td>FP_C</td></td>	A	TN_B	FP_B	TN_B	B <th>A</th> <td colspan="2">TN_C</td> <td>FP_C</td>	A	TN_C		FP_C		
	B	N_{BA}	N_{BB}	N_{BC}		B	FP_A		TN_A		B	FN_B	TP_B	FN_B	B	TN_C		FP_C			
	C	N_{CA}	N_{CB}	N_{CC}		C	FP_A				C	TN_B	FP_B	TN_B	C	FN_C	TP_C				

Cookie Slides – Eager Learning + Lazy Learning

- ❑ Retrieval-augmented generation (RAG) with k-nearest neighbors
 - Given an **eagerly learned** pre-trained model and a query, it **lazily adapts** with a few relevant examples



Cookie Slides – LLM-powered Few-shot Classification

1. Tabular data with k labeled rows

age	education	gain	Income
39	Bachelor	2174	≤50K
36	HS-grad	0	>50K
64	12th	0	≤50K
29	Doctorate	1086	>50K
42	Master	594	?

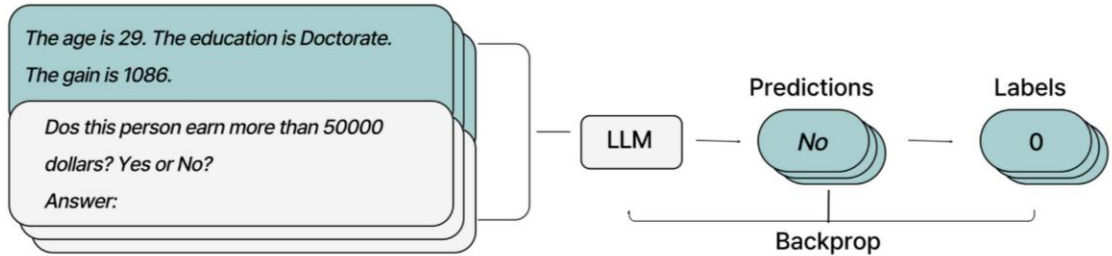
2. Serialize feature names and values into natural-language string

*The age is 42. The education is Master.
The gain is 594.*

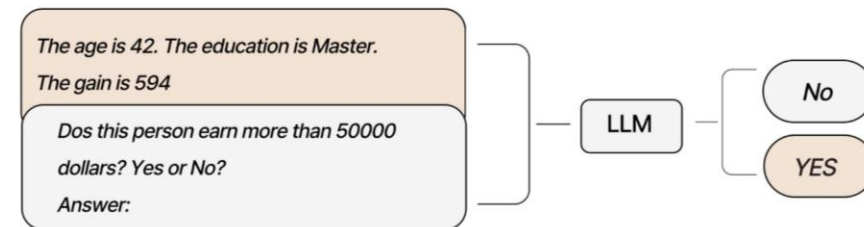
3. Add task-specific prompt

*Does this person earn more than 50000
dollars? Yes or No?
Answer:*

4a. Fine-tune LLM using labeled examples



4b. Use LLM for prediction on unlabeled examples



Cookie Slides – LLM-powered Few-shot Classification

